

# **Water Treatment Manual**

Site Operations Reference

Created by Oliver Peters · June 2026

# Section 0 — Site Orientation

**Before you touch anything:** read this section completely. Understanding the big picture makes every other procedure make sense.

## 0.1 What This System Actually Does

This is **not** a municipal drinking water plant. It is an **industrial site water treatment system** — its job is to clean contaminated site water (stormwater, process water, and groundwater collected from the industrial site) to a standard that meets discharge regulations before that water is released or reused.

The source water is not river or lake water — it is water that has been in contact with industrial operations. That means it carries:

| What's in it                         | Why it's a problem                                                                   |
|--------------------------------------|--------------------------------------------------------------------------------------|
| Dissolved iron (Fe <sup>2+</sup> )   | Turns water orange/brown; clogs filters; must be oxidized and filtered out           |
| Petroleum hydrocarbons (oils, fuels) | Toxic to aquatic life; regulated discharge limit; requires oil/water separation      |
| High suspended solids / turbidity    | Carries other contaminants; must be settled and filtered out                         |
| pH imbalance (too high or too low)   | Affects chemical treatment steps; affects downstream ecology                         |
| Fine silts and colloidal particles   | Too small to settle on their own — need coagulants and flocculants to clump together |

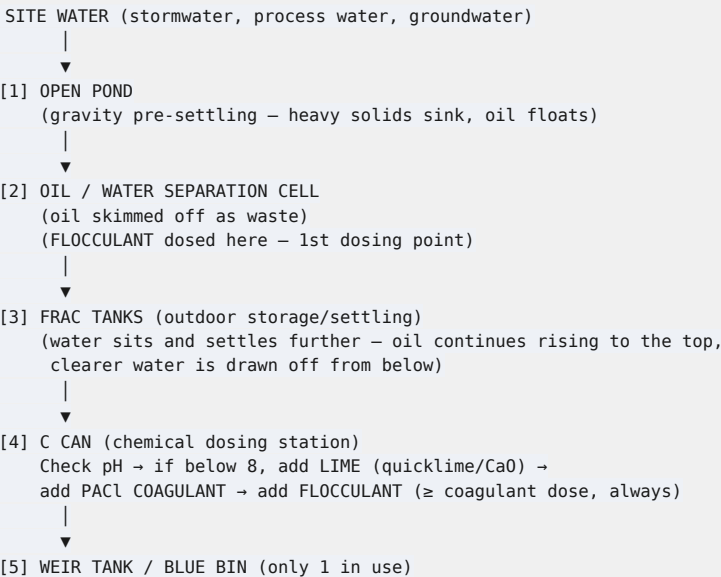
**The goal:** treated water clear enough to go into the storage tanks, where it's reused on site (e.g., dust control) — **the operator's job ends once treated water reaches the storage tank; what happens to it afterward is not the operator's responsibility to track.** **Note — CONFIRMED BY SUPERVISOR (Brendan):** This site does **not have a permit to discharge to the drain.**

 **CORRECTED BY OPERATOR:** Earlier notes here described a final "trucked to the lagoon for disposal" step after the storage tanks. **"The lagoon" is just another name for the Oil/Water Separation Cell (Section A) — there is no separate final-disposal lagoon step in the operator's job.** See Section I for the operator's actual last task (sludge removal).

 **WHY GETTING IT CLEAR STILL MATTERS:** In Brendan's words: **"The main focus is getting the water clear."** Clarity is still the operational target — partly because the carbon vessel at the end of the line "will grab anything [ozone/upstream steps don't]," so getting the water visually clear upstream protects that last line of defense, and partly because clear water is what the dust-control reuse actually needs.

## 0.2 The Treatment Train (Overview)

 **CONFIRMED** — current, live process



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    (settles – visual check: as clear as tap water)
    |
    ▼
[6] OZONE ON, then SUBMERSIBLE PUMP starts (in that order)
    (pump pulls water mid-depth into the intake pipe;
     ozone injected into the water as it travels that pipe indoors)
    |
    ▼
[7] INDOOR SETTLING BINS / WET CELL
    (settles again – same visual check)
    |
    ▼
[8] BAG FILTER
    (sand filter is NOT used at this site – this is the first
     filtration step the water actually hits)
    |
    ▼
[9] CARBON VESSEL (SRS-CV-15)
    (activated carbon – last line of defense for dissolved
     fuel smell/colour)
    |
    ▼
[10] STORAGE TANKS (Tank 1 / Tank 2)
    (also used on-site for dust control – THE OPERATOR'S JOB ENDS HERE.
     What happens to the water after this point is not the operator's
     concern/responsibility.)
    |
    ▼ (parallel process, not part of the main line – the operator's
       actual last task each cycle)
       SLUDGE MANAGEMENT
       (AODD diaphragm pump → Kontek filter press, to clear
        sludge out of the settling bins/weir tank after treated
        water has been pumped to the storage tanks)

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**i** TERMINOLOGY NOTE – “the lagoon”: **“Lagoon” is just another name for the Oil/Water Separation Cell, step [2] above — it is NOT a separate final-disposal destination.** There is no confirmed step where treated water is trucked or dosed at a separate lagoon after the storage tanks; that idea came from an earlier, unconfirmed account and is corrected here.

**⚠** CONFIRMED BY OPERATOR (Oliver) – flow rate vs. settling time: Don't let the incoming flow rate run too high. If too much water comes in at once, steps [3]–[7] (frac tanks, C Can dosing, weir tank, indoor settling bins) don't get enough time to actually settle before water gets pushed forward — the whole train is a balancing act between flow rate in and settling time needed at each stage, not a fixed setting. This system was never built as a purpose-designed treatment plant, so this balancing has to be actively managed, not assumed.

## 0.2a Plain-English Walkthrough — How This System Actually Works, Section by Section

**✓** This is the current, confirmed, live process. Read this top to bottom for the whole picture. Photo requests are listed at the end of each section — send what you can, when you can, so each section can be checked against what's actually on site.

### Section A — Outdoor: Pond → Oil/Water Separator → Frac Tanks

1. **Open pit/pond.** Raw dirty water collects here first. It just sits — oil floats (it's lighter than water), solids sink. No equipment, gravity does the work.
2. **Oil/Water Separation Cell.** A concrete, two-sided channel — one side holds the water, the other side holds the oil that's separated out (shows up as a dark maroon/red liquid, easy to tell apart from the water side at a glance). Floating oil is skimmed off and disposed of as waste — it never reaches the clean side. **A bit of flocculant (CL-200 polymer) is added here too** — the first of the dosing points, so particles already start clumping this early.
3. **Frac Tanks** (large portable steel tanks, ~100,000 L each, borrowed from the oil/gas industry). More settling time. Clearer water is drawn off the middle/below; oil and solids stay behind. Tanks are labelled clean / dirty / in treatment — know what's in each one before you move anything.

**i** WHY THIS ORDER: Oil has to come out first — it's toxic to discharge and fouls everything downstream. Starting the floc dose this early means you're not waiting until indoors to begin clumping fine particles.

**📷 Send photos of:** the pond, the oil/water separator, and the frac tanks (with their clean/dirty/in-treatment labels if visible).

## Section B — How Water Actually Moves (no fixed pipes)

Nothing between outdoor tanks is permanently plumbed. A **submersible pump** (motor + pump sealed in one waterproof housing, e.g. the blue Sunsun-style unit on site) is physically carried to whichever tank needs to be emptied, dropped in fully underwater, and a **2" hose** is run to the tank that needs to receive the water. The same pump and 2" hose setup is used in both the frac tank and the weir tank — same equipment, just moved to wherever it's needed. Once it's sitting in the bottom of a tank, all you see from outside is the hose fitting at the corner where it connects — that's normal, the pump itself is fully submerged.

**i WHY THIS MATTERS:** Before assuming water is flowing somewhere, check — is a pump actually sitting in a tank right now, and where does its hose go? Nothing moves on its own here.

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## Section C — The C Can: Making and Adding the Chemicals

Water gets pumped from the frac tank into the **C Can** (the white sea can / shipping container, parked right beside the weir tank and hose-connected to it). This is where the real chemistry happens.

Inside the C Can, three drums hold the made-up stock chemicals, side by side: **far left = flocculant** (white drum), **middle = PACI coagulant** (white drum), **right = lime** (blue drum).

### Order of operations:

1. **Check the pH first.** Below 8 is not good for the next steps.
2. **If pH is under 8, add lime.** This is quicklime (CaO) — confirmed by Brendan, even though the bag just says "lime." Lime is only added when the pH reading actually comes back low — not every batch automatically.
3. **Add PACI coagulant.** Neutralizes the charge on fine particles so they can stick together.
4. **Add flocculant (CL-200) right after.** Nets the coagulated particles into bigger, heavier clumps ("floc") that can settle. **Rule: flocculant dose must always be equal to or more than the coagulant dose — never less.**

### How to make each chemical (mixed as stock solutions, not dosed dry):

- **Lime:** 1 bag (50 lbs) per 200L of water. Must be kept **continuously mixing** the whole time — it settles out and hardens if you stop. Gets hot when wetted (it's caustic) — goggles, chemical gloves, dust mask.
- **PACI (coagulant):** 1 bag of dry powder per 100L of clean water (not the dirty water you're treating). Mix it, then **let it sit for 3 hours before using it** — don't dose straight from a fresh batch.
- **Flocculant (CL-200):** Ratio is 1 kg polymer per 1000L of clean water (or 500g per 500L at the smaller batch size you actually use). The mixing order matters:
  1. Pour in the **first half** of the water.
  2. **Turn the mixer on.**
  3. Pour in the **polymer** and the **second half of the water** at the same time, while the mixer is running.
  4. Keep mixing until it's fully dissolved.
  5. **Let it sit for 24 hours before using it** — plan a batch a day ahead so you don't run out mid-shift.

**i WHY POUR IT THIS WAY:** Adding the polymer with water flowing in around it (instead of dumping it into still water) keeps it from clumping into unusable lumps — this is the same reason dry polymer is never added straight into the treatment stream.

**i WHY NO FIXED DOSE NUMBER:** The dose changes daily depending on what's actually come into the system that day. You dose by watching for floc actually forming, not a fixed setpoint — adjust the dosing pump's stroke rate/length to get there.

**📷 Send photos of:** inside the C Can, the three dosing pumps, the lime bags/PACI bags/flocculant drum, and the mixing totes.

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## Section D — Outdoor Weir Tank (Blue Bin): Settling

**i TERMINOLOGY:** There are two tanks with "weir" in the name at this site — this **outdoor** one (the blue bin) and an **indoor** one (Section F). Always say "outdoor" or "indoor" when logging or talking about them so it's clear which one you mean.

Dosed water flows from the C Can into the outdoor blue weir tank (only 1 bin in use) — a blue intermodal-style bin/tank parked right next to the C Can, marked "NO GARBAGE" (it's not a waste bin, just repurposed equipment). It sits and settles.

**i HOW LONG:** No fixed time — it's visual. Water should look **as clear as tap water** before you move on. Without the chemical dosing from Section C, this tank does almost nothing on its own — it just drops the biggest particles.

Sludge collects on the tank floor over time and needs the diaphragm (AODD) pump to remove it — a regular submersible pump would clog on it.

**📷 Send photos of:** the weir tank itself, and the sludge/diaphragm pump if you can get a shot of it in use.

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## Section E — Ozone, Then Pump Indoors

**Order matters here: ozone ON first, then start the pump.** Turning the pump on first means untreated water gets pulled in before ozone is actually doing anything to it.

1. Turn the ozone generator **on**.
2. Then start the 2" submersible pump, placed mid-depth in the outdoor weir tank (avoids the sludge on the bottom, avoids the oil layer on top — the cleanest water is in the middle).
3. As water gets pulled through the intake pipe into the building, ozone is injected into it along the way. The intake pipe itself is white PVC — it's the pipe entering the building wall where the outdoor weir tank pump's hose connects to it.

 **WHY:** Ozone is a strong oxidizer — it breaks down dissolved hydrocarbons and turns dissolved iron into a solid form, while the water is in transit through the pipe.

 **Send photos of:** the ozone generator, the submersible pump, and the intake pipe.

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## Section F — Indoor Weir Tank / Settling Bins

Water arrives here through a white PVC pipe coming down from overhead — that pipe is the delivery point for water that's already been through the C Can, outdoor weir tank, and ozone. It lands and settles again — same visual rule as Section D: clear as tap water before moving on, and check NTU here too. Once it settles, a submersible pump is hooked into this tank to push the water onward through the bag filter and carbon filter to the storage tank. Sludge builds up here too, and gets handled the same way (AODD pump → Kontek filter press, which squeezes the water out; the dry leftover is waste, the squeezed-out water rejoins the system).

 **Send photos of:** the indoor weir tank / settling bins.

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## Section G — Filters: Bag Filter, Then Carbon Filter

**We only use the bag filter and the carbon filter — the sand filter is not used at this site.**

1. **Bag filter** — black cylindrical housing with a pressure gauge, 100-micron pre-filter bag, then a 10-micron polishing bag. Since the sand filter isn't used, this bag filter alone is catching everything ahead of carbon — check and change bags more often because of that.  **CONFIRMED BY OPERATOR:** the submersible pump's hose connects to the camlock elbow at the base of this housing — water is pushed up into the housing, through the bag, and back down, exiting through a **black hose** that carries it on to the carbon vessel.
2. **Carbon filter (vessel, SRS-CV-15)** — blue pressure vessel, 75 PSIG max, tagged "PV-SERIES SRS-CV-15." Activated carbon grabs dissolved fuel smell/colour that filters can't catch (dissolved = mixed in like sugar, not a particle). Backwashed weekly as routine maintenance; checked every 1.5 months for exhaustion (smell/colour returning means the carbon needs replacing — backwashing can't fix exhausted carbon).  **CONFIRMED BY OPERATOR:** the black hose enters near the base of this vessel; the **green hose**, also near the base, carries water onward via a PVC pipe to the storage tank.

 **BEFORE STARTUP:** This carbon vessel currently has **sludge/buildup inside with old, spent carbon**. It needs a full clean-out before any water treatment starts — this is a one-time job, separate from the routine weekly backwash above.

**Turbidity (NTU) is checked at both the bag filter outlet and the carbon vessel outlet** — these are the site's real compliance check points.

 **Send photos of:** the bag filter housing (with bags), and the carbon vessel.

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## Section H — Storage Tanks

Treated water goes into Tank 1 / Tank 2 — two large white cylindrical tanks, side by side, each with its own ladder, labelled on the side. It's also the source used for on-site dust control (sprayed on roads/yard).

 **Send photos of:** Tank 1 and Tank 2.

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## Section I — Final Task: Sludge Removal

 **CONFIRMED BY OPERATOR:** **The job ends once treated water is pumped into the storage tank.** What happens to the water after that is not the operator's responsibility — that's somebody else's call, not something to track or worry about here.

The actual last task in the cycle is cleaning up: once treated water has been pumped out of the indoor weir tank/settling bins into the storage tank, use the AODD diaphragm pump to pull the settled sludge out of the settling bins (and the outdoor weir tank) and send it through the Kontek filter press (Section 6.11/6.12).

**i** TERMINOLOGY NOTE: **"The lagoon" is just another name for the Oil/Water Separation Cell from Section A** — it is not a separate final-disposal pond. Older notes in this manual describing a final "discharge to the lagoon" step with an extra flocculant dose were based on an earlier, unconfirmed account and are corrected here — there is no such step in the operator's actual job.

 **Send photos of:** the sludge pump (AODD) in use, and the filter press during a cycle.

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## Section J — Logbook and Testing

Every shift, you log: date, time, operator, sample point (C Can / weir tank / bag filter / carbon vessel / storage tank), pH, NTU, whether lime was added (and how many bags), how much PACl and flocculant were added (or the stroke setting used), whether the intake pipe valve and storage tank valve were opened, whether ozone pressure/AERQUS flow looked OK, plus your observations and whatever action you took.

**i** WHY THESE COLUMNS: pH and NTU are the two real test numbers on this site. The chemical columns track what was actually dosed, since the dose changes daily by observation, not a fixed number. The valve columns exist because forgetting to open them lets pressure sit on the pipes. Observations + action taken is what makes the log useful to whoever reads it after you.

 Ready-to-print version: [ LOGBOOK.md ](../LOGBOOK.md) — print one page per shift, starting Monday.

 **Send a photo of:** the logbook itself, the pH meter, and the NTU/turbidity meter.

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**i** OPERATING PHILOSOPHY — worth remembering: "Make sure the flow coming in isn't too much or there won't be enough time for settling. It's all a balancing process because they never really made a proper treatment facility." This site runs on careful adjustment, not a fully automated system — watching, adjusting, and using judgment at every step is the actual job.

**Your job:** Keep each step running correctly so the water leaving the system is clear. If one step fails, the steps after it work harder — and sometimes can't compensate. That's why you monitor everything.

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## 0.3 A Note on Iron — What the Staining Means

You will notice heavy orange/rust staining on virtually every concrete surface in this facility — the settling tanks, wet cell walls, floors, and pipe connections. **This is normal for this site.** It is not a sign of neglect or failure.

The source water contains high levels of **dissolved iron (Fe<sup>2+</sup>)** — iron that is invisible in solution but turns orange-brown the moment it is oxidized (exposed to oxygen or hydrogen peroxide). When this oxidized iron [Fe<sup>3+</sup>] precipitates out of solution as iron hydroxide [Fe(OH)<sub>3</sub>], it coats everything it touches with that characteristic rusty stain.

**i** WHY: Understanding iron chemistry is central to this operation. The treatment system is specifically designed to first oxidize dissolved iron so it precipitates, then filter that iron precipitate out. The staining on the walls is just a record of years of iron removal doing its job.

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## 0.4 A Note on Oil/Water Separation

Petroleum hydrocarbons are less dense than water — they float. The Shell Canada oil/water separator exploits this: site water enters the separator, oil and fuel float to the top and are skimmed off, while cleaner water is drawn from below. Sludge (heavier-than-water solids) settles to the bottom.

The recovered petroleum product and sludge are managed as regulated waste — they do NOT go to the discharge point.

**i** WHY: Hydrocarbons in discharge water are toxic to fish and aquatic invertebrates. Even small concentrations (parts per million) violate discharge permits. The oil/water separator is a critical protection step.

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## 0.5 Site-Specific Information

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| Field                                  | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Site type                              | Industrial/commercial site water treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Associated operation                   | Shell Canada petroleum operations                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Location                               | Kamloops, BC Interior                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Water treatment supplier               | SUMAS Environmental Services Inc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| SUMAS contact number                   | <b>1-250-374-4151</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Regulatory framework                   | <i>BC Environmental Management Act</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Discharge permit                       | <b>CONFIRMED BY SUPERVISOR (Brendan): site has NO permit to discharge to the drain.</b> Treated water goes to the storage tanks (also used for on-site dust control).  <b>CORRECTED BY OPERATOR:</b> There is no separate final "lagoon" disposal step — "the lagoon" is just another name for the Oil/Water Separation Cell (Section A). The operator's job ends at the storage tank; what happens after is not tracked here. |
| Regulatory authority                   | BC Ministry of Environment and Climate Change Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Testing equipment limitation           | <b>CONFIRMED BY SUPERVISOR (Brendan): "They don't even have the proper equipment to test for dissolved contents."</b> This is a known, named gap, not an oversight in this manual — site testing currently relies on visual clarity/turbidity rather than dissolved-analyte lab testing on site. Brendan noted he's requested this repeatedly without it being provided.                                                                                                                                        |
| Site supervisor                        | Brennan ( <i>confirm full name and contact</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| On-call operators                      | Rob, Glen ( <i>confirm contact numbers</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Emergency contact — regulatory hotline | ( <i>confirm with supervisor</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## 0.6 Site Layout Walkthrough

On your first day, physically walk the entire site with a supervisor and locate each item below. Check it off when you know exactly where it is and what it looks like.

### Outdoor area:

- ☐ Primary settling pond (turbid green/grey water — first pond)
- ☐ Secondary/stabilization pond (dark red/iron-rich water — second pond)
- ☐ Frac tanks (large portable above-ground steel storage tanks — note which are "clean" vs. "dirty")
- ☐ Shell Canada above-ground oil/water separator
- ☐ Underground holding tank access port (for dipping the OWS underground tank)
- ☐ Site drainage and stormwater collection points

### Indoor building (wet cell / chemical room):

- ☐ Wet cell / separation cell (concrete multi-compartment indoor settling tanks)
- ☐ Chemical dosing room — PACI drum (white, Class 8 label) + blue diaphragm pump
- ☐ Chemical dosing room — CL-200 polymer drum (white, "FLOCCULANT" label) + yellow LMI pump
- ☐ Other chemical drums (hydrogen peroxide, CO2 cylinder, pH+ powder pail, HaloKlear)
- ☐ SRS-SF-11 sand filter (green pressure vessel on legs, pump below — present on site but not used, see Section 6.4)
- ☐ SRS-CV-15 carbon vessel (blue PV Series vessel on pallet)
- ☐ Bag filter housing (blue cartridge housing with Les Hall filter bags)
- ☐ CD30TIX ozone generator (stainless steel cabinet with pressure gauge)
- ☐ AERQUS ozone diffuser (in-line or in-tank)
- ☐ Accu-Tab chlorinator and Cal-Hypo tablet storage
- ☐ Kontek APC5 filter press (blue/white frame, ~30 filter plates, blue control panel)
- ☐ AODD pump (aluminum body, air-operated, for sludge transfer)
- ☐ Portable pH meter (grey case) and calibration buffer solutions (pH 4.0, 7.0, 10.0)

### Safety locations (memorize these — do not just check them off):

- ☐ Safety shower and eyewash station(s) — know the closest one to the chemical room
- ☐ Chemical spill kit(s)
- ☐ Fire extinguisher(s)
- ☐ First aid kit
- ☐ All Safety Data Sheets (SDS binder or posted location) — know where these are for EVERY chemical on site
- ☐ Electrical panels and LOTO (Lockout/Tagout) points for each major piece of equipment

## 0.7 The "Brennan's List" — Understanding the Basic Operational Flow

A handwritten operational note on site (referred to as "Brennan's List") captures the day-to-day task sequence:

*"Transfer dirty water from least full frac tank to most full frac tank. Transfer water from Wet Cell through carbon or sand filter to clean frac tank. Get wet cell as water free as possible to transfer solids out. Treat water, fill up storage tanks, collect sample. Transfer oil and dirty water from wet cell to Stabilization pit."*

In plain terms, the operating cycle is:

1. **Consolidate dirty water** — move raw/untreated water so it's in the right tanks for processing.
2. **Filter the wet cell water** — push water from the wet cell through the bag and carbon filter to the clean frac tank.
3. **Empty the wet cell** — get it as dry as possible so the accumulated sludge/solids can be pumped out.
4. **Treat and store** — dose chemicals, run the full treatment train, fill clean storage tanks, collect a compliance sample.
5. **Move oil and dirty residuals** — transfer separated oil and dirty wet cell water to the stabilization pit for further management.

**i** WHY: This system operates in **batch mode** — it processes water in batches rather than as a continuous flow. Understanding which tanks hold what type of water, and in what order to process them, is the core operational skill on this site.

## 0.8 Who Does What

| Role                        | Responsibilities                                             | Name / Contact                                            |
|-----------------------------|--------------------------------------------------------------|-----------------------------------------------------------|
| Site Supervisor             | Operational decisions, compliance, directing daily tasks     | Brennan ( <i>confirm full contact</i> )                   |
| Operator                    | Daily rounds, chemical dosing, equipment checks, log entries | Rob ( <i>confirm contact</i> )                            |
| Operator                    | Daily rounds, chemical dosing, equipment checks, log entries | Glen ( <i>confirm contact</i> )                           |
| Chemical/Equipment Supplier | Chemical deliveries, equipment support, technical advice     | SUMAS Environmental Services Inc. — <b>1-250-374-4151</b> |

## 0.9 How to Read This Manual

- **Bold text** = a term defined in the Glossary (Section 12).
- **[ACTION]** boxes = something you physically do.
- **[CHECK]** boxes = something you observe and record.
- **⚠ WARNING** = potential for injury, equipment damage, or compliance violation.
- **i WHY** callouts = explains the reason behind a step (don't skip these — understanding why helps you catch problems).
- (*confirm with supervisor*) = information specific to this site that was not confirmed at time of writing — ask Brennan or call SUMAS before proceeding if this is unclear.

Last updated: 2026-06-27 | Source: Direct operator confirmation (Brendan/Brennan) — supersedes earlier former-operator accounts where they conflicted.



# Section 1 — Daily Operations

Daily operations are the heartbeat of the plant. Every shift, you walk a predictable route, check the same things, and record what you see. Consistency is what catches problems early.

## 1.1 The Golden Rule of Shift Work

Leave the plant in better shape than you found it.

Before you leave, your relief operator should be able to open your logbook and understand exactly what happened — and what to watch.

## 1.2 Shift Start Routine (First 30 Minutes)

### Step 1 — Review the logbook and task list

[ACTION] Pick up the logbook and any task notes (e.g., "Brennan's List") from the previous shift. Read every entry.

Look for:

- Any equipment that is offline, locked out, or bypassed
- Frac tank levels and status (clean / dirty / in treatment)
- Any chemical shortages flagged
- Ongoing treatment batches in progress — what stage are they at?
- Unusual water quality (high turbidity, pH out of range, petroleum odour in effluent)
- Tasks carried over to your shift

**i WHY:** This system runs in batches. Water is staged across multiple tanks at different treatment stages. If you don't read the handoff, you won't know where each batch is, and you could mix clean water with dirty water — a costly mistake.

### Step 2 — Visual check of control panel and equipment indicators

[ACTION] Walk to each piece of powered equipment and note its status.

[CHECK]

| Equipment                           | Expected status                         | Actual |
|-------------------------------------|-----------------------------------------|--------|
| PACI dosing pump (blue)             | Running / standby per dosing plan       | _____  |
| CL-200 flocculant pump (yellow LMI) | Running / standby per dosing plan       | _____  |
| CD30TIX ozone generator             | ON indicator lit; pressure gauge normal | _____  |
| AERQUS ozone diffuser               | Operating (bubbling in contact water)   | _____  |
| Bag filter housing                  | Flow OK, bag not overloaded             | _____  |
| Kontek filter press (if in cycle)   | Status per previous shift note          | _____  |
| H <sub>2</sub> O <sub>2</sub> drums | Level — note on log                     | _____  |
| Cal-Hypo tablets in chlorinator     | Level — confirm not empty               | _____  |

**i WHY:** This system does not have centralized SCADA — you are the monitoring system. Each piece of equipment must be individually checked.

### Step 3 — Physical rounds (indoor and outdoor)

Walk the full site in the same order every time. Use all your senses.

**Outdoor:**

- [ ] Settling ponds: level, oil sheen, debris blocking outlet?
- [ ] Frac tanks: levels, labels current (clean/dirty/in treatment)?
- [ ] OWS (oil/water separator): any visible petroleum around the housing?
- [ ] Any hoses or connections showing leaks?

**Indoor chemical/treatment room:**

- [ ] Any chemical spills on the floor?
- [ ] Strong chemical smell other than faint ozone or faint chlorine?
- [ ] PACI and CL-200 pump tubing — cracked, kinked, or spraying?
- [ ] Bag filter housing — any drips or leaks at the lid seal?
- [ ] SRS-CV-15 carbon vessel — petroleum odour coming from effluent?
- [ ] Wet cell — sludge accumulation visible; water clarity normal?

**Smell specifically for:**

- Petroleum/fuel smell in treated water → carbon vessel likely exhausted
- Rotten egg (H<sub>2</sub>S) smell → possible biological activity in an anaerobic sludge pocket
- Strong bleach → possible Cal-Hypo over-dose or spill
- Sharp metallic/bleach smell (ozone) → normal near ozone system; strong ozone elsewhere = ventilate

Record everything you observe, normal or not.

**Step 4 — Take daily samples**

(See Section 4 for full procedures)

| Sample Point                             | Tests to Run                                   | Frequency                                   |
|------------------------------------------|------------------------------------------------|---------------------------------------------|
| Settling pond outlet / wet cell inlet    | pH, turbidity (visual OK in field)             | Each shift                                  |
| After chemical dosing (wet cell outflow) | pH, turbidity                                  | Each shift                                  |
| Bag filter outlet                        | pH, turbidity (NTU)                            | Each shift                                  |
| Carbon vessel (SRS-CV-15) effluent       | pH, turbidity, petroleum odour check           | Each shift                                  |
| Final treated water (pre-discharge)      | pH, free chlorine, turbidity                   | Each shift — this is your compliance sample |
| Oil/water separator — weekly dip         | Fuel thickness, sludge thickness, liquid level | Weekly (Shell Canada log sheet)             |

**[ACTION]** Run tests immediately after collection. Record all results in the log with time.

**1.3 Continuous Monitoring During the Shift****Every 1–2 hours:**

- [ ] Check chemical drum/tank levels — are the dosing pumps consuming at the expected rate?
- [ ] Check frac tank levels and update status labels if any transfers occurred
- [ ] Check bag filter housing — is flow rate maintained, or is the bag loading up?
- [ ] Check ozone system — CD30TIX indicator still on? Pressure gauge normal?
- [ ] Test free chlorine in treated water — confirm residual is maintained
- [ ] Log everything — even “no change” is worth noting

**If something is out of range:**

1. Confirm it's real (re-test manually, don't just trust one reading).
2. Protect the discharge point first — if treated water quality is in question, stop the discharge valve.
3. Find the cause (Section 8 — Troubleshooting).
4. Fix or escalate to supervisor.
5. Record: time of problem, reading that flagged it, what you did, result.

**i WHY:** BC Ministry of Environment can inspect at any time and ask to see your records. A discharge that exceeds permit limits AND has no log entry looks much worse than one that was caught, logged, and corrected.

**1.4 Key Parameters and Target Ranges**

Operational targets below are typical for industrial discharge treatment. Confirm your site-specific permit limits with your supervisor — your actual permit limits govern.

| Parameter                          | Typical Discharge Limit           | Operational Target | Why It Matters                                              |
|------------------------------------|-----------------------------------|--------------------|-------------------------------------------------------------|
| Turbidity (final discharge)        | ≤ 25 NTU (typical BC permit)      | ≤ 5 NTU            | Turbidity carries other contaminants and harms fish habitat |
| pH (final discharge)               | 6.5–9.5 (typical BC permit)       | 7.0–8.5            | Too acidic or alkaline is toxic to aquatic life             |
| Total suspended solids (TSS)       | ≤ 25–75 mg/L (varies by permit)   | Minimize           | Settled solids can smother fish spawning areas              |
| Free chlorine (final discharge)    | ≤ 0.1 mg/L (protect aquatic life) | ≤ 0.05 mg/L        | Chlorine is toxic to fish even at low levels                |
| Total petroleum hydrocarbons (TPH) | ≤ 1–5 mg/L (varies)               | Not detectable     | Petroleum is toxic to aquatic life                          |
| Iron (total)                       | ≤ 0.3–1 mg/L (varies by permit)   | Minimize           | Iron precipitation in receiving waters harms fish           |

## 1.5 Shift End Routine (Last 30 Minutes)

[ACTION] Before your relief arrives:

- ☐ Complete all log entries and test results — no blanks
- ☐ Note any equipment that is offline, in manual, or acting abnormally
- ☐ Check chemical inventory — flag anything running low
- ☐ List any tasks you couldn't complete (for the next shift)
- ☐ Prepare a verbal briefing for your relief operator (5 minutes face-to-face is worth more than a page of notes)

[ACTION] Brief the incoming operator:

- What happened during your shift
- Current status of all major processes
- Anything to watch closely
- Any ongoing issues or anomalies

[ACTION] Sign the logbook. The moment you sign, you own that shift's record.

### 1.6a Log Sheet Columns (Site-Specific)

**i** Built from confirmed site procedure — use these columns going forward:

| Date | Time | Operator | Sample point (C Can / outdoor weir tank / indoor weir tank—settling bins / bag filter / carbon vessel / storage tank) | pH | Clear/cloudy? (or NTU if you have the meter) | Lime added? (Y/N + bags) | PACI added (amount or stroke setting) | Flocculant added (amount or stroke setting) | Ozone turned on before pump? | Intake pipe valve opened? | Storage tank valve opened? | AODD pump used (which tank)? | Observations / issues | Action taken |
|------|------|----------|-----------------------------------------------------------------------------------------------------------------------|----|----------------------------------------------|--------------------------|---------------------------------------|---------------------------------------------|------------------------------|---------------------------|----------------------------|------------------------------|-----------------------|--------------|
|      |      |          |                                                                                                                       |    |                                              |                          |                                       |                                             |                              |                           |                            |                              |                       |              |

**i** WHY these columns: pH and clarity (or NTU) are your two real test numbers at this site (Section 4.3). Lime/PACI/flocculant columns track what was actually dosed, since the dose changes daily by observation, not a fixed number. The ozone and valve columns exist because forgetting them either skips treatment or lets pressure sit on the pipes (Section 6.1a, 6.7). The AODD column tracks sludge removal — your actual last task each cycle (Section I). Observations + action taken is what makes the log useful to the next person, not just a list of numbers.

 A ready-to-print version of this table is in [ LOGBOOK.md ] (../LOGBOOK.md) — print one page per shift starting Monday.

---

# 1.6 Sample Logbook Entry

---

DATE: 2026-06-29     SHIFT: 07:30–16:00     OPERATOR: [Name]

07:35 Reviewed previous shift notes. Carbon vessel clean-out [done/in progress].  
07:45 Morning rounds complete. No leaks. C Can drums checked (lime/PACl/flocculant levels OK).  
08:00 C Can check: pH 7.1 (below 8 – added lime, 1 bag). PACl and flocculant dosed per usual setting.  
09:15 Outdoor weir tank: visually clear, settled well overnight. Ozone turned ON, then started intake pump. AERQUS flow meter ball moving, reading [X] SCFH.  
10:30 Indoor weir tank/settling bins: clear as tap water, NTU checked – within range.  
Submersible pump started, pushing water to bag filter and carbon vessel.  
12:00 Midday check – all within targets.  
13:30 Pumped treated water into storage tank. Job on this batch ends here.  
13:45 Used AODD pump to clear settled sludge from indoor settling bins and outdoor weir tank, ran through Kontek filter press. 1 cake bin filled.  
15:45 Briefed incoming operator – sludge bin needs pickup, PACl drum getting low.

Signature: \_\_\_\_\_

# Section 2 — Startup Procedures

Starting the plant (or restarting after a shutdown) is one of the highest-risk moments in operations. Rushing startup causes more problems than almost anything else. Work slowly, verify each step, never skip.

## 2.1 Types of Startup

| Type                 | When it happens                                                                      | Key difference                                        |
|----------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------|
| Cold start           | Plant has been fully offline (power outage, extended maintenance)                    | All systems from scratch; longest and most methodical |
| Warm start           | Plant briefly offline (a few hours); systems still primed                            | Shorter checklist; equipment likely still primed      |
| Unit process restart | A single unit (e.g., one filter, one clarifier) coming back online after maintenance | Only the checklist for that unit                      |

If you're unsure which type applies, treat it as a cold start. It's always safer.

## 2.2 Pre-Startup Checklist (All Types)

Complete this before turning anything on.

### Safety checks

- ☐ Confirm all lockout/tagout (LOTO) locks have been removed by the person who placed them
- ☐ Verify no personnel are inside basins, tanks, or vessels
- ☐ Confirm all manhole covers, hatches, and access points are closed and secured
- ☐ Check that all safety shower and eyewash stations are operational
- ☐ Review the logbook — any open work orders or known issues?

### Mechanical checks

- ☐ Inspect all pump packing glands / mechanical seals — no significant leaks
- ☐ Check all valve positions — confirm they match the startup configuration (refer to your P&ID)
- ☐ Verify all motor coupling guards are in place
- ☐ Confirm all chemical injection points are clear (no blockages)

### Chemical checks

- ☐ Verify chemical tanks have sufficient inventory for at least one full day of operation
- ☐ Confirm chemical dosing pump tubing is not cracked or kinked
- ☐ Prime dosing pumps if they have been dry (follow Section 2.5 below)

### Electrical / instrumentation checks

- ☐ Confirm all control panels are energized
- ☐ Check that all online instruments (turbidimeters, chlorine analyzers, flow meters) are reading and not in fault
- ☐ Confirm SCADA communications are live (if applicable)

## 2.3 Cold Start — Sequence

(This sequence follows a conventional coagulation/flocculation/sedimentation/filtration/disinfection plant. Your site-specific sequence will be refined once you provide your process flow diagrams and SOPs.)

## Phase 1: Raw Water Intake

- [ACTION] 1. Slowly open the raw water intake valve (if manual) — open no more than 25% initially.
- [ACTION] 2. Start the raw water (low-lift) pumps at minimum speed/flow.
- [CHECK] 3. Confirm flow on the raw water meter. Wait for stable flow before proceeding.
- [CHECK] 4. Watch for air release at high points in the suction line — open air release valves as needed.

**i** WHY: Starting at low flow prevents water hammer (a pressure surge that can crack pipes and damage valves). Letting air escape prevents air locks and inaccurate flow readings.

---

## Phase 2: Chemical Dosing — Coagulant

- [ACTION] 5. Start the coagulant dosing pump (typically alum or polyaluminium chloride/PACl).
- [ACTION] 6. Set the dose to your baseline starting point (from your dosing table — see Section 5).
- [CHECK] 7. Confirm the pump is actually injecting: watch the injection point for the chemical plume, or verify the tank level is dropping.

**i** WHY: You must be dosing coagulant BEFORE water enters the flocculation basin. If water passes through without coagulant, it will carry suspended solids straight to the filters — and turbidity spikes can take an hour or more to clear.

---

## Phase 3: Flocculation

- [ACTION] 8. Start flocculation mixers at low speed.
- [CHECK] 9. Observe the forming floc in the basin. Healthy floc looks like small white or grey snowflakes that grow and settle. Poor floc is pinpoint-sized or stays in suspension.

**i** WHY: Flocculation uses gentle mixing to help coagulated particles collide and stick together into larger, heavier "floc" that can settle.

---

## Phase 4: Sedimentation / Clarification

- [ACTION] 10. Confirm sedimentation basin outlet valve is open.
  - [CHECK] 11. As flow enters the basin, watch for even distribution across the inlet. Uneven flow causes short-circuiting (water bypasses settling and exits too fast).
  - [CHECK] 12. Do NOT expect good settled water quality for the first 1–2 hours of startup. The basin needs time to reach steady state.
- 

## Phase 5: Filtration

- [ACTION] 13. Before opening a filter to flow, confirm it has been backwashed recently (within the last backwash cycle). If not, run a backwash first (see Section 6).
- [ACTION] 14. Slowly open the inlet valve to the filter — no more than 25% initially.
- [CHECK] 15. Monitor the filter effluent turbidity continuously. During startup, turbidity may be elevated for 10–30 minutes ("filter ripening").
- [ACTION] 16. Do NOT send filter effluent to the distribution system until turbidity is below your target (typically  $\leq 0.3$  NTU, often  $\leq 0.1$  NTU per your permit).

**i** WHY: Filter ripening is normal — the filter media needs time to re-establish its biological and physical filtering layer. The first-flush water can contain elevated turbidity and should go to waste or back to the head of the plant.

---

## Phase 6: Disinfection

- [ACTION] 17. Start the disinfection system (chlorine dosing pump, UV system, or both — per your plant design).
- [ACTION] 18. Set the dose to your target CT value (contact time  $\times$  concentration — see Section 5).
- [CHECK] 19. DO NOT allow water to enter the distribution system until you have confirmed:

- Finished water turbidity  $\leq$  permit limit
- Free chlorine residual  $\geq$  regulatory minimum at the entry point to distribution

**⚠** WARNING: Sending under-disinfected water to the distribution system is a public health emergency. If you are unsure, recirculate or send to waste. Call your supervisor.

---

## Phase 7: Ramp Up to Normal Flow

- [ACTION] 20. Gradually increase raw water flow to the operational target over 30–60 minutes.
  - [ACTION] 21. Adjust chemical doses proportionally as flow increases (see dosing table in Section 5).
  - [CHECK] 22. Take a set of startup samples (raw, settled, filter effluent, finished water) and record in logbook.
  - [CHECK] 23. Confirm all online instruments are reading correctly and agree with your grab samples (within  $\pm 10\%$ ).
- 

## 2.4 Unit Process Restart

---

When restarting just one unit (e.g., a filter or clarifier that was taken offline for maintenance):

1. Confirm maintenance work order is signed off and LOTO removed.
  2. Run the relevant portion of the Pre-Startup Checklist above.
  3. Bring the unit online slowly while keeping the rest of the plant at stable operation.
  4. Monitor the effluent quality of that unit for 30 minutes before returning it to normal service.
  5. Record in the logbook: time offline, reason, time returned to service.
- 

## 2.5 Priming Chemical Dosing Pumps

---

If a dosing pump has been dry (empty tube/tank), you must prime it before use:

- [ACTION] 1. Ensure the chemical tank has liquid.
- [ACTION] 2. Disconnect the discharge tubing from the injection point (hold a rag over it to catch chemical).
- [ACTION] 3. Run the pump until chemical flows steadily from the discharge — no bubbles.
- [ACTION] 4. Reconnect the tubing.
- [ACTION] 5. Verify the pump is injecting correctly at the injection point.

 **WARNING:** Wear appropriate PPE (gloves, goggles, face shield) when working with dosing chemicals. See Section 9.

---

## 2.6 Startup Completion Checklist

---

Before declaring startup complete, confirm all of the following:

- ☐ All processes are running at target flow
  - ☐ Chemical doses are set and confirmed injecting
  - ☐ Finished water turbidity is within target
  - ☐ Finished water chlorine residual is within target
  - ☐ Finished water pH is within target
  - ☐ All online instruments are reading and calibrated
  - ☐ Startup logged with start time and initial readings
  - ☐ Supervisor notified if this was an unplanned startup
-

# Section 3 — Shutdown Procedures

Shutdown is the reverse of startup — but with some important differences. The goal is to leave the plant in a safe, stable, known state that can be restarted cleanly.

## 3.1 Types of Shutdown

| Type                          | Cause                                               | Urgency                     | Procedure   |
|-------------------------------|-----------------------------------------------------|-----------------------------|-------------|
| Planned shutdown              | Scheduled maintenance, holiday, seasonal            | Low — time to prepare       | Section 3.2 |
| Controlled emergency shutdown | Equipment failure, power issue, contamination event | Medium — minutes, not hours | Section 3.3 |
| Emergency stop (E-stop)       | Imminent safety hazard to people                    | Immediate                   | Section 3.4 |

## 3.2 Planned Shutdown Procedure

### Preparation (day before or hours before)

- ☐ Notify your supervisor and any downstream users if water supply will be interrupted
- ☐ Fill the clear well / storage reservoir to maximum capacity — this buffers supply during the shutdown
- ☐ Confirm chemical inventory is sufficient for restart (no point fixing equipment and having no alum)
- ☐ Document the current state of all processes (flow rates, doses, pressures) — this is your baseline to return to

### Shutdown Sequence

Work in reverse order from the distribution system back to intake. This ensures you are never pushing untreated water forward.

- [ACTION] 1. Reduce raw water flow gradually over 15–30 minutes (don't shut off suddenly — water hammer).
- [ACTION] 2. Reduce chemical doses proportionally as flow decreases.
- [ACTION] 3. Stop the raw water pump(s). Close the intake valve.
- [ACTION] 4. Stop coagulant and any pre-treatment chemical dosing pumps.
- [ACTION] 5. Allow flocculation and sedimentation basins to drain to waste or remain filled (per your SOP — keeping basins full prevents structural damage from differential pressure on walls).
- [ACTION] 6. Close filter inlet valves. Note filter head levels.
- [ACTION] 7. Reduce chlorine dosing to zero OR maintain a minimal dose if water will sit in contact chamber for an extended period.
- [ACTION] 8. Close the entry point valve to the distribution system (only when clear well level is sufficient for the shutdown period).
- [ACTION] 9. Record shutdown time, reason, and plant state in the logbook.
- [ACTION] 10. Apply LOTO to all equipment that maintenance will work on (see Section 9).

### After Shutdown

- ☐ Walk the plant — confirm all equipment is stopped that should be stopped
- ☐ Secure chemical rooms (locked, ventilated)
- ☐ Confirm clear well level is logged
- ☐ Post a "PLANT SHUTDOWN — DO NOT OPERATE" notice on the main panel if you will be away

## 3.3 Controlled Emergency Shutdown

These situations require faster action but still allow a methodical sequence:

**Triggers:** Major equipment failure (pump cavitation, motor failure), chemical feed system failure, pipe break, power fluctuation.



[ACTION] 1. Immediately assess: Is the problem affecting finished water quality?

- If YES: Close the distribution entry valve NOW, then troubleshoot.

- If NO: You may have time for an orderly shutdown.

[ACTION] 2. Stop the raw water pump(s).

[ACTION] 3. Stop chemical dosing.

[ACTION] 4. Record the time, the triggering event, and all current readings in the logbook.

[ACTION] 5. Notify your supervisor immediately.

[ACTION] 6. Follow emergency response procedures if the event involves a chemical spill, contamination, or safety hazard (Section 10).

---

### 3.4 Emergency Stop (E-Stop)

---

Use only when there is an immediate threat to human safety (fire near chemical storage, someone in the water, gas leak, electrical arc).

[ACTION] Press the E-stop button on the main panel (red, mushroom-head button).

This cuts power to all process equipment simultaneously.

 **WARNING:** E-stop does NOT close valves — water may continue to flow by gravity. After pressing E-stop, manually close the intake and distribution valves if safe to do so.

[ACTION] After the E-stop: call emergency services if needed, then notify your supervisor. Do NOT restart the plant after an E-stop without supervisor authorization and a full inspection.

---

### 3.5 Extended Shutdown (More Than 24 Hours)

---

If the plant will be offline for more than a day:

- ☐ Drain and flush chemical dosing lines to prevent crystallization (especially alum and phosphoric acid lines)
  - ☐ Leave filter media wet (drying out damages the bed structure and biological layer)
  - ☐ Maintain at least a minimal chlorine residual in the clear well to prevent bacterial regrowth
  - ☐ Set up a monitoring schedule — someone should check the plant daily even during extended shutdowns
  - ☐ If temperatures will drop below freezing: drain any lines or instruments that could freeze and crack
- 

### 3.6 Post-Shutdown Checklist

---

Before leaving the plant after any shutdown:

- ☐ Logbook entry complete (time, reason, final readings, who was notified)
  - ☐ All relevant LOTO applied and documented
  - ☐ Chemical rooms secured
  - ☐ All exterior doors locked
  - ☐ Supervisors and relevant contacts notified
  - ☐ Clear well level recorded
  - ☐ Any active alarms acknowledged and documented
- 

*Last updated: 2026-06-04 | Source: Initial framework*

# Section 4 — Sampling & Lab Testing

Your samples are the evidence that the plant is working. A bad sample gives you bad data. Bad data leads to bad decisions. Learn to sample and test correctly — it protects public health and protects you legally.

## 4.1 The Cardinal Rules of Sampling

- 1. **Take samples consistently** — same location, same method, every time.
- 2. **Label immediately** — write the sample point, date, and time on the container before you fill it, not after.
- 3. **Analyze promptly** — most water quality parameters degrade quickly. Chlorine, for example, must be tested within minutes of collection.
- 4. **Record results immediately** — don't trust memory.
- 5. **Never manipulate results** — if a result looks off, re-sample to confirm. Falsifying records is a serious offence.

## 4.2 Sample Collection Procedures

### 4.2.1 Grab Sample (Most Common)

A single snapshot in time. Suitable for turbidity, pH, temperature, chlorine, and most routine parameters.

**Equipment needed:** Sample bottle (type depends on test — see table below), gloves, marker.

- [ACTION]
- 1. Run the tap or sample tap for at least 2–3 minutes before collecting (flush stagnant water from the line).  
- *Exception:* Lead and copper samples must NOT be flushed — check your protocol.
  - 2. Hold the bottle downstream of the flow (not directly under the tap).
  - 3. Fill to the required volume. Do not fill to the brim if preservatives are needed.
  - 4. Cap immediately.
  - 5. Label: location, date, time, your initials.
  - 6. Deliver to the lab or analyze within the holding time for that parameter.

### 4.2.2 Composite Sample

Multiple grab samples combined over time to get an average. Used for parameters that vary throughout the day (e.g., BOD, certain metals).

**WHY:** A grab sample is like a photograph. A composite sample is like a time-lapse — it captures variation.

*Your site may or may not require composite sampling. Update this section when you have your sampling schedule.*

## 4.3 Sample Points and What to Test

The table below is a generic framework. Your site-specific sampling plan (from your operating permit) will be filled in once provided.

| Sample Point                       | Parameter                  | Frequency       | Holding Time        | Notes                                  |
|------------------------------------|----------------------------|-----------------|---------------------|----------------------------------------|
| Raw water intake                   | Turbidity, pH, temperature | Every shift     | Analyze immediately | Baseline for process control           |
| Raw water intake                   | Alkalinity                 | Daily           | 24 hours            | Determines coagulant dose              |
| Post-coagulation (rapid mix)       | pH                         | Each jar test   | Analyze immediately | Optimal coag pH varies by chemical     |
| Settled water (clarifier effluent) | Turbidity, pH              | Every 2–4 hours | Analyze immediately | Should be significantly lower than raw |

| Sample Point                                                            | Parameter                                                 | Frequency                                    | Holding Time        | Notes                                                                                                                                                                |
|-------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filter inlet                                                            | Turbidity                                                 | Every 2–4 hours                              | Analyze immediately | Should match settled water                                                                                                                                           |
| Filter effluent (each filter)                                           | Turbidity                                                 | Continuously (online) + manual grab 1×/shift | Analyze immediately | Key compliance parameter                                                                                                                                             |
| Finished water (entry to distribution)                                  | Turbidity, free chlorine, total chlorine, pH, temperature | Every 2–4 hours (min 1×/shift)               | Chlorine: immediate | Primary compliance point                                                                                                                                             |
| Clear well                                                              | Free chlorine, total chlorine                             | Every 4 hours                                | Chlorine: immediate | Verify residual for CT calculation                                                                                                                                   |
| Distribution system (remote points)                                     | Free chlorine, pH                                         | Per your sampling plan                       | Chlorine: immediate | Verify residual to end of system                                                                                                                                     |
| Backwash waste                                                          | Turbidity                                                 | During backwash                              | Analyze immediately | Verify when backwash is complete                                                                                                                                     |
| <b>Bag filter outlet / Carbon vessel outlet — CONFIRMED BY OPERATOR</b> | Turbidity (NTU meter)                                     | (confirm frequency)                          | Analyze immediately | Site-specific compliance point — this site checks turbidity at the bag filter or carbon vessel (sand filter is bypassed, Section 6.4), not at a sand filter effluent |

## 4.4 How to Run Each Test

### 4.4.1 Turbidity (NTU)

**Instrument:** Bench-top or portable turbidimeter (e.g., Hach 2100Q, Hach TU5).

#### [ACTION]

1. Rinse the sample cell twice with sample water (discard rinses).
2. Fill the sample cell to the fill line. Do not overfill.
3. Wipe the outside of the cell with a lint-free cloth (fingerprints affect readings).
4. Place in the instrument, cap, and press READ.
5. Record the result in NTU.

#### Common errors:

- Air bubbles in the cell → give a falsely high reading. Tap the cell gently to release bubbles before reading.
- Cell scratches → cause light scatter. Replace scratched cells.
- Instrument not calibrated → calibrate before shift per instrument manual (Section 6 will cover instrument calibration once provided).

### 4.4.2 Free and Total Chlorine Residual (mg/L)

**Method:** DPD colorimetric (most common). DPD = N,N-Diethyl-p-phenylenediamine.

#### [ACTION]

1. Collect the sample directly in the test vial — do NOT let the sample sit before testing. Chlorine degrades in minutes.
2. Fill to the correct volume (usually 10 mL).
3. Add one DPD-1 tablet (or DPD-1 powder pillow) for **free chlorine**. Cap and invert gently 3× to mix.
4. Within 60 seconds, compare the colour to the comparator wheel OR read in the colorimeter.
5. Record **free chlorine** result.
6. For **total chlorine**, add a DPD-3 tablet to the same vial. Read again within 60 seconds.
7. Record **total chlorine** result.
8. **Combined chlorine** = total chlorine – free chlorine.

**i WHY:** Free chlorine is the active disinfectant. Total chlorine includes combined chlorine (chloramines), which is a weaker disinfectant. If your free chlorine is very low but total chlorine is high, you may have a chloramine problem.

#### Common errors:

- Waiting too long to read → chlorine dissipates → falsely low result.
- Using old/damp DPD reagent → colour doesn't develop properly.
- Very high chlorine bleaches the indicator → gives a falsely low result. Dilute the sample 1:10 and multiply by 10.

### 4.4.3 pH

**Instrument:** Bench-top or portable pH meter (e.g., Hach HQ series, YSI).

**⚠️ RULE:** Rinse the probe with clean/distilled water immediately before EVERY test — before each buffer during calibration, and before every single sample. No exceptions — skipping this carries residue between readings and throws off the result.

**[ACTION]**

1. Calibrate the electrode with pH 7.0 and pH 4.0 (or 10.0) buffer solutions before each shift or as specified, rinsing the electrode between each buffer.
2. Rinse the electrode with distilled water. Blot dry (do not rub).
3. Immerse in sample. Wait for the reading to stabilize (typically 30–60 seconds).
4. Record the result.
5. Rinse electrode with distilled water before taking the next sample, if any.
6. Rinse electrode with distilled water and store in electrode storage solution (NOT distilled water — this damages the electrode).

**i WHY:** pH affects almost every process in the plant. Coagulant works best at a specific pH range. Chlorine disinfection is much more effective at lower pH. Too-low or too-high pH corrodes pipes.

---

### 4.4.4 Temperature

**Instrument:** Thermometer or probe on the pH/DO meter.

**[ACTION]**

1. Immerse the probe in the sample or in-situ at the sample point.
2. Wait for the reading to stabilize.
3. Record in °C.

**i WHY:** Temperature affects chlorine demand (colder water needs more contact time), coagulant dose (cold water coagulates poorly), and biological activity.

---

### 4.4.5 Alkalinity (mg/L as CaCO<sub>3</sub>)

**Method:** Titration with sulphuric acid (H<sub>2</sub>SO<sub>4</sub>).

*Detailed titration method will be added once your site's specific procedure is provided.*

**i WHY:** Alkalinity is the water's ability to buffer pH changes. It's critical for coagulation — alum consumes alkalinity, so water with low alkalinity may need alkalinity supplementation (e.g., lime or sodium bicarbonate added) to allow coagulation to work.

---

### 4.4.6 Jar Test (Coagulant Optimization)

The jar test is a miniature version of the coagulation/flocculation process. You use it to find the optimal coagulant type and dose before changing the full-scale process.

**Equipment:** Jar test apparatus (gang stirrer), 6 identical 1-L beakers, dosing syringes.

**[ACTION]**

1. Fill 6 beakers with 1 L of raw water each.
2. Set up the jar test apparatus and insert all 6 paddles.
3. Add different doses of coagulant to each beaker (e.g., 5, 10, 15, 20, 25, 30 mg/L).
4. **Rapid mix:** Stir at high speed (100–150 RPM) for 1 minute to simulate rapid mixing.
5. **Slow mix (flocculation):** Reduce to low speed (20–30 RPM) for 15–20 minutes. Observe floc formation.
6. **Settling:** Stop stirrers and allow to settle for 10–15 minutes.
7. Draw off settled water from each jar (do not disturb settled floc).
8. Measure turbidity and pH of each sample.
9. The jar with the lowest settled turbidity at the correct pH is your optimal dose.
10. Record results. Adjust full-scale dose accordingly.

**i WHY:** Raw water quality changes constantly — rain events, seasonal algae, temperature shifts all affect how much coagulant you need. A jar test takes 30–40 minutes and saves hours of poor treatment.

---

## 4.5 Sample Containers and Preservation

---

| Parameter           | Container Type                                                                   | Preservative               | Max Holding Time          |
|---------------------|----------------------------------------------------------------------------------|----------------------------|---------------------------|
| Turbidity           | Plastic or glass, dark                                                           | None                       | 24 hours (ASAP preferred) |
| pH                  | Plastic or glass                                                                 | None                       | Analyze immediately       |
| Free/Total Chlorine | Plastic                                                                          | None                       | Analyze immediately       |
| Temperature         | N/A (in-situ)                                                                    | N/A                        | N/A                       |
| Alkalinity          | Plastic                                                                          | Ice (4°C)                  | 24 hours                  |
| Coliform bacteria   | Sterile plastic bottle (Na <sub>2</sub> S <sub>2</sub> O <sub>3</sub> pre-dosed) | Ice (4°C)                  | 6–24 hours (check lab)    |
| Metals (Fe, Mn)     | Plastic                                                                          | HNO <sub>3</sub> to pH < 2 | 28 days                   |

---

## 4.6 Quality Assurance

---

- **Calibrate instruments** before each shift. Log calibration check results.
  - **Run a blank** (distilled water) and a known-concentration standard periodically. If the standard reads outside ±5%, the instrument needs service.
  - **Duplicate samples** — occasionally take two samples from the same point at the same time and analyze both. Results should agree within ±10%.
  - **Never re-use sample containers** for bacteriological testing without re-sterilizing.
-

## Section 5 — Chemical Dosing

Every chemical in this system has one job. Learn what that job is, what "working correctly" looks like, and what the warning signs of a problem are. Never mix chemicals together — several combinations on this site are dangerous.

### 5.1 The Chemicals On This Site and Why Each One Is Here

This is an iron-removal and hydrocarbon-removal system, not a drinking water plant. The chemicals reflect that:

| Chemical                                                                                                                                           | Container                                               | Job in this system                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PACI</b> (Poly Aluminum Chloride)                                                                                                               | White 200L drum, Class 8, SUMAS label                   | Coagulant — neutralizes particle charge so solids clump                                                                                                                                                    |
| <b>CL-200</b> (polymer flocculant)                                                                                                                 | White drum, "FLOCCULANT" label                          | Flocculant — grows small clumps into big, heavy floc. <b>Added at 2 confirmed points: the oil/water separation cell (also called "the lagoon") and the C Can</b>                                           |
| <b>Flopham</b> (dry polymer)                                                                                                                       | White bags, "NON-REGULATED" label                       | Backup/supplemental flocculant — same role as CL-200                                                                                                                                                       |
| <b>HaloKlear LiquiFloc 2%</b>                                                                                                                      | (chitosan acetate solution)                             | Natural chitosan flocculant — used for fine silts, especially stormwater                                                                                                                                   |
| <b>Lime — quicklime (CaO), confirmed by supervisor</b> (bag label says TEXTLIME Hydrated Lime, but supervisor's word overrides — see Section 5.6a) | 50 lb bags, mixed 1 bag per 200L in a tank at the C Can | Added at the C Can before PACI + flocculant — raises pH to 8+ for coagulation, also precipitates dissolved metal ions. <b>Conditional on pH — confirmed by supervisor: only used if the pH is too low.</b> |
| <b>Sodium Hydroxide (NaOH)</b> — NEW, see Section 5.6b                                                                                             | Big tote bag (location TBD)                             | Alternative pH RAISER to lime — confirmed present on site, not yet confirmed in routine use                                                                                                                |
| <b>Hydrogen Peroxide</b>                                                                                                                           | Blue 200L drums, Class 5.2/8, SUMAS label               | Oxidizer — converts dissolved iron (invisible) to iron hydroxide (filterable)                                                                                                                              |
| <b>CO<sub>2</sub> (Carbon Dioxide)</b>                                                                                                             | Air Liquide gas cylinder                                | pH REDUCER — lowers pH when water is too alkaline                                                                                                                                                          |
| <b>pH+ powder</b> (Summer Smiles Sani Marc)                                                                                                        | 8 kg pail                                               | pH RAISER — raises pH when water is too acidic                                                                                                                                                             |
| <b>Accu-Tab Blue</b> (Cal-Hypo tablets)                                                                                                            | White pails, Class 5.1, SUMAS label                     | Disinfectant — final chlorination before discharge                                                                                                                                                         |

#### 5.1a pH Drives Coagulant/Flocculant Dosing — CONFIRMED BY OPERATOR

**i** CONFIRMED: PACI coagulant and flocculant dosing are not "set and forget" — when the pH of the water changes, the coagulant/flocculant treatment has to be adjusted in response.

**i** TARGET RANGE — CONFIRMED BY OPERATOR: **pH 8 or higher is good for the coagulant. Below 8 is not good** — lime is used to bring pH up to 8+ before PACI coagulant and flocculant go in. Operator also noted PACI coagulant is itself acidic and lowers pH on its own. Supervisor (Brendan) confirmed lime is only needed when pH is too low (see Section 5.6a) — it is not a fixed every-batch step.

**i** WHY (general water treatment principle, to be replaced with site-specific detail): Coagulants like PACI work by neutralizing the electrical charge on suspended particles, and that charge-neutralizing reaction is pH-dependent — PACI has an optimal pH window, and outside that window it works poorly even at the correct dose. So if incoming water pH drifts below 8, the coagulant dose (and sometimes the flocculant dose right after it) needs to be re-tuned, not just left on the same setting. This is part of why pH is tested every day — it's not just a compliance number, it's a direct input into how much coagulant/flocculant to run.

**i** CONFIRMED BY SUPERVISOR (Brendan): **There is no fixed daily target dose (mg/L) for PACI/flocculant.** In his words: "the dose is going to change daily depending what they dump. So you're just going to have to play with it and wait until you see it floc." Raw water composition varies day to day depending on what's been dumped into the system, so the dose is set by observation each time — watch for floc forming, not a fixed setpoint. This replaces the earlier open question about a fixed target.

#### 5.1b Ongoing Testing Drives Dose Adjustment — CONFIRMED BY OPERATOR

**i** **CONFIRMED:** Dosing at the C Can is not a one-time "set it and walk away" step. The operator tests the water on an ongoing basis and adjusts dosing based on what the results show:

- **If results show under-dosing** (e.g., poor floc formation, turbidity not dropping — see Section 4 for signs), **add more coagulant or flocculant at the C Can into the weir tank.**
- **Pump speed/stroke adjustment:** if the dose itself needs to change, this is done by adjusting the **stroke rate and/or stroke length** on the relevant dosing pump (coagulant, flocculant, or lime) — not just by adding chemical by hand.
- Operator is sending photos of the actual dosing pumps used to inject coagulant, flocculant, and lime at the C Can — once received, this section will be updated with pump make/model and where the stroke rate/length controls are on each one. **Flocculant pump confirmed (Section 5.4) — coagulant and lime pumps still pending.**

**i** **RULE — CONFIRMED BY OPERATOR:** **Coagulant (PACl) dose must never be more than the flocculant dose.** Flocculant ≥ coagulant, always — never the other way around. Keep this in mind any time you're adding more of either one.

## pH check during testing at the C Can

While testing at the C Can, **check the pH of the water.** Operator will read out the pH value live — based on the target range in Section 5.1a (pH 8+ is good, below 8 is not), confirm whether the reading is off and what to do about it (add lime to bring it up to 8+, per Section 5.6a).

(still pending: photos of the 3 dosing pumps at the C Can; current stroke rate/length setting on each; what specific test result triggers a dose increase vs. a pump adjustment)

## 5.1c Full Chemical Mixing & Dosing Sequence — Step by Step (CONFIRMED BY OPERATOR, Oliver)

This ties together Sections 5.3, 5.5, and 5.6a into the order they're actually done at the C Can, before water moves on to the weir tank, ozone, bag filter, and carbon vessel (see the full Treatment Train, Section 0.2). If you're new, read the "why" line under each step before you do it — knowing why catches mistakes a checklist alone won't.

### [ACTION]

#### Step 1 — Test the pH first.

This single reading decides everything that follows. It tells you whether lime is needed at all — lime is *conditional*, not an every-batch default (Section 5.6a).

**i** **WHY:** PACl coagulant only works well at pH 8 or above, and PACl itself is acidic — it drags pH down further once added. If you skip the pH check and the water is already low, you'll dose coagulant into water where it barely works, see no floc forming, and waste chemical chasing a problem that was actually a pH problem.

#### Step 2 — Decide on lime.

- **If pH reads below 8:** mix lime in the blue lime tote/drum at the C Can — **1 bag (50 lbs) per 200L of water.** Keep the pneumatic mixer running the *entire time* — both while you're first mixing it in and while you're pumping it back out. Lime settles and hardens fast the moment the blade stops turning.
- **If pH already reads 8 or higher:** skip lime completely and go straight to Step 3.

**i** **WHY:** Lime does two things at once — it raises pH into the range PACl needs to work, and separately, the higher pH itself can force dissolved metal ions (like iron) to turn solid so they can be filtered out, acting as a backup to ozone. But it's only needed when the pH reading actually calls for it — adding it every time regardless of pH just means pushing the pH too high (Brendan's caution: "both raise pH so don't add too much").

#### Step 3 — Mix the PACl coagulant stock.

**2 bags of dry PACl powder per 200L of clean water** — clean water, not the water you're treating. Let the mixed stock **sit for 3 hours** before you start dosing from it.

**i** **WHY:** Coagulant works by neutralizing the electrical charge on suspended particles so they can stick together (Section 5.3). A freshly mixed batch hasn't fully dissolved/activated yet — dosing too early means you're injecting partially-mixed powder instead of an effective coagulant, and you won't see good floc form even though you added the "right" amount.

#### Step 4 — Mix the flocculant stock.

**100g of flocculant per 200L of water** — measured as half a scoop from the white pail per 200L batch. Add it **slowly, sprinkled in, never dumped**, while the water is mixing. **Keep mixing for about an hour** before it's ready to use.

**i** WHY: Flocculant is a long-chain polymer that works like a net, bridging small coagulated clumps into bigger, faster-settling floc (Section 5.4). Dump it in too fast and it clumps into masses on contact with water — those masses don't disperse, and they're exactly what clogs the chemical dosing pumps downstream. Slow and steady gets it evenly dissolved instead.

**Step 5 — Dose both into the water together**, right before the weir tank.  
**Flocculant dose must always be  $\geq$  the coagulant dose — never less** (Section 5.1b).

**i** WHY: Flocculant's whole job is to bridge the floc that coagulant just created. If there's more coagulant than flocculant to bridge it, you get lots of small, weak clumps that never grow big enough to settle quickly.

**Step 6 — Watch for floc forming visually, and adjust from there.**

There is no fixed daily mg/L target for either chemical — raw water composition changes day to day depending on what's gone into the system, so the dose is tuned by what you actually see, not a setpoint you dial in once and forget. If you're not seeing good floc, adjust the dosing pump's stroke rate/length rather than assuming the ratio above is wrong.

**i** WHY: In Brendan's own words: "the dose is going to change daily depending what they dump. So you're just going to have to play with it and wait until you see it flocc."

**Step 7 — Don't overload the front end.**

Keep the incoming flow rate in check. If too much water comes in at once, the floc you just worked to create doesn't get enough time to actually settle before it's pushed forward, and every stage downstream inherits a problem it can't fix either.

**i** WHY: This site was never built as a purpose-designed treatment plant — there's no engineered buffer capacity to absorb a flow spike. Managing flow rate is as much a part of "dosing correctly" as getting the chemical ratios right; a perfect dose into water that never had time to settle still fails.

**Step 8 — Hand off to the rest of the train.**

From here the water continues exactly as mapped in Section 0.2: weir tank settling → ozone → indoor settling bins → bag filter → carbon vessel → storage tank. This site does not run a sand filter — the bag filter and carbon vessel are the only two filtration steps the water actually passes through, so anything that gets past your chemical dosing here is relying on just those two to catch it.

## 5.2 Critical Compatibility Warning — Read Before Handling Any Chemical

**⚠ WARNING:** Several chemicals on this site will react violently if mixed together.

| Never mix...             | With...                                        | What happens                                      |
|--------------------------|------------------------------------------------|---------------------------------------------------|
| Hydrogen Peroxide        | Calcium Hypochlorite (Cal-Hypo)                | Violent decomposition, fire, chlorine gas release |
| Hydrogen Peroxide        | Any organic material (rags, wood, oils)        | Can ignite spontaneously                          |
| Cal-Hypo                 | Any acid (including PACl, which is pH 2.2–2.8) | Chlorine gas released                             |
| CO <sub>2</sub> (as gas) | Enclosed space without ventilation             | Displaces oxygen — asphyxiation risk              |

**Rule:** Store each chemical in its designated location. Never move containers without supervisor approval. Never decant into an unlabelled container. Never combine leftover chemicals.

## 5.3 Chemical 1 — PACl (Poly Aluminum Chloride)

**Product:** CK-311 Coagulant (Aluminum Hydroxide Chloride Solution), supplied by SUMAS Environmental Services

**SDS supplier:** CARBONeT Holdings Inc., 604-761-2308

**Container:** White 200L drum, Class 8, UN 3264, PG III, pH 2.2–2.8

**Pump:** Blue diaphragm dosing pump

**What it does**



Water contains tiny particles that are too small to settle on their own. These particles carry a negative electrical charge, which keeps them suspended and apart — like tiny magnets repelling each other.

PACI is a positively charged aluminum compound. When it's added to the water and mixed, its positive charge neutralizes the negative charge on the particles. Once the charges are neutralized, particles can approach and stick together — this is **coagulation**.

**i WHY:** Without coagulation, fine particles pass straight through every filter in this system. Coagulation is always the first chemical step.

## How to dose

- Typical dose range: 5–40 mg/L (highly variable based on water quality — start with your baseline and adjust)
- Dose point: add to water with good mixing (turbulent mixing zone or rapid mix)
- Check: is the pump running and consuming chemical? Watch the tank level — it should visibly drop over a shift.

## Preparing the dry PACI stock solution

**i CONFIRMED BY CURRENT OPERATOR:** A dry PACI powder (25kg bags) is mixed into a stock/working solution before being metered into the process — this is separate from (or an alternative to) the liquid CK-311 drums described above.

- **Ratio: 1 bag of dry PACI powder per 100L of clean water — NOT the water being treated** (confirmed by current operator; matches the earlier former-operator figure of 2 bags per 200L — same ratio, now confirmed current)
- **Let it sit for 3 hours before using — CONFIRMED BY OPERATOR.** Don't dose straight from a freshly mixed batch. Plan ahead like the flocculant stock (Section 5.4, 24-hour rest) so you don't run out of usable stock mid-shift.
- Mix in the tank, then meter the resulting solution into the process via the dosing pump
- Indoors, coagulant + flocculant are injected together right before the water reaches the white settling tanks (see Section 0.2/0.2a) — the combination is what causes contaminants and dirt to clump and settle to the bottom of the tank
- You may need to adjust the dosing pump's stroke rate/length to get the right amount going in — watch visually for good floc formation as your guide

## Signs of under-dosing

- Tiny pinpoint particles that don't clump
- Water stays turbid/cloudy after settling
- Filters get dirty quickly (loading too much fine solids)

## Signs of over-dosing

- Turbidity gets WORSE, not better (charge reversal — particles go positive and repel again)
- Unusually high aluminum in discharge water
- pH drops sharply (PACI is acidic, pH 2.2–2.8)

## Safety

- Corrosive (Class 8). Causes skin and eye burns.
- **PPE:** Chemical splash goggles, chemical-resistant gloves, apron
- Eye contact: flush immediately at eyewash for 15 minutes

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## 5.4 Chemical 2 — CL-200 Polymer Flocculant

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**Product:** CL-200 Water Treatment Flocculant, aqueous solution of water-soluble polymer

**Container:** White drum, "FLOCCULANT / NON-REGULATED MATERIAL" label

**Pump (indoor injection point):** Yellow LMI diaphragm dosing pump

**Pump at the C Can — CONFIRMED BY PHOTO (IMG\_2807, IMG\_2808):** Blue **JCMB Solenoid Dosing Pump**, Model **JCMB55-20/2.5**, Capacity 20 L/H, Pressure 2.5 bar, Power 30W, AC100–240V. Dial sets stroke length; « » buttons adjust stroke rate; **SET** and **ON/OFF** buttons; green MOTOR light confirms it's running.

**Current setting — CONFIRMED BY OPERATOR:** flocculant pump speed adjusted to **48.0**.

## What it does

Coagulation (PACI) creates small particle clusters. Flocculation grows those small clusters into larger, heavier floc that settles faster and is easier to filter. CL-200 polymer is a long-chain molecule that acts like a net — it bridges between coagulated particles, connecting them into larger aggregates.

**i WHY:** Coagulation alone produces fine, slow-settling floc ("pinpoint floc"). Adding polymer produces large, fast-settling floc. This is the difference between waiting hours for settling and waiting minutes.

## How to dose — TWO confirmed injection points on this site

 **CORRECTED BY OPERATOR:** This section previously listed four injection points, including "the lagoon" as a separate final-dosing/disposal location. **That's been corrected — "the lagoon" is just another name for the Oil/Water Separation Cell (point 1 below), not a separate fourth point.** The operator's job ends once treated water reaches the storage tank — there is no dosing step after that. The current confirmed dosing points are:

1. **Oil/Water Separation Cell (outdoor, Step 2 — also called "the lagoon")** — added early to start clumping fine particles before the water even reaches the frac tanks
2. **The C Can (Section 0.2a, Section C)** — dosed alongside PACl coagulant, right before the outdoor weir tank

(unconfirmed, needs checking with operator: the "two Blue Separator Bins" and "peroxide oxidizer" dosing points listed in older notes — these haven't been confirmed as part of the current live process.)

 **WHY MULTIPLE POINTS, NOT ONE:** Each point is treating water that has already lost some solids to the step before it, but is also picking up disturbance/mixing from pumping. Re-dosing flocculant at each major transfer point keeps floc forming continuously instead of relying on one dose to last the whole way through. Think of it like reapplying glue at each stage of an assembly line, rather than gluing everything once at the very end.

- General rule wherever it's added: dose **AFTER** any PACl/coagulant has had time to work, and during gentle mixing — not violent/rapid mixing (the gentle tumbling lets the polymer bridge particles together without tearing the floc apart)
- Typical dose range: 0.1–2 mg/L per injection point (much lower than coagulant — polymers are very active)
- Over-dosing polymer is easy and wastes money. Start conservative at each point.

## Preparing the flocculant stock solution

 **CONFIRMED BY CURRENT OPERATOR:** Ratio is **1 kg polymer per 1000L of clean water (1 g/L)**, added in two half-volume pours:

1. Add the **first half of the water** (NOT the water being treated) to the tote.
2. **Turn on the mixer.**
3. Add the **polymer** while adding the **second half of the water** at the same time — pouring the water and polymer in together while mixing lets the polymer disperse instead of clumping.
4. Keep mixing well until fully dissolved.
5. **Let it sit for 24 hours before use — CONFIRMED BY OPERATOR.** Don't dose straight from a freshly mixed batch; the stock solution needs a full day to rest/cure before it's ready to inject. Plan batches a day ahead so you don't run out of usable stock mid-shift.

**Batch size confirmed by operator: 500L, not 1000L** — scaled to the same ratio, that's 250L water → mixer on → **500g polymer** added with the second 250L of water. (scaling assumes the same 1 g/L concentration applies at the smaller batch size — confirm with operator if the polymer amount doesn't simply halve with the batch.)

 **MOVED — CONFIRMED BY OPERATOR (Oliver):** The 100g-per-200L, scoop-from-the-white-pail ratio previously logged here is **the dry Flopham polymer, not CL-200** — moved to Section 5.5 where it now lives with its mixing technique. (This 1kg/1000L two-pour procedure right above describes a powder being poured in alongside water too — possibly also Flopham rather than liquid CL-200. Not reattributed yet since it hasn't been specifically confirmed either way — flagging for a future check.)

## Signs of under-dosing

- Floc forms but stays small and slow to settle
- Settled water still turbid

## Signs of over-dosing

- Floc may break apart or become sticky ("restabilization")
- Water can become viscous or stringy

## Safety

- Non-regulated, pH 6.5–7.5. Low hazard.
- Can make floors very slippery if spilled — clean up immediately
- **PPE:** Gloves recommended; mop up spills right away

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## 5.5 Chemical 3 — Flopham Dry Polymer

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**Product:** Flopham water soluble polymer (polyacrylamide-based)

**Container:** White bags, "NON-REGULATED MATERIAL" SUMAS label

## What it does

Same job as CL-200 — a flocculant. Flopham is a dry polymer that must be dissolved in water before use.

 **CORRECTED BY OPERATOR (Oliver):** This is **not** a backup/supplement to CL-200 — this dry polymer, scooped from the white pail, is the one actually in routine use at the C Can for flocculant dosing (see Section 5.1c for where it fits in the full mixing sequence).

## Ratio and mixing technique — CONFIRMED BY OPERATOR (Oliver)

**Ratio: 100g of dry Flopham per 200L of water (~500 mg/L)** — measured in practice as **half a scoop from the white pail** per 200L batch.

### [ACTION]

1. Start with water in the mixing container, mixer running.
2. **Add the polymer very slowly — sprinkle it in, never dump it.** Adding it too fast (or adding water to powder instead of powder to water) forms clumps/fish-eye lumps that will clog the chemical dosing pumps downstream.
3. **Keep mixing for about an hour** until fully dissolved and the solution looks clear/uniform.
4. This stock solution is then metered into the process alongside the PACI coagulant — together they cause contaminants and dirt to clump and settle.

 **NOTE:** This 100g/200L ratio (~0.05%) runs lighter than the generic 0.1–0.5% solution-concentration range sometimes quoted for polyacrylamide flocculants — that generic range is kept below for reference, but **100g/200L is the live, operator-confirmed ratio to use.**

(reference only – generic guidance, not site-specific):

- Typical solution concentration range quoted for this polymer type: 0.1–0.5% (1–5 g polymer per litre of water)
- Add the prepared solution to the treatment stream via dosing pump or measured addition

 **WHY:** Pre-dissolving eliminates clumping and ensures even distribution. Undissolved polymer lumps pass through the system without doing anything useful.

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## 5.6 Chemical 4 — HaloKlear LiquiFloc 2% (Chitosan Flocculant)

**Product:** HaloKlear LiquiFloc 2%, HaloSource Inc., Bothell WA

**Chemical:** Chitosan Acetate Solution — a natural biopolymer derived from crustacean shells

**Appearance:** Clear to pale yellow viscous liquid with a vinegar odor

## What it does

Chitosan is a natural, biodegradable flocculant that is particularly effective for removing fine clay and silt particles from stormwater. It works similarly to synthetic polymers but is preferred in some applications because it is non-toxic and biodegradable.

## When to use it

- During high turbidity events (heavy rain, surface runoff)
- When stormwater contains fine clay that doesn't respond well to PACI alone
- When an environmentally sensitive application requires a natural flocculant

## Dose

- Typical: 1–10 mg/L (varies by turbidity and particle type)
- 100% biodegradable and nontoxic to aquatic life

## Safety

- Very low hazard. May be mildly irritating to eyes.
- Smells like vinegar — this is normal.
- **PPE:** Gloves recommended as good practice.

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### 5.6a Chemical 4a — Lime

 **CONFIRMED BY OPERATOR:** This is the "3rd chemical" at the C Can — lime. (Operator briefly second-guessed the chemical name in favor of "sodium hydroxide," then retracted that and confirmed lime is correct.)

**Product label (photographed, IMG\_2803, IMG\_2743):** TEXTLIME High Calcium Hydrated Lime

**Chemical —**  **RESOLVED, CONFIRMED BY SUPERVISOR (Brendan, text message):** Brendan, asked directly, said: **"So it's quicklime which is calcium oxide, the bag we should around and it just says lime."** This is **quicklime (CaO)**, not hydrated lime (Ca(OH)<sub>2</sub>) as the photographed bag label (IMG\_2803) suggests — the supervisor's direct word overrides the label here; the bag itself is apparently just generically marked "lime" with no further specifics. Treat this as **CaO going forward**.

**Ratio — CONFIRMED BY SUPERVISOR (Brendan, text message):** **1 bag (50 lbs) per 200L of water. Must be kept mixing continuously — it will settle out and harden if you stop.** This matches the dry-bag-mixed-into-a-stock-tank pattern used for PACl/flocculant (Sections 5.3/5.4); the blue drum/tote staged at the C Can (Section 6.1b) is this mixing tank.

 **CRITICAL — CONFIRMED BY OPERATOR:** **The mixer blade on the blue lime drum must be running while you're pumping the lime out, not just while you first mix it.** Lime settles out fast — if the blade stops turning while you're dosing, it won't work well. The mixer is the pneumatic (air-driven) tote mixer, Section 6.16 — open the air valve and keep it running for the whole time you're drawing lime out.

 **Photo-illustrated quick steps for mixing lime, PACl, and flocculant (and running the mixer):** [CHEMICAL\\_MIXING\\_GUIDE.pdf](#)

**Alternative chemical — CONFIRMED BY SUPERVISOR (Brendan, text message):** "You can also use sodium hydroxide as well. Which I think they have a big tote bag worth of." Sodium hydroxide (NaOH, caustic soda) is a separate, more hazardous strong base that can substitute for lime as a pH-raiser. Confirmed present on site in a large tote bag, but not yet confirmed as in current routine use — see new chemical entry, Section 5.6b.

 Brendan's caution, applies to both: **"Both raise pH so don't add too much."**

**Dosing logic at the C Can —**  **RESOLVED, CONFIRMED BY SUPERVISOR (Brendan, text message):**

Brendan's direct answer: **"Lime you only need to use if the pH is too low. If it's too low sometimes the chemicals won't work."**

This confirms lime is **conditional on the pH reading**, not pumped on a fixed every-batch schedule. Earlier statements on this — one matching this conditional logic, and one saying "we pump the lime every time because the coagulant lowers the pH" — are reconciled this way: check pH every time, and add lime whenever that reading comes back below 8. In practice this may often look like "every time" if the pH is consistently low, but the decision point is the reading, not the clock — there is no fixed schedule independent of the pH check.

**How low is "too low":** The working threshold is **pH 8**. Anything less than 8 is not good and needs lime — treat 8 as the hard cutoff, not just a target to aim near.

## What it does

Lime raises pH. PACl coagulant works well at pH 8 and above, and is itself acidic — so without lime to counteract it, the coagulant would drag the water's pH down out of its own optimal range.

**Second function — CONFIRMED BY SUPERVISOR (Brendan, text message):** "If you add the lime it will force any metal ions dissolved to transform into solid particles that can be filtered out in the treatment process if you don't trust the ozone." This is a real, separate mechanism from pH-correction-for-coagulant: raising pH with lime can push dissolved metal ions (like iron) past their solubility limit, so they precipitate out as solid metal hydroxide particles that the filters can then catch — similar end result to what ozone/peroxide does by oxidation (Section 5.7), but via a pH shift instead of an oxidizing reaction. Brendan's framing suggests this is a backup/parallel pathway for metals removal, not a replacement for ozone.

 **SAFETY:** This is quicklime (CaO) — caustic/corrosive, can burn skin and eyes, and the dust is a respiratory irritant. **Quicklime reacts exothermically (releases real heat) when wetted** — on top of the causticity, mixing this generates heat, which is a bigger handling hazard than ordinary hydrated lime. Handle with confirmed PPE (chemical splash goggles, chemical-resistant gloves, dust mask) and never add water to a sealed/closed container of it.

(still needed: PPE specifics from the SDS; where the lime is stored/staged at the C Can)

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## 5.6b Chemical 4b — Sodium Hydroxide (NaOH, Caustic Soda) — NEW, NOT YET FULLY DOCUMENTED

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 **CONFIRMED BY SUPERVISOR (Brendan, text message):** Sodium hydroxide is on site as an alternative pH-raiser to lime — "they have a big tote bag worth of" it. Not yet confirmed whether it's in current routine use or just available as a backup option.

**Chemical:** Sodium hydroxide (NaOH) — a strong, fast-acting base, more hazardous to handle than lime.

 **SAFETY (general — site-specific SDS still needed):** Caustic soda is highly corrosive to skin, eyes, and respiratory tract — more aggressively so than lime. Dissolving it in water is also exothermic (releases heat). Standard handling requires chemical splash goggles, face shield, chemical-resistant gloves and apron, and care to avoid generating heat too fast in a closed container.

 Brendan's caution: Like lime, this raises pH — **don't add too much**.

(still needed: exact product/concentration, container type and location, current dose/ratio if actually in use, SDS, PPE confirmation.)

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## 5.7 Chemical 5 — Hydrogen Peroxide (H<sub>2</sub>O<sub>2</sub>)

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**Product:** Hydrogen Peroxide, All Grades, Brenntag Canada

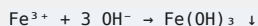
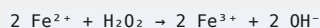
**Container:** Blue 200L drums, SUMAS label "ORGANIC PEROXIDE TYPE E, LIQUID", Class 5.2 + Class 8

**Concentration:** Likely 25–50% (confirm with supervisor and SDS)

### What it does — Iron Oxidation

The source water contains dissolved iron in its **reduced form (Fe<sup>2+</sup>)** — ferrous iron. In this form, iron is invisible in water (clear). You can't filter out dissolved iron — it passes straight through any filter because it's not a particle.

Hydrogen peroxide oxidizes Fe<sup>2+</sup> to Fe<sup>3+</sup> (ferric iron):



Fe(OH)<sub>3</sub> is iron hydroxide — a reddish-brown solid that precipitates out of solution and can now be filtered. This is where the orange staining on every surface comes from.

**i WHY:** You cannot filter dissolved iron — you must convert it to a particle first. H<sub>2</sub>O<sub>2</sub> does this conversion, and then your sand filter catches the iron particles that result.

### Dose guideline

- Rule of thumb: **1 mg H<sub>2</sub>O<sub>2</sub> per mg of Fe<sup>2+</sup>** (1:1 ratio)
- In practice, 1–3 mg/L H<sub>2</sub>O<sub>2</sub> is common for iron oxidation (confirm actual Fe<sup>2+</sup> concentration in your source water)
- Excess H<sub>2</sub>O<sub>2</sub> is not harmful in small amounts — it decomposes to water and oxygen

### How to check it's working

- Orange/rusty precipitate forming in the settling tanks is a good sign
- If no precipitate forms after adding H<sub>2</sub>O<sub>2</sub>, check: Is the pump actually running? Is the tank level dropping? Is the pH in the right range (H<sub>2</sub>O<sub>2</sub> iron oxidation works best at pH 5–8)?

**⚠ WARNING — Hydrogen Peroxide is a serious hazard**

- **Class 5.2 Organic Peroxide** and **Class 8 Corrosive** — both apply
- Concentrated H<sub>2</sub>O<sub>2</sub> causes severe skin and eye burns on contact
- Strong oxidizer — reacts violently with organics (rags, wood, fuels), metals (iron, copper, manganese), and strong bases
- **NEVER mix with Cal-Hypo** — violent reaction
- Decomposition produces oxygen — sealed/hot containers can over-pressurize
- **Storage:** Cool, dry, away from all combustibles. No direct sunlight. Keep away from other chemicals. Do not store near Cal-Hypo.
- **PPE:** Full face shield, chemical splash goggles, chemical-resistant gloves (neoprene or nitrile, check SDS), apron

**First aid — skin contact:** Flush with large amounts of water for 20 minutes. Remove contaminated clothing while flushing.

**First aid — eye contact:** Eyewash station for 15 minutes minimum. Seek medical attention.

**First aid — inhalation (vapour/mist):** Move to fresh air. Call 911 if breathing affected.

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## 5.8 Chemical 6 — CO<sub>2</sub> (Carbon Dioxide)

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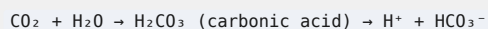
**Product:** Carbon Dioxide Gas/Liquid, Air Liquide Canada

**Container:** Pressurized cylinder (high-pressure gas — handle with care)

**Emergency contact:** Air Liquide Canada, (514) 878-1667

### What it does — pH Reduction

When CO<sub>2</sub> dissolves in water, it forms carbonic acid (H<sub>2</sub>CO<sub>3</sub>), which lowers pH:



More  $\text{H}^+$  ions = lower pH.  $\text{CO}_2$  is a safe, precise pH reducer — unlike strong acids (sulphuric acid),  $\text{CO}_2$  is self-limiting (the more you add, the more equilibrium pushes back) and leaves no harmful residual.

### When to use it

- When incoming water pH is too high (too alkaline) — for example, after lime or concrete contact
- When you need to bring pH down before a treatment step that requires a certain pH range

#### **WARNING — $\text{CO}_2$ Cylinder Safety**

- **High pressure gas** — cylinders must be chained to wall or fixed support. A falling cylinder can rupture and become a projectile.
- $\text{CO}_2$  gas is heavier than air — in an enclosed/low space, it displaces oxygen without warning (no smell, no colour). **Asphyxiation risk.**
- Always work in ventilated areas; never enter a  $\text{CO}_2$  storage vault alone
- **PPE:** Safety glasses, gloves. For  $\text{CO}_2$  liquid contact: thermal gloves (rapid evaporation causes frostbite)

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## 5.9 Chemical 7 — pH+ Powder (Summer Smiles Sani Marc)

**Product:** Summer Smiles pH+ (pool-grade sodium carbonate/bicarbonate blend)

**Container:** 8 kg pail

### What it does — pH Raising

Sodium carbonate ( $\text{Na}_2\text{CO}_3$ ) is a mild base. When added to water, it raises pH. This is the same product used in swimming pools to raise water pH.

### When to use it

- When water pH is too low (too acidic) — for example, after heavy PACl dosing (which is acidic, pH 2.2–2.8)
- When discharge requirements specify a minimum pH (usually  $\geq 6.5$ )

### How to add it

#### **[ACTION]**

1. Wear gloves — the powder can irritate skin and eyes.
2. Add the powder slowly to a turbulent area (flowing water, not a stagnant tank).
3. Add a small amount, mix thoroughly, test pH, then add more if needed.
4. Do NOT dump in a large amount all at once — pH can overshoot.

 **WHY:** pH adjustment requires patience. A little, mix, test, repeat. Overcorrecting costs you time and extra chemicals to correct back.

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## 5.10 Chemical 8 — Accu-Tab Blue (Calcium Hypochlorite Tablets)

**Product:** Accu-Tab Blue, Calcium Hypochlorite

**Container:** White pails, SUMAS label, UN 2880, Class 5.1 (Oxidizer)

**Concentration:** Hydrated mixture, 5.5–16% water content

### What it does — Disinfection

Calcium hypochlorite dissolves in water to release hypochlorous acid ( $\text{HOCl}$ ) — the same active disinfectant as liquid bleach. The tablets dissolve slowly as water passes over them in a tablet chlorinator (or contact chamber with tablets).

Cal-Hypo is the final disinfection step — it kills any remaining bacteria before discharge and provides a measurable free chlorine residual to confirm treatment is complete.

### How to use tablets

#### **[ACTION]**

1. Confirm the contact vessel/chlorinator is clean and dry before adding new tablets.
2. Wear gloves and do NOT inhale dust — the tablets are strong oxidizers and the dust irritates lungs and eyes.

3. Add tablets per your dosing SOP — do NOT add to a vessel that has wet residue from a previous chemical batch (especially peroxide residue). Dry vessel = critical.
4. Allow water to flow through at the designed rate — tablets dissolve proportionally to flow.
5. Test free chlorine residual in the effluent with a DPD test kit to confirm correct dose.

### Target free chlorine for discharge

(Confirm with supervisor and permit) — typical range: 0.1–0.5 mg/L at discharge point.

#### WARNING — Cal-Hypo Hazards

- **Class 5.1 Oxidizer** — strong oxidizer. Can cause fire on contact with organics (wood, paper, clothing)
- **NEVER mix with acids** (including PACl solution) — releases chlorine gas
- **NEVER mix with hydrogen peroxide** — violent decomposition
- Store in a cool, dry, well-ventilated location, away from all other chemicals
- **PPE:** Chemical splash goggles, gloves, apron. If handling dusty tablets: N95 or half-face respirator.

## 5.11 Chemical Dose Calculation

All doses are proportional to the volume of water being treated. In a batch system (frac tanks):

$$\text{Dose (mg/L)} \times \text{Tank volume (L)} = \text{Total chemical mass needed (mg)}$$

Convert to volume:

$$\text{Total mass (mg)} \div \text{Chemical concentration (mg/mL)} = \text{Volume to add (mL)}$$

#### Example — adding PACl to a frac tank:

- Target dose: 20 mg/L PACl
- Frac tank volume: 50,000 L (50 m<sup>3</sup>)
- PACl concentration:  $\approx 10\% = 100 \text{ g/L} = 100,000 \text{ mg/L} = 100 \text{ mg/mL}$

$$\begin{aligned} 20 \text{ mg/L} \times 50,000 \text{ L} &= 1,000,000 \text{ mg total} \\ 1,000,000 \text{ mg} \div 100 \text{ mg/mL} &= 10,000 \text{ mL} = 10 \text{ L of PACl solution} \end{aligned}$$

 **WHY:** In a batch system you're adding a fixed amount to a known volume, not running a continuous feed. Getting the math right matters — too little = poor treatment, too much = wasted chemical and potential exceedances.

## 5.12 Chemical Inventory Checklist

Check at the start of every shift:

| Chemical                       | Current Level | Reorder Threshold | Supplier                   |
|--------------------------------|---------------|-------------------|----------------------------|
| PACl (white drum)              | —             | < 20% remaining   | SUMAS 1-250-374-4151       |
| CL-200 polymer (white drum)    | —             | < 20% remaining   | SUMAS 1-250-374-4151       |
| Flopham polymer (bags)         | —             | < 2 bags          | SUMAS 1-250-374-4151       |
| Hydrogen peroxide (blue drums) | —             | < 1 full drum     | SUMAS 1-250-374-4151       |
| Cal-Hypo tablets (white pails) | —             | < 1 full pail     | SUMAS 1-250-374-4151       |
| CO <sub>2</sub> cylinder       | —             | Cylinder < ¼ full | Air Liquide 1-800-817-7697 |
| pH+ powder                     | —             | < 2 kg            | (confirm supplier)         |

Record inventory in the daily log.

## Section 6 — Equipment Guide

This section describes every major piece of equipment on this site — what it does, how to operate it, what normal looks like, and what warning signs to watch for. Learn the name and location of each piece before your first solo shift.

### 6.1 Outdoor Settling Ponds

**What they are:** Large concrete block-walled open basins outdoors.

**There are two distinct ponds:**

| Pond                           | Typical water appearance       | Purpose                                                                                 |
|--------------------------------|--------------------------------|-----------------------------------------------------------------------------------------|
| Primary settling pond          | Turbid green or grey           | First pond receiving raw site water. Heavy solids settle here.                          |
| Stabilization / secondary pond | Dark red or purple (iron-rich) | Receives more heavily contaminated water. Oil/water separation and additional settling. |

**What they do:** Gravity pre-settling. Given time (hours to days), large solids sink to the bottom, free oil floats to the surface, and the clearer middle layer is drawn off for further treatment.

[CHECK] Each shift:

- [ ] Water level in each pond — note if rising or falling
- [ ] Oil sheen on the surface — any visible oily rainbow or dark film?
- [ ] Debris (trash, vegetation) blocking the outlet pipe?
- [ ] Are the concrete block walls intact — any erosion or instability?

**i WHY:** The settling ponds are the first treatment step and absorb the "shock loads" from rain events and site activity. If they're overloaded or short-circuiting, the rest of the system gets hammered.

### 6.2 Indoor Storage Tanks — "Tank 1" and "Tank 2" (Large White Cylindrical Tanks)

**i CONFIRMED BY OPERATOR:** site operators refer to these two tanks as **Tank 1** and **Tank 2**. Use these names going forward instead of generic "white tanks."

**What they are:** Two large cylindrical above-ground tanks inside the treatment building. White poly or fibreglass construction, approximately 3–4 metres tall and 1.5–2 metres in diameter. Each has a metal ladder attached to the outside for level checking. Outlet piping and ball valves at the base. One tank is labelled "ANKO2" (*confirm whether this is Tank 1 or Tank 2, and capacity, with supervisor*). Photo IMG\_2757 shows both tanks together, with white tote chemical containers and blue chemical drums staged on pallets next to them.

**What they do:** **i CONFIRMED** – these are STORAGE tanks, not settling tanks. They hold water at different stages (e.g., raw/incoming vs. treated) rather than being where chemical floc actually forms and settles. Settling happens in a separate baffled structure (see Section 6.2a / IMG\_2741, IMG\_2742) — do not confuse the two. (*confirm with supervisor: what stage of water is in each of these two tanks – raw, dosed, or treated?*)

**i ADDITIONAL USE – CONFIRMED BY OPERATOR:** Water held in the storage tank is also used on-site for **dust control** (sprayed on roads/yard). The operator's job ends once water reaches this tank — what happens to it after that isn't tracked here (see Section 0.2a, Section I).

**What NTU should this water be?** There's no separate, stricter turbidity spec for dust control — it's a non-discharge reuse, not a permitted release to the environment, so the discharge-permit turbidity limit (Section 3/4, still pending from supervisor/SUMAS) doesn't apply here. By the time water reaches the storage tank it has already passed through the bag filter and/or carbon vessel — the same NTU reading checked at that outlet (Section 4.3) is what you'd expect in the tank. Practically, for dust control the only real requirement is staying low enough turbidity that it doesn't clog the spray nozzles/pump — there's no compliance number to hit. (*if you take an actual NTU reading from the tank, log it here so we have a real baseline instead of an estimate.*)

**i PRACTICAL TARGET** (general judgment, not a confirmed site number): Given there's no permit limit and the main risk is nozzle clogging, **aim for under 10 NTU** — visually clear with no cloudiness. If a reading comes back higher than that: don't send the water on yet. First check whether the weir tank actually finished settling (give it more time if floc is still visibly suspended), then check the dose (more coagulant/flocculant if floc is weak), then check the bag filter for loading (change if it's been sitting near its 1–2 week limit). Re-test after whichever fix applies before calling the batch done.

[CHECK] Each shift:

- [ ] Water level in each tank (visual from ladder or sight glass — (*confirm method*))
- [ ] Tank status label current (clean / dirty / in treatment)



- [ ] No leaks at the base valves or outlet connections
- [ ] Chemical dosing lines are injecting into the correct tank

**i WHY:** Mixing water from a "clean" tank into a "dirty" tank means retreating that water from scratch. Always know which tank contains what.

## 6.1a Blue Separator Bins (Outdoor, 2 Units)

**What they are:** Two large rectangular steel bins, painted blue, sitting outdoors on raised concrete ecology blocks (for spill containment/drainage underneath). Each has a small viewing window in the door and a round access/vent port on top. At least one has internal angled baffle plates inside (not a flat-bottom bin) and shows heavy orange-brown iron staining — consistent with iron-rich water having flowed through it. One bin has a "NO GARBAGE" sign — these are dedicated process equipment, not waste containers.

**Where they sit in the process:** Confirmed by site operator — water flows: oil/water separation cell → frac tanks (settles, oil rises to top, water drawn from below) → **these 2 blue bins** → submersible intake pump → the two white indoor tanks (Section 6.2).

**What they do:** Final outdoor polishing/separation step before water is pumped indoors for chemical dosing. The internal baffle plates force water to travel a longer path, giving any remaining oil sheen and solids more time and surface area to separate out before the water goes inside.

**[CHECK]** Each shift:

- [ ] Water level in each bin
- [ ] Any oil sheen visible through the viewing window
- [ ] Submersible intake pump (see below) running and not airlocked
- [ ] No leaks at the base or around the access port

**i WHY:** Anything not separated out here gets carried into the indoor tanks and chemical dosing stage — catching oil/solids at this outdoor stage reduces the load on every step downstream.

### Bin-to-bin configuration — CONFIRMED BY BRENNAN (series, gravity-fed)

- The two blue bins run **in series**, not parallel — confirms the settle-then-transfer model, not dual-parallel dosing.
- One bin sits at a slightly higher (or equal) elevation than the other and **gravity-feeds** into it through a 4" hose — a 2" pump's flow rate cannot overflow a 4" gravity line, so this transfer is passive/safe by design once set up.
- Brennan also described his typical full design for this kind of series setup: a "**sea can**" (shipping container) housing chemical totes and dosing pumps, with chemical injected via a **flow switch** in the piping, feeding into **weir tanks** where the actual settling happens, then discharging to an **open tote** with a submersible pump inside that pumps water into the building.

**✓ CONFIRMED BY PHOTO (IMG\_2802, IMG\_2806):** The C Can IS physically present, sitting right next to the blue bin marked "**NO GARBAGE.**" This confirms Brennan's design is, at least in part, what's actually built here — it is not purely his idealized/typical setup.

**✎ CORRECTED BY OPERATOR (IMG\_2806):** The hose/check valve at the base of the C Can (visible in both photos) is the **INFLOW from the frac tank into the C Can** — not the C Can's discharge line to the blue bin as earlier noted. So the confirmed path is: frac tank → check valve → into the C Can (dosing happens here) → ??? → blue bin (weir tank). The actual discharge connection from the C Can into the blue bin still needs to be traced/confirmed — it may be a separate, shorter line given how close the two sit together. See Section 6.1b for the C Can itself.

### Transfer Pumps (Portable Submersible)

**Confirmed by operator:** transfers are NOT done with fixed, permanently-plumbed pumps at each stage. Instead, portable submersible pumps (blue housing, single discharge port, standard electrical plug) are physically dropped into whichever tank needs to be moved next — **both the frac tanks and the blue bins** use the same style of pump — with a hose run to the destination tank.

- **Frac tank → blue bin:** pump dropped into the frac tank, hose run to the blue bin
- **Blue bin 1 → blue bin 2:** gravity-fed via 4" hose (see above) — no pump needed for this hop if elevation allows it
- **Blue bin 2 → indoors (Tank 1 / Tank 2):** per Brennan, discharge lands in an **open tote** with a submersible pump inside, which then pumps water into the building (confirm: is this open tote actually present on site, or is the pump currently dropped directly into blue bin 2 as earlier documented?)
- **Weir tank discharge — CONFIRMED BY OPERATOR:** a 2-inch submersible pump is placed in the **middle of the weir tank** (not too close to the floor — avoids the settled sludge; not too high — avoids the oil layer near the top) once the water has settled. It connects directly into the **intake pipe** running indoors. Ozone is turned on as the water flows through this intake pipe (see Section 6.7) — so this pump is also the trigger point for starting ozone treatment.
- **Sludge at the weir tank — CONFIRMED BY OPERATOR:** settled sludge accumulates at the bottom of the weir tank and needs a **diaphragm pump** to remove it (thick sludge will clog/destroy a standard submersible pump). **Confirmed: it's the same AODD pump used for both the outdoor weir tank and the indoor weir tank / settling bins (Section 6.12)** — not a separate dedicated unit.

**[CHECK]** Before starting any transfer:

- [ ] Pump and cord are in good condition (no cracked housing, no exposed wiring)

- [ ] Hose is connected securely at both ends, no kinks
- [ ] Destination tank has room for the incoming volume
- [ ] Pump is fully submerged before powering on (running dry can damage it)

⚠️ **VALVE SEQUENCING – CONFIRMED BY SUPERVISOR:** Open the **intake pipe valve** before turning on the submersible intake pump that pushes the weir tank water indoors, and open the **storage tank pipe valve** before water actually reaches the storage tank stage at the end of the process. Reason given: starting a pump against a closed valve lets pressure build up and sit on the pipe with nowhere to go. (still needed: exact physical location/identity of each valve – confirm with Brennan next time you're there.)

i **WHY:** Because the same pumps move between locations, always confirm which tank a pump is currently sitting in and where its hose is routed before assuming a transfer is happening automatically — nothing runs continuously here; every transfer is a deliberate, set-up step. (confirm exact pump model/spec with supervisor)

⚠️ **NOTE – needs confirmation:** It is unclear whether the "Wet Cell / Separation Cell" described below (Section 6.2a) is a **separate, additional** indoor concrete tank, or whether earlier site notes were actually describing these same blue bins. Confirm with supervisor before treating them as two different pieces of equipment.

i **PARTIALLY RESOLVED** (photos IMG\_2741, IMG\_2742): These show rusted, orange-iron-stained, baffled steel compartments **indoors** (OSB stud-wall construction visible behind them, with totes/drums nearby) — NOT outdoors on ecology blocks like these blue bins. That rules out these specific compartments being the blue bins described above; the staining and indoor location instead match the Wet Cell description in Section 6.2a. So there are 3 distinct structures on site: outdoor blue bins (this section), the indoor baffled "settling bins" (6.2a, operator's term), and the indoor white storage tanks Tank 1 / Tank 2 (6.2) — not 2.

## 6.1b The "C Can" (Chemical Dosing Shipping Container)

✅ **CONFIRMED BY PHOTO** (IMG\_2802): A tan/white shipping container, serial markings "**CPIU175014**" visible on the side, sitting right next to the blue bin marked "NO GARBAGE."

✅ **CONFIRMED BY OPERATOR** (IMG\_2806): The hose/check valve at the base of the container is the **inflow** — water comes from the frac tank, through this check valve, into the C Can.

**What it does:** Houses the chemical dosing step (pH check, then lime if needed, then PACI coagulant + flocculant — see Section 5.6a) before water flows on to the blue bin/weir tank for settling.

✅ **RESOLVED – CONFIRMED BY OPERATOR:** Water treated/dosed at the C Can discharges into the blue weir tank — confirmed as the live operating path. (The exact physical hose/fitting used for this discharge hasn't been visually traced in photos, but the connection and direction — C Can → blue weir tank — is confirmed.)

### Interior — CONFIRMED BY PHOTO (IMG\_2803, IMG\_2805)

Inside the C Can:

- **Two white 200L drums** with SUMAS labels, sitting side by side — one clearly labelled "**FLOCCULANT.**" ✅ **CONFIRMED BY OPERATOR:** the second drum is the **PACI coagulant**.
- **One blue 200L drum** to the right, with a hose/tubing dipped into the top. ✅ **CONFIRMED BY OPERATOR:** this is the **lime** (TEXLIME High Calcium Hydrated Lime) — kept here as a liquid/slurry in a blue drum, not as a dry bag. ⚠️ **NOTE:** Hydrogen peroxide is also stored in blue 200L drums elsewhere on this site (Section 5.7) — at the C Can specifically, the blue drum is lime, not peroxide. Go by location, not just colour.
- A **blue diaphragm-style dosing pump** mounted on a horizontal grey PVC pipe manifold above the drums, feeding into two large horizontal PVC pipe runs along the wall.
- An electrical control box and timer mounted on a plywood backing above the drums (red trays visible — possibly drip trays under fittings, not confirmed).
- IMG\_2805 shows the chemical injection detail: small clear tubing taps into the main PVC pipe runs through small fittings/check valves — this is where the dosing pump actually injects chemical into the flow line.
- A spill kit / eyewash-style station is mounted on the wall to the right (red bottle, white wall-mounted box).

(still needed: which pipe run is which – frac tank inflow vs. weir tank discharge)

## 6.2a Indoor Weir Tank / Settling Bins — "Wet Cell" / Separation Cell (Indoor Settling Tanks)

i **TERMINOLOGY:** The operator calls this the **indoor weir tank / settling bins** — distinct from the **outdoor weir tank** (the blue bin, Section 6.1a). Always specify indoor/outdoor when referring to either one.

i **CONFIRMED BY OPERATOR** (IMG\_2740, IMG\_2741, IMG\_2742, IMG\_2745): These are **the settling tanks** referred to elsewhere in this manual — physically separate from, but located right next to, the white cylindrical storage tanks (Tank 1 / Tank 2, Section 6.2) (a white tank is visible at the edge of the

frame in these photos/video). Do not confuse the two: storage tanks just hold water; this is where the actual settling happens.

**i** TERMINOLOGY — CONFIRMED BY OPERATOR: site operators call these the **"settling bins"** (plural — there are 2), and describe them as used **"in conjunction with the filters."** Walkthrough videos IMG\_2791/IMG\_2792 support this: the rusted/baffled settling bin interior, the Kontek filter press, and the green SRS-SF-11 sand filter vessel all sit in the same room within a few steps of each other, with pipe runs (including red-handled ball valves) connecting the area. No new/different equipment was identified in these videos — they confirm physical adjacency and sequence (settling bins feed into the sand filter, not separate equipment) rather than introducing anything new.

**What they are:** Multi-compartment tanks inside the building, baffled partitions dividing the flow into sections. Construction appears to be welded/rusted **steel**, not poured concrete as earlier notes assumed — confirm which with supervisor.

**Appearance:** Heavy dark brown/black staining (oil residue mixed with iron staining) on all walls and floor — this is normal, not neglect. It's the visual record of treatment doing its job. Currently shown empty/dry in the latest photos/video.

The baffled partitions divide the flow path into compartments, forcing water to travel a longer path and giving solids more time to settle.

**What they do:** Secondary settling — but **✓** CONFIRMED BY OPERATOR: this is also where water lands right after ozone treatment (Section 6.7), not just after chemical dosing. Sequence: weir tank (outdoor blue bin) → ozone-fed via the "down" pipe → these settling bins, to settle further → submersible pump pushes it out via the "up" pipe → bag filter → carbon vessel. Floc formed by PACl and polymer settles to the floor here. The sludge accumulates and must be periodically removed.

**✓** RESOLVED — how long to settle here before moving to the storage tanks, CONFIRMED BY SUPERVISOR (Brendan, text message): There's no fixed settle time — **it's visual/judgment-based**. Brendan's exact words: *"If there's no chemicals in the water it's not going to settle out so they haven't been settling just sitting outside in the tanks. Maybe the giant particles but that doesn't matter. The water should be as clear as tap water by the end of settling so it's visual. It depends how much chemicals you put in, how dirty the water is and how long you let it settle."*

**i** WHAT THIS MEANS IN PRACTICE:

- Settling only really happens **after chemical dosing** (PACl + flocculant) — without chemicals in the water, it won't form floc and settle out here; sitting in the outdoor weir tank alone only drops the largest particles, which doesn't matter much.
- The bar to clear before pumping on to storage: **water should look as clear as tap water**. If it's still cloudy/turbid, give it more time.
- No stopwatch number to hit — settle time depends on how much chemical was dosed, how dirty the water was going in, and how long it's been given. Use the same visual check each time rather than a fixed clock.

**[CHECK]** Each shift:

- ☐ Water level — is it at the normal operating level?
- ☐ Sludge on the floor — is it accumulating faster than usual? Can you see the bottom?
- ☐ Water clarity — does the outflow look noticeably clearer than the inflow?
- ☐ Any oil floating on top inside the wet cell?

**Sludge removal:** Pump the accumulated sludge from the bottom of the wet cell using the AODD pump (Section 6.12) to the filter press (Section 6.11) or to the stabilization pond as directed.

**⚠ WARNING:** The wet cell is a **confined space**. Never enter without a confined space entry permit, atmospheric testing, and a trained attendant outside. The iron-laden environment can generate hydrogen gas and deplete oxygen. See Section 9.6.

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## 6.3 Chemical Dosing Room

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The chemical dosing room contains drums and dosing pumps for each treatment chemical. Each pump draws chemical from its drum and injects it into the water stream at the correct dose.

### PACl (Coagulant) Pump

- **Drum:** White 200L SUMAS drum, Class 8 label, "POLY ALUMINUM CHLORIDE"
- **Pump:** Blue diaphragm metering pump
- **Injection point:** Into the water flow before the wet cell (or as directed)

### CL-200 (Flocculant) Pump

- **Drum:** White drum, "NON-REGULATED MATERIAL — FLOCCULANT" label
- **Pump:** Yellow LMI diaphragm dosing pump
- **Injection point:** After PACl, before or during slow mixing in the wet cell

### Injection manifold — CONFIRMED (video IMG\_2767)

A walking video of the treatment room confirms the full physical path: the dosing drums (flocculant + PACl, same station as IMG\_2758, sitting next to a white storage tank) feed a wall-mounted PVC manifold that runs horizontally along the wall with several T-fittings/valves branching off it. At one of those branches,

the video clearly shows a pipe **dropping straight down into the settling tank** (the rusted/baffled trough from Section 6.2a). This confirms the injection sequence: **dosing drums** → **wall manifold** → **settling tank**. The earlier sighting of a similar black manifold on video (IMG\_2775, IMG\_2776) is the same system, just a different angle/section of the same wall run.

### Dosing drum station confirmed via photo (IMG\_2758) and video (IMG\_2767)

Flocculant drum (white, SUMAS label) with its yellow LMI metering pump mounted directly on the drum lid, plus a second blue drum (likely the PACI/coagulant stock mix) right beside it, sitting next to a white storage tank for scale. The floor around this station has heavy **white** crusting — distinct from the orange/iron staining seen everywhere else on site, and consistent with PACI forming a white precipitate where it's mixed/handled.

### Checking a dosing pump is actually working

#### [CHECK]

1. Is the pump making a rhythmic clicking/ticking sound? (Diaphragm pump = audible with each stroke)
2. Is the tank level dropping over the course of a shift?
3. Is chemical visible at the injection point? (Hold a rag near the injection — you should see a small squirt/drip with each stroke)
4. If the pump sounds right but nothing is coming out → check for a broken or cracked injection check valve, or a clogged injection point.

### Priming a dry dosing pump

If a pump has run dry or been disconnected, it needs priming:

#### [ACTION]

1. Disconnect the discharge tubing from the injection point (hold a rag over the end).
2. Run the pump until liquid (not air bubbles) flows continuously from the discharge.
3. Reconnect to the injection point.



**WARNING:** Wear chemical-resistant gloves and splash goggles when working on any dosing pump — chemical may spray from the disconnected line.

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## 6.4 Sand Filter (SRS-SF-11)



**RESOLVED – NOT USED, CONFIRMED BY OPERATOR:** **"We don't use the sand filter."** This is the current, settled operating reality at this site — **water flows straight from settling to the bag filter and carbon vessel only**, and that's the normal process, not a temporary workaround. (Background, for context only: the vessel has a known solenoid fault on the backwash control box — see "Backwash control box" below — but fixing that is not treated as an active to-do here, since the sand filter step isn't part of how this site runs the train.) Because the sand filter step is skipped, the bag filter (Section 6.6) is taking on the full solids load it wasn't designed to handle alone — watch it closely and change it more often.

**Appearance:** Green cylindrical pressure vessel(s) on a metal stand. Tagged "SRS-SF-11" (confirmed via close-up photo, IMG\_2778, on a vessel with a multi-port control valve and actuator on top, plus a pump mounted below/beside it).



**RESOLVED – there really are two green vessels:** Photo IMG\_2777 confirms two green cylindrical pressure vessels standing side by side, each with its own round bolted inspection/access hatch on the side, connected together by a shared pipe manifold across the top. This looks like a **duplex (twin) sand filter arrangement** — a common design where two vessels share the load, or one runs while the other is backwashed/serviced, or one is a standby spare for the other. Visible right next to them: the blue/white filter-press-looking unit (the one you said "we don't use") and a white bag at the base that's likely sand filter media stock (matches the Play Sand/Sakrete bags spotted near the H<sub>2</sub>O<sub>2</sub> drums in earlier photos).

(low priority – sand filter isn't used at this site, so this is background curiosity rather than an open operational question: is this a true duplex setup where both vessels are meant to run, or was one of these two the vessel that failed, with the other being the spare/backup? Which one is tagged SRS-SF-11 specifically – IMG\_2778's multi-port valve head looks different from the simpler top connections on the pair in IMG\_2777, so there may be a 3rd vessel, or IMG\_2778 is a closer shot of one of the same two.)

### Backwash control box (Alex-Tronix Controls, Model F2AC/DC-D, Serial F2-04519)

Confirmed via photo (IMG\_2782, IMG\_2783): a separate control timer box mounts near the vessel and automates the backwash cycle on the multi-port valve/actuator. Front panel has 3 timing dials (days between backwash, backwash duration in seconds, a third timed stage), alarm/dwell indicator lights, a digital backwash counter, and three buttons — **MANUAL START/ADVANCE**, **COUNTER RESET**, **ALARM RESET** — plus a **POWER** switch and a **1.6 AMP output fuse**.



#### TROUBLESHOOTING – filter/valve not responding:

1. Confirm the **POWER switch** is set to ON.
2. Check the **1.6A output fuse** — pull and inspect, or test with a multimeter.
3. Press **MANUAL START/ADVANCE** to force the valve to move right now, bypassing the timer dials. Motor responds → power/board are fine, only the schedule was off. No response → continue.
4. Confirm the **DAYS dial** (leftmost) isn't set to OFF (this only blocks automatic cycling — manual start should still work regardless).

5. With power OFF, open the case (IMG\_2782) and check the wiring at the transformer and green circuit board for loose, disconnected, or corroded connections.

6. If the valve motor still doesn't move after all of the above with confirmed power reaching the box, the actuator/motor itself has likely failed. **This may be the actual root cause of the sand filter being reported "broken"** — not the vessel or the sand bed itself. Flag this distinction to your supervisor/SUMAS, since a control box or actuator repair is a much smaller job than vessel replacement.

**i** CONFIRMED BY BRENNAN — likely root cause identified:

- **Water must be actively flowing through the vessel for the backwash cycle to work** — the cycle is meant to be triggered while water is running through the filter, not with flow stopped.

- A correctly-working cycle should produce **an audible click** as the **bottom valve** closes, plus visible/audible actuation of the **small grey solenoids with black tubing** coming out of them (these solenoids are the actual valve actuators — distinct from the Alex-Tronix timer board itself, which only sends the signal).

- Brennan's assessment: **the grey solenoids themselves are most likely the failed component**, not the timer board (consistent with the timer board's logic/counter working fine in earlier testing, but nothing physically moving at the valve).

- Brennan does not know how to repair the solenoids — at his previous site (Burnaby plant) this was handled by **bringing in an electrician**.

**i** NOTE: This diagnostic background is kept for reference only — since the sand filter isn't used at this site (see top of Section 6.4), getting an electrician in to fix the solenoids is not an active task. If that ever changes, this is where troubleshooting left off.

**What it does:** Pressurized filtration through a bed of graded sand. Water flows down through the sand, which traps iron hydroxide particles, fine suspended solids, and any remaining floc.

**i** WHY: After hydrogen peroxide oxidizes dissolved iron into iron hydroxide particles, those particles need to be removed. The sand filter is their primary removal point. The orange staining visible on the exterior of all equipment around the filter confirms how much iron this filter catches.

## Normal operation checks

[CHECK]

- [ ] Water is flowing through (inlet pressure gauge reading)
- [ ] Outlet turbidity is lower than inlet turbidity
- [ ] No unusual vibration or noise from the pump below

## Why backwashing is necessary

Over time, water flowing through the sand carves the same paths over and over, forming **channels** — narrow worn-in routes that water flows through easily while bypassing the rest of the sand bed. Once channels form, the filter looks like it's running fine (water still flows) but it's actually filtering less and less, because most of the water is taking the path of least resistance instead of passing through clean sand. Backwashing (reversing the flow) breaks up those channels and resets the sand bed. Beyond backwashing, the sand itself also needs to be fully replaced periodically as it gets dirty — how often depends on how much solids loading it sees.

## When does it need backwashing?

Backwash the sand filter when any of the following:

1. Outlet turbidity starts climbing above your target
2. Flow rate through the filter drops significantly (filter is clogged)
3. Pressure drop across the filter exceeds the setpoint

## Backwash procedure (generic — confirm exact valve positions with supervisor)

[ACTION]

1. Close the inlet valve.
2. Close the outlet (forward flow) valve.
3. Open the backwash drain valve.
4. Open the backwash supply valve (clean water in from the bottom, up through the sand).
5. Run until the backwash discharge runs clear — typically 5–15 minutes.
6. Close backwash valves, re-open inlet and outlet to return to normal operation.
7. Redirect first 5–10 minutes of post-backwash effluent back to the wet cell (not forward to clean storage) — filter needs to "ripen" after backwash.
8. Record backwash in the log: time, reason, duration.

---

## 6.5 Carbon Vessel (SRS-CV-15)

**Appearance:** Blue cylindrical pressure vessel, "PV SERIES" label, tagged "SRS-CV-15". Sits on a wooden pallet. **Maximum working pressure: 75 PSIG** (marked on the vessel — never exceed this). Matches photo IMG\_2796 (yellow "...SERIES" tag visible, hose fitting at base, mounted against the wood-stud

wall).

**i** CONFIRMED – currently in active use: the sand filter isn't used at this site (Section 6.4), so water flows straight from settling to the **bag filter**, then this **carbon vessel** — this is one of the two filtration steps the water actually sees.

**✓** CONFIRMED BY OPERATOR (photos IMG\_2795, IMG\_2796) – hose path, corrected: The **black hose** is the line between the bag filter and this carbon vessel — it's the carbon vessel's inlet, not the pump's connection to the bag filter (an earlier note had this backwards). Once water has passed through the carbon vessel, the **green hose** carries it via a PVC pipe to the **storage tank** (Section 6.2) — that's the last step before storage.

Full confirmed path: submersible pump (in settling bins, Section 6.2a) → hose → elbow/camlock fitting at the **bottom** of the bag filter housing (Section 6.6) → water flows up into the bag filter, through the bag, back down → **black hose** → carbon vessel (enters near the base) → **green hose** → PVC pipe → storage tank. The sand filter (Section 6.4) and the Kontek filter press (Section 6.11, used only for sludge dewatering) are **not** part of this water path.

**i** NOTE on backwashing: Both visible connections on this vessel (the black hose inlet and the green hose/PVC outlet) are at the **base** of the vessel, near each other — no separate backwash supply/drain valves were visible in photo IMG\_2796. The only other opening is the top manway/access lid. This supports the theory that backwashing this vessel (if done at all) means manually disconnecting and swapping these two hoses to reverse flow, rather than turning dedicated valves — still needs confirming with whoever has actually backwashed this vessel before.

**⚠** CONFIRMED BY OPERATOR – CURRENT CONDITION (as of this writing): This vessel has **sludge and buildup inside, with old/spent carbon** still in it. **It must be cleaned out before starting any water treatment.** This is a one-time clean-out of the vessel's current state, separate from the routine weekly backwash and 1.5-month check described below — don't treat this as already covered by those.

**What it does:** Activated carbon adsorption. The vessel is filled with granular activated carbon (GAC) — a highly porous material with enormous surface area. Dissolved petroleum hydrocarbons, colour compounds, and odour-causing organics bind to the carbon surface as water passes through.

**i** WHY: Suspended solids are removed by sand filtration. But dissolved hydrocarbons (the ones that give the water a fuel smell or create a downstream sheen) cannot be filtered — they pass through sand. Carbon adsorption is the only way to remove them.

## Normal operation checks

### [CHECK]

- [ ] Water is flowing through (inlet valve open, no unusual pressure)
- [ ] Outlet water does not smell of petroleum or fuel
- [ ] No colour in the effluent (should be essentially clear)
- [ ] **CONFIRMED BY OPERATOR:** check turbidity here with the NTU meter (the bag filter and carbon vessel outlets are the site's turbidity check points — see Section 4.3)

## Check frequency — CONFIRMED BY OPERATOR

Check the carbon vessel every 1.5 months.

## Signs the carbon is exhausted (needs replacing)

- Petroleum odour returns in the treated water output
- Colour (brown/yellow tint) appears in the effluent
- The vessel has been in service for its rated operational volume ((confirm with supervisor) — carbon life depends on hydrocarbon loading)

**⚠** WARNING: Backwashing carbon does **not** restore its adsorption capacity. Once the carbon is chemically saturated (it's adsorbed all it can), no amount of backwashing brings that back — the media must be physically removed and replaced with fresh carbon. Exhausted carbon is a regulated waste — dispose of through an approved waste contractor. Backwashing below is a separate, routine maintenance step — see why below.

## Backwash frequency — CONFIRMED BY OPERATOR

**Backwash the carbon vessel every week.** This is separate from the 1.5-month check above: the weekly backwash is routine bed maintenance, not a check for exhaustion.

**i** WHY (general principle): Even though backwashing can't "recharge" spent carbon, it still matters — water flowing through a carbon bed over time packs fines and trapped solids into the same flow paths, the same channeling problem described for the sand filter (Section 6.4). A weekly backwash fluffs the bed back up, breaks up channels, and flushes out accumulated debris so the full bed surface stays in contact with the water — keeping the carbon that's still active working as effectively as possible until it's actually exhausted and needs replacing.

## Backwash procedure (generic — confirm exact valve positions/timing with supervisor)

### [ACTION]

1. Close the inlet (forward flow) valve.
2. Close the outlet valve.
3. Open the backwash drain valve.

4. Open the backwash supply valve (clean water in from the bottom, up through the carbon bed, lifting and expanding it).
5. Run until the backwash discharge runs clear of fines/debris.
6. Close backwash valves, re-open inlet and outlet to return to normal operation.
7. Record the backwash in the log: time, duration.

(still pending: exact valve layout on SRS-CV-15, target backwash duration, and whether a "ripening" period like the sand filter's is needed before sending flow forward again.)

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## 6.6 Bag Filter Housing

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**Appearance:** Blue/dark cylindrical filter housing, approximately 1 metre tall, mounted on a stand directly above a white IBC tote (matches photo IMG\_2795 — bolted ring clamp on the housing body, pipe fitting dropping down toward the tote underneath).

✅ **CONFIRMED BY OPERATOR** (photo IMG\_2795) — **hose path:** The **inlet port** is the elbow fitting with the camlock connector at the bottom of the housing, right above the IBC tote. A hose from the submersible pump in the settling bins (Section 6.2a) connects here — water is pushed up into the housing through this port, passes through the filter bag, then flows back down and out through a separate **black hose** at the bottom of the housing, which carries it onward to the carbon vessel (Section 6.5).

**Filter bags:** Les Hall Filter brand

- Pre-filter bags: LHFSPE100PJNBW (100 micron) — removes larger particles
- Polishing bags: LHFSPE10PJNBW (10 micron) — removes fine particles

i **CONFIRMED — currently in active use:** the sand filter isn't used at this site (Section 6.4), so this bag filter is the **first** filtration step the water hits, ahead of the carbon vessel (Section 6.5) — carrying the full solids load alone. Check/change bags more often as a result.

**What it does:** Acts as a safety screen — its job is making sure no dirt reaches the carbon vessel (Section 6.5) downstream. The 100-micron bag catches coarse particles; the 10-micron bag polishes the rest. i **ORDER CORRECTION:** A former operator placed this filter **before** the carbon vessel (sand filter → bag filter → carbon vessel), not after it as earlier photo-based notes suggested. Carbon can't be backwashed or easily cleaned (see Section 6.5), so protecting it with a filter ahead of it makes more sense than filtering after it — treat the bag filter → carbon order as the current best understanding until confirmed otherwise. **Since the sand filter (Section 6.4) isn't used, this bag filter is the main line of defense before the carbon vessel — check and change bags more frequently.**

### How to change a filter bag

#### [ACTION]

1. **Close the inlet and outlet valves** on the bag filter housing.
2. Open the **pressure relief/vent valve** to release pressure (there will be water in the housing — have a bucket ready).
3. Undo the top closure ring and lift the lid.
4. Pull the old bag out by the collar — hold it over a bucket, it will drip.
5. Inspect the old bag: How much solids are on it? Is it iron-coloured or hydrocarbon-coloured? Note in the logbook.
6. Place old bag in waste container.
7. Install new bag — ensure the collar seats fully in the housing ring. A poorly seated collar allows water to bypass the bag entirely.
8. Replace lid and tighten closure ring.
9. Slowly open the inlet valve first (let housing fill), then open outlet.
10. Check for leaks at the top closure.
11. Record bag change: date, time, which bag (100-micron or 10-micron), appearance of old bag.

### When to change bags

- **Check the bag filter every 1–2 weeks — CONFIRMED BY OPERATOR.**
  - Also check sooner if flow rate through the housing drops noticeably (bag is loaded with solids)
  - After any event that puts unusual solids loading through the system (major rain event, wet cell cleanout)
  - **CONFIRMED BY OPERATOR:** check turbidity here with the NTU meter (the bag filter and carbon vessel outlets are the site's turbidity check points — see Section 4.3)
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## 6.7 Ozone System (AERQUS + CD30TIX)

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✅ **RESOLVED — CONFIRMED BY OPERATOR:** Ozone is the **first thing that happens once water leaves the weir tank and enters the building** — not a polishing step at the end. This confirms the former operator's account over the earlier photo-based guess that placed it after carbon.

**Confirmed sequence at this transition:**

1. Water finishes settling in the weir tank (blue bin).



2. A **2-inch submersible pump** is placed **in the middle of the weir tank** — deliberately not too close to the floor (avoids sucking up settled sludge) and not too high (avoids the oil layer near the top).
3. That pump connects to the **intake pipe** running indoors.
4. **Ozone is turned on as the water flows through the intake pipe** — so ozone treats the water on its way in, before the bag filter and carbon vessel.

✓ **RESOLVED — ORDER OF OPERATIONS, CONFIRMED BY SUPERVISOR (Brendan, text message):** **Turn the ozone on *before* starting the pump that connects the outdoor blue weir tank to the intake — not after.** Brendan's exact words: *"Turn the ozone on first before pumping."* So the correct startup order is: ozone generator ON → then start the weir tank submersible pump/intake. Don't reverse this — starting the pump first means raw water flows into the intake before ozone is actually treating it.

i **WHY pump placement matters:** The weir tank has settled sludge at the bottom and an oil layer near the top — the cleanest water is the middle layer. Placing the pump intake there pulls the best water and avoids re-suspending sludge or pulling in oil.

i **WHY ozone first:** This means ozone is oxidizing raw, just-settled water — breaking down hydrocarbons/iron and disinfecting before the bag filter and carbon vessel do their polishing. The carbon vessel is the deliberate backstop for whatever ozone doesn't fully catch (per Brendan, Section 6.7 below).

✓ **CONFIRMED BY OPERATOR — what happens right after ozone:** Ozone-fed water doesn't go straight to the bag filter. It first goes into the **indoor settling bins/wet cell** (Section 6.2a) to settle further. Two pipes are involved at this transition:

- **The "down" pipe** — carries water from the weir tank, gets fed ozone along the way, and delivers it into the indoor settling bins.
- **The "up" pipe** — once the water has settled in the indoor settling bins, a submersible pump pushes it through this pipe, on to the bag filter and carbon vessel.

⚠ **STILL OPEN:** Reviewed all 3 walkthrough videos (IMG\_2791, IMG\_2792, IMG\_2748) frame-by-frame — found two bag/carbon-style filter housings (same design, different rooms) but couldn't conclusively trace which specific pipe is the "down" vs. "up" pipe due to motion blur in the footage. A close-up photo of the two pipes themselves (not the whole room) would resolve this.

#### Equipment:

- **CD30TIX Ozone Generation System** — stainless steel cabinet with pressure gauge. Generates ozone (O<sub>3</sub>) from an oxygen feed gas (see AERQUS below). **Gas concentration — CONFIRMED BY OPERATOR: 10% ozone gas.**

- **AERQUS — CORRECTED BY PHOTO (IMG\_2823, IMG\_2824):** This is **not** a diffuser. The unit is labelled "**AERQUS<sup>2</sup> — Reliable On-Site Oxygen**" — it's an **on-site oxygen generator/concentrator** that supplies oxygen feed gas to the CD30TIX. i **WHY THIS MATTERS:** Corona-discharge ozone generators fed with pure oxygen (instead of ambient air) produce roughly double the ozone concentration for the same unit — this is almost certainly why this site can run 10% ozone gas. Earlier manual entries calling this a "diffuser" were wrong and are corrected here.

- **AERQUS flow meter / adjustment knob — CONFIRMED BY PHOTO (IMG\_2823):** This is the dial Brendan described ("a little dial on one that you can turn to increase or decrease the amount"). It's a rotameter-style flow meter, scale **0–20 SCFH** (standard cubic feet per hour), with a **–/+ knob** on the left side of the tube to adjust oxygen flow to the CD30TIX. Turning it up increases oxygen feed gas (and therefore ozone output); turning it down decreases it.

- A small digital hour-meter display and a green indicator light also sit on the AERQUS panel — likely run-hours and power/status, not yet confirmed.

✓ **RESOLVED — "0%" READING, CONFIRMED BY SUPERVISOR (Brendan, text message):** **"The ball should move, if it's not moving there might be a blockage somewhere. But as long as it's not at 0 it's injecting into the system."** So: check the float ball position on the flow meter — if it's moving and reads above 0 SCFH, oxygen is actively feeding the CD30TIX and the system is working, regardless of what a separate "0%" display elsewhere might suggest. If the ball is genuinely stuck at 0 or not moving at all, that indicates a blockage to troubleshoot.

i **CONFIRMED BY SUPERVISOR (Brendan) — why exact ozone dosing isn't the main worry:** "The carbon at the end of the system will grab anything it doesn't [catch]. The main focus is getting the water clear." The carbon vessel (Section 6.5) is the deliberate backstop for whatever ozone doesn't fully oxidize — so small variations in ozone output aren't treated as a crisis the way a clarity problem would be.

#### What ozone does:

Ozone is a powerful oxidizer — much stronger than chlorine or hydrogen peroxide. In this system, it:

1. **Breaks down organic contaminants** — hydrocarbons and other organics
2. **Breaks down inorganic contaminants** — including oxidizing dissolved iron (Fe<sup>2+</sup>) into filterable Fe<sup>3+</sup> particles
3. **Disinfects** — kills bacteria and other pathogens

i **WHY:** Whether it runs first or last, ozone's underlying job doesn't change — it's one of the strongest oxidizers available on site, breaking down contaminants that gravity and filters alone can't touch, and handling odour elimination by destroying the molecules responsible for petroleum smell.

#### Normal operation checks

##### [CHECK]

- [ ] CD30TIX power indicator light is on
- [ ] Pressure gauge on CD30TIX reads in normal range (*confirm target with supervisor*)
- [ ] AERQUS flow meter float ball reads above 0 SCFH (oxygen is actually feeding the CD30TIX)
- [ ] Faint ozone smell in the room — this is normal and confirms ozone is being generated
- [ ] No alarm lights on either unit

⚠ **WARNING — Ozone is Toxic to Breathe**



- Ozone has a sharp, bleach-like smell. At low concentrations it irritates the respiratory tract. At higher concentrations it damages lungs.
- **Never smell ozone directly or put your face near the diffuser.**
- The room where ozone is used must be well-ventilated.
- **Workplace exposure limit (Canada):** 0.1 ppm (8-hour TWA) — very low
- If you smell strong ozone, ventilate the area and stand back.
- Ozone destroys rubber — inspect any rubber hoses or gaskets near the ozone system regularly for cracking/degradation.

## 6.8 Chlorination (Accu-Tab Blue Cal-Hypo Tablets)

**Product:** Accu-Tab Blue Calcium Hypochlorite tablets in a tablet chlorinator or contact vessel.

**What it does:** Final disinfection step before discharge. As water flows over the tablets, they dissolve slowly and release free chlorine. The target is a measurable free chlorine residual at the discharge point, confirming the water has been adequately disinfected.

### How to add tablets

#### [ACTION]

1. Wear chemical splash goggles and chemical-resistant gloves.
2. Confirm the contact vessel is dry or free of other chemical residues (especially H<sub>2</sub>O<sub>2</sub> residue — see compatibility warning in Section 5.2).
3. Add tablets as directed — do not add a full bucket at once if the vessel is not designed for it.
4. Ensure water is flowing through at the correct rate (too fast = under-dosing; too slow = over-dosing).
5. Test free chlorine in the effluent with a DPD test kit (see Section 4.4.2).
6. Record: tablets added (approximate number or weight), test result, date, time.

### Tablet level monitoring

[CHECK] Each shift: check tablet level in the chlorinator. Tablets dissolve continuously. Running out = no disinfection = potential permit violation.

## 6.9 Oil/Water Separator (Shell Canada System)

**System:** Above-ground separator vessel + underground holding tank

**Documentation:** Shell Canada log sheet and weekly inspection flow chart on file

**What it does:** Petroleum products (oil, fuel) are less dense than water, so they float. The separator allows water to sit long enough for:

- Petroleum to rise and accumulate at the top (where it is skimmed off)
- Sludge and heavy solids to sink to the bottom
- Cleaner water to be drawn from the middle layer

The recovered petroleum and sludge are **regulated wastes** — they must be drummed, labelled, and removed by a licensed waste contractor. They do NOT go to the discharge point.

### Weekly Inspection (per Shell Canada Flow Chart)

[ACTION] Each week:

1. **Inspect the slab and drainage points** — any spills or leaks visible?
2. **Dip the underground holding tank** for fuel thickness and sludge thickness.
  - Record: total liquid depth, fuel thickness, sludge thickness
3. **Check that liquid level is within acceptable range:** 0.8m to 1.1m from tank bottom (25,000 L = 0.8m, 44,000 L = 1.1m from bottom — *(confirm your tank specs)*)
4. **Dip the above-ground separator** for fuel and sludge thickness.

### Action thresholds

| Measurement                                     | Action                                                                     |
|-------------------------------------------------|----------------------------------------------------------------------------|
| Underground tank sludge > 15 cm                 | Call to arrange sludge removal                                             |
| Underground tank fuel (L.P. petroleum) > 2.5 cm | Arrange skimming                                                           |
| Above-ground separator fuel > 15 cm             | Skim manually (call maintenance contractor if it re-occurs)                |
| Liquid level outside 0.8–1.1m range             | Check pump switch is set to automatic; call contractor if problem persists |

| Measurement                            | Action                                                            |
|----------------------------------------|-------------------------------------------------------------------|
| Any visible petroleum bypass at outlet | Inspect separator; skim; notify Shell field engineer if re-occurs |

## Completing the Shell Canada log sheet

Every weekly inspection must be logged on the Shell Canada log sheet including:

- Date, location, inspector name
- Visual inspection results
- All dip measurements
- Any corrective actions taken
- Sludge/petroleum removed (litres)
- Contractor name and manifest number for any waste removal

## 6.10 Frac Tanks

**What they are:** Large portable above-ground steel water storage tanks — the same type used in oil-and-gas hydraulic fracturing ("frac") operations for water storage. Typically 100,000 L (100 m³) capacity each.

### Their role in this system:

- Store raw site water before treatment
- Store treated water before discharge
- Used to transfer water between treatment stages
- Buffer capacity — allow you to treat in batches even when incoming flow is variable

### The clean/dirty rule

**⚠ WARNING:** Mixing clean (treated) water with dirty (untreated) water means starting treatment over. Know the status of every frac tank at all times.

**[ACTION]** Maintain a visible label or tag on each frac tank:

- **DIRTY / UNTREATED** — raw site water
- **IN TREATMENT** — water currently being processed
- **CLEAN / TREATED** — water that has passed all treatment steps and is ready for discharge or reuse

### Level monitoring

- Check levels each shift — manual dip tape or sight glass (*confirm method with supervisor*)
- Record in logbook: tank ID, level, content status (clean/dirty)
- Alert supervisor if: a clean tank is getting dangerously low (need to treat more); a dirty tank is overflowing (need to treat faster or manage incoming flow)

### Transferring between frac tanks (per "Brennan's List" task note)

**[ACTION]**

1. Confirm source and destination tanks and their current status.
2. Connect transfer hose — check for cracks, leaks, loose fittings before starting.
3. Start transfer pump; monitor for leaks at all connections.
4. When source tank is at the desired level, stop the pump and disconnect hoses.
5. Update tank labels and log the transfer.

## 6.11 Kontek APC5 Filter Press (Sludge Dewatering)

**Manufacturer:** Kontek Ecology Systems Inc., 3250 Harvester Road Unit #1, Burlington, Ontario L7N 3W8

**Phone:** (905) 332-8365 | [www.kontekecology.com](http://www.kontekecology.com)

**Model:** APC5 | **Serial:** K2012190 | **Electrical:** 120V, 1-phase, 3A

**What it does:** A filter press squeezes water out of sludge, producing:

1. **Filter cakes** — semi-dry solids that fall into collection bins below the press. These are regulated waste — removed by licensed contractor.
2. **Filtrate water** — the water pressed out of the sludge. Usually returned to the head of the treatment system for re-treatment.

**i WHY:** The wet cell and settling ponds accumulate sludge continuously. You cannot just pump sludge to a drain — it must be dewatered and disposed of as regulated waste. The filter press is how you turn a liquid sludge into a manageable solid waste.

## Control Panel

- **Keyed ON/OFF switch** (top) — must be in the ON (I) position for the press to operate
- **Green START button** (middle) — starts the hydraulic pump / pressing cycle
- **Red RESET button** (bottom) — stops the cycle / resets after a fault

## Basic Operation Sequence

### [ACTION]

#### Before starting:

1. Confirm all filter plates are properly closed and aligned.
2. Confirm the hydraulic ram has clamped the plate stack tight (check that there is no gap between plates).
3. Place collection bins beneath the press to catch filter cakes.
4. Confirm filtrate collection is set up (where will the pressed water go?).

#### Running the press:

5. Key switch to ON.
6. Press the green START button.
7. Start the sludge feed pump — pump the sludge slurry into the press inlet. The press fills from one end and fills each chamber between plates.
8. Monitor the filtrate outlets on each plate — you should see water draining from each plate's corner outlets.
9. The press cycle is complete when: filtrate flow slows to almost nothing (press is full, all water has been squeezed out).
10. Stop the sludge feed pump.
11. Press RESET (red button) if needed to release pressure.
12. **Slowly** open the press by releasing the hydraulic clamping pressure (per your specific press design — confirm with supervisor).
13. Open the plate stack — pull plates apart one by one; filter cakes will fall (or need to be manually dislodged) into the collection bins below.
14. Inspect each filter cloth — any torn or damaged cloths will allow sludge to bypass. Replace damaged cloths.
15. Close plates, clamp, and repeat for next batch.

**⚠ WARNING:** **Never open the plates while the press is under feed pump pressure.** The pressure will force liquid and sludge out explosively. Stop the feed pump and confirm pressure is zero before opening.

## Collection bins

Two IBC-style bins sit below the press to collect filter cakes. When full, arrange waste removal through a licensed waste contractor. Record: date, approximate volume/weight, waste type (iron-bearing sludge), contractor name and manifest number.

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## 6.12 AODD Pump (Air-Operated Double-Diaphragm)

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**Appearance:** Silver/aluminum body with two large circular end caps, air exhaust port in the centre, bolted flange connections. Labeled "OUT" with handwritten note.

**Likely make/model — identified via image lookup (IMG\_2838), not yet confirmed off the physical nameplate:** ARO air-operated double-diaphragm pump, possibly model 6662A3-344-C — 2-inch fluid connections, polypropylene body, PTFE diaphragms. (confirm against the actual nameplate next time you're at the pump — image-based ID can be wrong.)

**What it is:** An air-powered pump that uses two flexible diaphragms alternating back and forth to push liquid. No electric motor — runs on compressed air from the shop air supply.

**i CONFIRMED BY OPERATOR:** **This is the one diaphragm pump used to pump out sludge for both the outdoor weir tank and the indoor weir tank / settling bins** — not separate dedicated units for each.

**What it's used for:** Pumping thick sludge and slurries that would destroy a standard centrifugal pump. The AODD can handle:

- Indoor weir tank / settling bins sludge transfer to the filter press
- Outdoor weir tank sludge (Section 6.1a)
- Any thick, gritty slurry

## Using it to pump sludge into a tote

### [ACTION]

1. Drop/position the suction hose into the sludge at the bottom of the tank you're clearing (outdoor weir tank or indoor weir tank / settling bins).
2. Run the discharge hose into the tote you're collecting sludge in.

3. Connect the air supply hose, open the air valve — pump starts cycling ("chunk-chunk-chunk").
4. Increase air pressure to get a steady flow (up to ~100 psi max).
5. Watch the source — once you're pulling mostly air/no more sludge coming up, the tank is as clear as this pump will get it. Close the air valve to stop.
6. Disconnect hoses in reverse order.

**i** How often: No fixed daily schedule confirmed for this site — sludge removal is based on how much has actually built up, not a clock. Check visually (look at the bottom of the tank) at the start or end of your day; if there's a meaningful layer, pump it out. **Confirm with Brendan what frequency he actually wants** — daily, weekly, or "whenever you see it."

**!** Where the sludge goes – does NOT go back into the Oil/Water Separation Cell ("the lagoon"). Sludge is settled-out solids and contaminants — exactly what you removed from the water on purpose. Putting it back into the Oil/Water Separation Cell defeats the treatment — it's the start of the process, so the same contaminants would just flow through the whole system again. The confirmed process for sludge from this system is: filter press dewater it → solid filter cake goes out as **regulated waste through a licensed contractor** (Section 6.11). If you're filling a tote directly instead of running it through the filter press, **that tote's contents should follow the same regulated-waste route — confirm this directly with Brendan before disposing of a full tote any other way.**

## How to operate an AODD pump

### [ACTION]

1. Connect the suction hose to the inlet and the discharge hose to the outlet. Confirm all clamps are tight.
2. Open the inlet valve (source) and outlet valve (destination).
3. Connect the air supply hose to the air inlet port.
4. Open the air supply valve — the pump will immediately start cycling with a characteristic "chunk-chunk-chunk" sound and air exhaust puffs.
5. Increase air pressure to achieve the desired flow rate (more pressure = faster pumping, up to about 100 psi max).
6. To stop: close the air supply valve.
7. Disconnect in reverse order.

## Common problems

| Symptom                                   | Cause                                                        | Fix                                                                                 |
|-------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Pump "stalls" (freezes mid-stroke)        | Air pressure too low, or pilot valve worn/stuck              | Increase air pressure; if stalled, quickly cycle the air pressure off/on to restart |
| Very slow, laboured pumping               | Inlet blocked (suction hose kinked or solids blocking inlet) | Check and clear suction line                                                        |
| Air blowing out of discharge (not liquid) | Diaphragm failure                                            | Take offline; replace diaphragm — have a spare on hand                              |
| Chattering/fast cycling but no flow       | Air-locked (air in the inlet line)                           | Prime by tilting the pump or submerging the inlet                                   |

## 6.13 pH Meter and Lab Setup

### Equipment visible on site:

- Portable pH meter (grey hard-sided case) — confirmed via photo IMG\_2798 to be an **Apera Instruments** digital pH pen/tester (pen-style probe, digital display showing pH and temperature)
- pH buffer solutions, confirmed via photo IMG\_2797: pH 4.0 (pink liquid), pH 7.0 (green liquid), pH 10.0 (blue liquid) — corrected from earlier note that guessed 7.0 was yellow
- Sample jars
- PVC primer (IPEX 5, and Weld-On/IPS purple primer per IMG\_2797) and cement (Oatey) for plumbing repairs
- Disposable gloves (Watson brand, 50-count boxes) stocked in the same area

**!** **RULE — RINSE WITH CLEAN WATER BEFORE EVERY SINGLE PH TEST, NO EXCEPTIONS:** Always rinse the probe tip with clean/distilled water immediately before dipping it into a buffer OR a sample — every time, between every reading. Skipping this carries residue from the last buffer/sample into the next one and throws off the reading.

### pH meter calibration — ONCE PER DAY/SHIFT, not before every single test

**i** **CONFIRMED:** Calibrate the pen **once at the start of each day/shift**. After that, test as many samples as needed that shift without recalibrating — just rinse the probe between every test. Recalibrate again if: it's a new shift, the pen has sat unused for a while, or a buffer check shows it's drifted off (e.g., no longer reading 7.0 in the 7.0 buffer).

### [ACTION]

1. Remove the pH probe from its storage solution. Rinse with distilled or clean water.

2. Blot dry (do not rub).
3. Place in pH 7.0 buffer (green liquid). Press CAL. Let it stabilize. Confirm it reads  $7.0 \pm 0.05$ .
4. **Rinse electrode with clean water**, blot dry.
5. Place in pH 4.0 buffer (pink liquid, for acidic water testing). Press CAL. Let stabilize. Confirm reads  $4.0 \pm 0.05$ .
6. **Rinse with clean water** and blot dry again. Meter is ready to use.
7. **Rinse with clean water** before dipping into your actual water sample (and again before any subsequent sample).
8. After use: rinse electrode, store in pH electrode storage solution (not distilled water — distilled water damages the electrode).

## 6.14 Safety Equipment in the Treatment Room

Identified from video (IMG\_2612):

| Item                                                    | Location                          | Purpose                                                                                                             |
|---------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Fire extinguisher</b> (red, ABC type)                | Wall near sand filter (SRS-SF-11) | General fire suppression                                                                                            |
| <b>AIR HORN</b> (blue compressed-air horn)              | Wall beside fire extinguisher     | Emergency alarm / evacuation signal                                                                                 |
| <b>Cylinder hand truck</b> (black frame, yellow wheels) | Treatment room floor              | Moving CO <sub>2</sub> and other gas cylinders safely — always use this; never drag or roll cylinders on their base |

[CHECK] Each shift:

- [ ] Fire extinguisher pressure gauge in the green zone
- [ ] Air horn is accessible and unobstructed
- [ ] Yellow inspection tag on fire extinguisher is current (within 1 year)

**i WHY:** In the event of a chemical fire (Cal-Hypo or H<sub>2</sub>O<sub>2</sub> near ignition sources) or an emergency requiring evacuation, you need to know exactly where these tools are. The air horn is your way of alerting everyone on site when verbal communication won't reach.

**Using the air horn for evacuation:**

- 3 short blasts = evacuate the building immediately

### A second safety station — workshop/maintenance area (confirmed via video IMG\_2774)

This is a separate area from the main treatment room — looks like a workshop or maintenance shed with a tool pegboard, work bench, and gas cylinders (appear to be welding gas, not CO<sub>2</sub> — confirm). Confirmed here:

| Item                                          | Location                                      | Purpose                                                                           |
|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Eyewash station</b> (yellow, wall-mounted) | Workshop area, near the exit door             | Eye flush for chemical/debris exposure — marked with an "EMERGENCY EYE WASH" sign |
| <b>Spill kit</b>                              | Workshop area, wall-mounted sign points to it | Containing/cleaning up chemical or fuel spills                                    |
| <b>Fire extinguisher</b>                      | Workshop area                                 | General fire suppression                                                          |

**i WHY THIS MATTERS:** The site has **at least two separate safety stations** in two different areas (treatment room, and this workshop area) — don't assume the nearest eyewash/extinguisher to where you're standing is the only one on site. Learn the route to BOTH.

(confirm with supervisor: is this workshop physically attached to/near the main treatment building, or a separate structure? What other eyewash stations exist elsewhere on site – outdoor pump areas, chemical storage?)

## 6.15 CO<sub>2</sub> Cylinder Handling

**Cylinder hand truck:** Always use the dedicated hand truck to move the CO<sub>2</sub> cylinder. CO<sub>2</sub> cylinders are heavy (~50 kg full) and top-heavy.

[ACTION] When moving the CO<sub>2</sub> cylinder:

1. Close the cylinder valve completely.
2. Attach the protective cap to the valve.
3. Tilt the cylinder onto the hand truck. Secure the chain or strap around the cylinder.
4. Tilt the hand truck back and wheel to the new location.
5. Secure the cylinder to the wall bracket or fixed support before removing from the hand truck.

**⚠ WARNING:** **Never move a CO<sub>2</sub> cylinder without the hand truck.** A falling cylinder can rupture the valve — the cylinder becomes a pressurized projectile. Also: CO<sub>2</sub> cylinders must always be chained or strapped to a fixed support when not on the hand truck.

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## 6.16 Pneumatic Tote Mixer (Coagulant/Flocculant/Lime Stock Mixing)

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**Appearance — confirmed via photo (IMG\_2820, IMG\_2821):** A mixer clamped to the top of an IBC tote (the 1000L/500L tote used to mix PACl and flocculant stock solutions, Sections 5.3/5.4). Mounted on a wood frame above the tote: a gray air-motor housing, a **red-handled valve** at the air motor's inlet, and a long angled shaft running down through the tote lid to a **propeller/impeller blade** submerged in the liquid. A black air supply hose with a quick-connect fitting feeds the motor from below.

### How it works

This is an **air-driven (pneumatic) mixer**, not electric — it runs off compressed air, not a power cord. The red handle is the air valve: opening it lets compressed air spin the motor, which turns the shaft and propeller inside the tote.

**[ACTION]** To start mixing:

1. Confirm the air hose is connected to the compressed air supply via the quick-connect fitting.
2. Open the **red-handled valve** (turn it — typically counter-clockwise opens a valve like this, but confirm the actual direction on this unit before relying on it) to let air into the motor. The propeller will start turning.
3. Open it gradually rather than all at once, especially when mixing flocculant — Section 5.4 calls for **gentle** mixing while the polymer is added; full-blast air can tear/clump the polymer instead of dissolving it.
4. To stop, close the red valve — this cuts air supply and the propeller stops.

(still needs confirming: which way the red handle actually turns to open vs. close on this specific valve; whether there's a separate speed/flow-rate adjustment beyond fully open/closed.)

**⚠ NOTE:** A second valve with a cross-shaped handle is visible nearby in the photo (lower part of the frame) — this looks like it belongs to the tote's drain/outlet plumbing, not the mixer itself. Don't confuse the two before opening anything.

**⚠ CRITICAL — CONFIRMED BY OPERATOR:** **For the blue drum where the lime is, the blade must be running while you're pumping it out — not just while you're initially mixing the batch.** If the blade isn't turning while lime is being pumped, the lime settles out and the dose won't work well (it's a thick slurry that drops out of suspension fast — see Section 5.6a). Same logic applies wherever this mixer is used: open the air valve and keep the propeller turning for as long as you're drawing liquid out, not just during mixing.

 **Photo-illustrated quick steps for mixing lime, PACl, and flocculant (and running this mixer):** [CHEMICAL\\_MIXING\\_GUIDE.pdf](#)

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## 6.17 Avoiding Trapped Pressure — Pump, Valve, and Hose Shutdown Sequencing

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**i WHY THIS SECTION EXISTS — CONFIRMED BY OPERATOR (Oliver):** Nothing on this site is a sealed, municipal-style pressurized pipe network. Every run is a **submersible pump pushing water through an open-ended hose or pipe** into the next tank. That means the risk isn't "the pipe is always under pressure" — it's narrower and more specific than that:

1. **Starting a pump against a closed valve** — the pump keeps pushing with nowhere for the water to go, and pressure builds and sits on the pipe.
2. **Closing a valve while the pump is still running** — same problem, caused in reverse.
3. **Disconnecting a hose or fitting right after shutoff, before residual water has had a chance to drain back out** — there's a moment after the pump stops where the line is still full and may still have some head pressure on it, even though the pump itself is off.

The fix for all three is the same idea: **pump and valve operations always happen in a deliberate order, never at the same time, and you give the line a moment to drain before you break a connection.** The two runs below are the ones you asked about — work through both the same way every time until it's automatic.

### Run 1 — Indoor PVC Intake Pipe (Weir Tank → Ozone → Indoor Settling Bins)

This is the pipe described in Section 6.1a and Section 6.7: a 2" submersible pump sits mid-depth in the outdoor weir tank and pushes water through the white PVC intake pipe into the building, with ozone injected into it along the way, landing in the indoor settling bins ("down" pipe).

**Starting up (already confirmed, Section 6.1a/6.7 — repeated here for the full sequence):**

1. **[ACTION]** Open the **intake pipe valve** at the building wall first.
2. **[ACTION]** Turn the **ozone generator** on.
3. **[ACTION]** *Then* drop in / power on the **weir tank submersible pump**.

**❗ WHY THIS ORDER:** Per Section 6.1a, starting the pump against a closed valve lets pressure build on the pipe with nowhere to go. Per Section 6.7, starting the pump before ozone means untreated water gets a head start past the point ozone is supposed to treat it. Doing both in this order avoids both problems at once.

**Shutting down once the indoor settling bins are full (the part you asked about):**

1. [ACTION] Turn the **weir tank submersible pump off first**. Do not close the intake pipe valve while the pump is still running — that's the same "closed valve, pump still pushing" problem as starting it backwards, just at the other end of the cycle.
2. [ACTION] Turn the **ozone generator off**.
3. [ACTION] **Wait a moment before touching the valve**. The intake pipe is open at the indoor end (it discharges into the settling bins) — once the pump stops pushing, whatever water is still in the pipe drains forward into the bins by gravity. Give it that moment; you'll typically hear/see the flow at the discharge end taper off to nothing.
4. [ACTION] *Then* close the **intake pipe valve** at the building wall. Closing it after the pump is off and the line has drained means you're sealing an already-depressurized pipe, not trapping pressure inside it.
5. [ACTION] If you ever need to disconnect the pump's hose from the intake pipe fitting itself (not just close the wall valve), do it only after steps 1–3 above. Crack the fitting open slightly first and pause — if any water sprays or sputters out, let it finish before fully separating it. That confirms the line is actually at atmospheric pressure, not just "probably fine."

**⚠ WARNING:** Never close the intake valve and walk away while the pump is still in the tank powered on. If you're stepping away mid-transfer, pump off first — every time, no exceptions.

## Run 2 — Outdoor Hose (Frac Tank Submersible Pump → C Can)

This is the hose described in Section 6.1a: a submersible pump dropped into the frac tank, hose run to the C Can, connecting at the check valve fitting at the base of the C Can (the confirmed *inflow* point — Section 6.1b).

**Starting up:**

1. [ACTION] Confirm the hose is connected securely at both ends (pump end in the frac tank, check valve end at the C Can) before powering anything on.
2. [ACTION] Drop in / power on the **frac tank submersible pump**. The check valve at the C Can only needs to be physically present and connected — it isn't a manual valve you open by hand, it just lets water in one direction and stops it from running back out.

**Shutting down once the transfer is done (or you're moving the pump to a new tank):**

1. [ACTION] Turn the **frac tank submersible pump off first** — same rule as Run 1, never disconnect a hose while the pump pushing it is still running.
2. [ACTION] **Disconnect at the pump end first, not the C Can end**. The check valve at the C Can is specifically designed to hold water back on that side — if you crack the connection there first, you're opening against whatever residual head the check valve was holding. The pump end has no check valve holding it shut, so breaking the connection there lets the hose gravity-drain back toward the frac tank with nothing trapped on either side.
3. [ACTION] Lift or lower the disconnected hose end so it drains fully before you coil it or set it aside — a hose left lying in a low loop can hold a surprising amount of standing water, which isn't a pressure risk but is worth draining so it's not sloshing around your feet or whoever moves it next.
4. [ACTION] Once you're confident the hose is empty, disconnect the check valve end at the C Can.

**❗ WHY DISCONNECT THE PUMP END FIRST:** The general rule for any hose with a check valve on only one end is: **break the connection on the end without the check valve first**. That end was never holding pressure back on its own — it's the safe side to open. The check valve end is the one built to resist backflow, so it's the side most likely to still be holding something back the moment after shutoff.

(still needed: exact physical location/identity of the intake pipe valve at the building wall – Section 6.1a flags this as unconfirmed. Confirm with Brennan so this section can reference it by name/location instead of just "the wall valve.")

## Section 7 — Preventive Maintenance

Preventive maintenance (PM) is doing the work before the equipment breaks. A pump bearing replaced on a PM schedule costs \$50 in parts and 30 minutes. A pump that fails at 2 AM during a storm costs thousands and puts the plant at risk. Do your PMs.

### 7.1 The PM Philosophy

**Reactive maintenance** = fix it when it breaks. High cost. Unpredictable downtime. Emergency repairs during bad weather.

**Preventive maintenance** = fix it before it breaks. Predictable. Cheaper. Keeps you in control.

**Predictive maintenance** = monitor equipment condition and intervene when data says it's degrading (vibration analysis, oil sampling, thermal imaging). This is the gold standard but requires specialized tools.

For most water treatment plants, a solid preventive maintenance program is the goal.

### 7.2 PM Frequencies and Tasks

*The following table represents a comprehensive generic PM schedule. Once you provide your equipment manuals, site-specific frequencies, and existing maintenance records, this will be refined for each piece of equipment.*

#### Daily (Every Shift)

| Task                               | Equipment                                | What to Check                                                          | Action if Problem                               |
|------------------------------------|------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------|
| Visual inspection                  | All pumps                                | Unusual noise, vibration, heat, leaks                                  | Log; escalate if abnormal                       |
| Seal/packing check                 | All pumps                                | Leakage amount (a small drip from packing is normal — a stream is not) | Adjust packing gland; schedule seal replacement |
| Lubrication check                  | Pump bearings                            | Temperature by touch; bearing housing level indicator if present       | If hot, stop pump, check oil level              |
| Chemical level check               | All dosing tanks                         | Levels recorded; reorder trigger                                       | Order if at reorder level                       |
| Instrument reading vs. grab sample | Turbidimeters, Cl <sub>2</sub> analyzers | Within $\pm 10\%$ of manual test                                       | Calibrate instrument                            |
| Clean flow-through cells           | Online turbidimeters                     | Algae/fouling                                                          | Clean with brush and DI water                   |

#### Weekly

| Task                            | Equipment                    | Procedure                                                                            |
|---------------------------------|------------------------------|--------------------------------------------------------------------------------------|
| Lubricate bearings              | Pumps, mixers                | Grease per manufacturer spec. Do not over-grease — it damages seals                  |
| Calibrate online turbidimeters  | All                          | Use StablCal or Formazin standard; document                                          |
| Calibrate chlorine analyzers    | All                          | Use DPD manual test on same stream; adjust if needed                                 |
| Calibrate pH meters/sensors     | All                          | 2-point calibration with pH 7 and pH 4 (or 10) buffers                               |
| Inspect dosing pump tubing      | All peristaltic pumps        | Look for cracks, kinks, soft spots — replace before failure                          |
| Check chemical injection points | All                          | Clear, no crystal buildup, injecting visibly                                         |
| Check all valve packing         | Manual valves used regularly | Adjust packing gland if leaking; do not over-tighten (valve will be hard to operate) |
| Test high-turbidity alarm       | Online turbidimeter          | Verify alarm triggers at setpoint; test notification path                            |
| Test low-chlorine alarm         | Chlorine analyzer            | Verify alarm triggers at setpoint                                                    |



| Task                         | Equipment   | Procedure                                                                      |
|------------------------------|-------------|--------------------------------------------------------------------------------|
| Inspect filter media surface | All filters | Even surface, no "rat holes" or cracking (these bypass water around the media) |

## Monthly

| Task                               | Equipment            | Procedure                                                                          |
|------------------------------------|----------------------|------------------------------------------------------------------------------------|
| Motor insulation check             | All motors           | Use megohmmeter; record reading and compare to baseline                            |
| Check coupling alignment           | Motor/pump couplings | Misalignment causes vibration and premature bearing failure                        |
| Inspect all electrical connections | Control panels, MCC  | Look for discolouration, corrosion, loose terminals — DO NOT touch live circuits   |
| Test all alarms end-to-end         | SCADA/panel          | Simulate fault conditions; verify alarm reaches intended recipients                |
| Inspect chemical storage area      | All chemical rooms   | Spills, container integrity, SDS current, ventilation working                      |
| Sludge blanket measurement         | Sedimentation basins | Record depth; compare to previous month; adjust drain schedule if rising           |
| Filter media inspection            | All filters          | Check for media loss, compaction, biological growth                                |
| Backwash pump flow test            | Backwash pump        | Confirm flow and pressure meet design spec                                         |
| Clear well inspection              | Clear well           | Check for cracks, algae growth, animal intrusion, disinfection residual uniformity |

## Quarterly (Every 3 Months)

| Task                               | Equipment                                     | Procedure                                                           |
|------------------------------------|-----------------------------------------------|---------------------------------------------------------------------|
| Change oil in gearboxes            | Mixer gearboxes, multi-stage pump gearboxes   | Use manufacturer-specified oil grade                                |
| Replace dosing pump diaphragms     | Diaphragm dosing pumps                        | Even if not failed — prevents chemical spills                       |
| Inspect and clean UV lamps/sleeves | UV disinfection unit                          | Wipe sleeves; measure lamp intensity; replace if below minimum      |
| Calibrate flow meters              | All                                           | Verify against a portable meter or timed fill                       |
| Full valve exercise                | All normally-closed or rarely-operated valves | Open/close fully to prevent them seizing; lubricate stem            |
| Inspect and test generator         | Backup generator                              | Load test: run on full generator power for 30 min; check fuel level |
| Inspect HVAC in chemical rooms     | Ventilation fans                              | Confirm adequate air changes; replace filters                       |

## Annually (Every 12 Months or Per Permit)

| Task                                                  | Equipment                                           | Procedure                                                                                   |
|-------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------|
| Pump curve verification                               | All major pumps                                     | Measure flow at multiple heads; compare to original curve — large deviation = worn impeller |
| Motor winding resistance                              | All motors                                          | Record and trend; significant drop = insulation failure approaching                         |
| Pressure vessel inspection                            | Hydropneumatic tanks, any pressurized vessels       | Check for internal corrosion; pressure test                                                 |
| Comprehensive electrical inspection                   | All panels                                          | Qualified electrician thermographic scan; check arc flash labeling                          |
| Calibrate all instruments against certified standards | All                                                 | Keep calibration certificates on file — regulators may ask for them                         |
| Review and update all emergency procedures            | All                                                 | Are contact numbers current? Is equipment still as described?                               |
| Confined space entry permit review                    | All confined spaces (clear well, basins, wet wells) | Confirm entry procedures, equipment, and trained staff are current                          |
| Chemical dosing pump overhaul                         | All                                                 | Replace all wetted parts on a schedule; cost far less than emergency repair                 |

### 7.3 LOTO (Lockout/Tagout) — The Non-Negotiable PM Step

Before ANY maintenance on equipment that could move, energize, or release stored energy:

- 1. **Identify** all energy sources (electrical, pneumatic, hydraulic, stored pressure, gravity).
- 2. **Notify** all affected personnel.
- 3. **Isolate** each energy source (turn off the breaker, close the isolation valve, etc.).
- 4. **Lock** the isolation device with YOUR personal lock. Each person working on the equipment puts on their own lock. No shared locks.
- 5. **Tag** the lockout point with your name, date, and reason.
- 6. **Verify** zero energy state: press the start button to confirm nothing happens; check pressure gauges for zero.

Only the person who placed the lock can remove it.

 **WARNING:** This is not bureaucratic paperwork. Workers are killed every year by equipment that was assumed to be de-energized. There are no exceptions to LOTO.

### 7.4 Maintenance Log

Every PM task should be logged:

DATE: 2026-06-04    TASK: Weekly bearing lubrication – Raw water pumps 1, 2, 3  
TECHNICIAN: [Name]  
FINDINGS: All bearings normal temperature. RW Pump #2 – slight increase in vibration vs. last week. Greased per spec. Scheduled vibration check for next week.  
PARTS USED: None (lubrication only)  
NEXT ACTION: Monitor Pump #2 vibration.

### 7.5 Spare Parts Inventory

Keep a minimum stock of:

| Item                                  | Why Critical                              | Recommended Stock                |
|---------------------------------------|-------------------------------------------|----------------------------------|
| Dosing pump tubing (peristaltic)      | Fails without warning; plant stops dosing | 2 sets per pump type             |
| Dosing pump diaphragms                | Same                                      | 2 per pump type                  |
| Pump mechanical seals                 | Failure causes flooding                   | 1 per pump type                  |
| UV lamp(s)                            | Must be replaced on schedule              | 1–2 per UV unit                  |
| UV quartz sleeve(s)                   | Can crack                                 | 1 per UV unit                    |
| DPD reagents (tablets or powder)      | Used daily                                | 2–4 week supply                  |
| Calibration standards (turbidity, pH) | Required for daily calibration            | Maintain per shelf life          |
| Chemical injection check valves       | Fail from chemical exposure               | 2–3 per injection point type     |
| Motor contactors                      | Electrical failure                        | 1 per motor type                 |
| Fuses and breakers                    | Electrical protection                     | As per electrical panel schedule |

Last updated: 2026-06-04 | Source: Initial framework — update with equipment-specific intervals from manufacturer manuals

# Section 8 — Troubleshooting

Something looks wrong. Before you panic, use this guide. Troubleshooting is structured detective work: observe → form a hypothesis → test it → fix it → verify the fix → record everything.

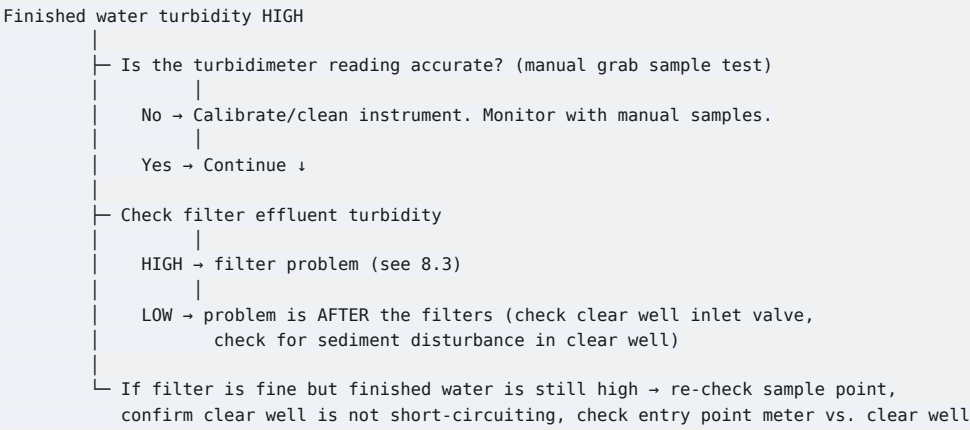
## 8.1 The Troubleshooting Mindset

1. **Stay calm.** Panicking makes it worse.
2. **Confirm it's real.** Is the instrument malfunctioning? Take a manual sample and re-test.
3. **Protect the public first.** If finished water quality is in question, stop distribution and call your supervisor before anything else.
4. **Work upstream.** Most finished water problems have causes upstream in the process. Start at the raw water and work forward.
5. **Change one thing at a time.** If you adjust multiple variables simultaneously, you won't know what fixed it.
6. **Document everything.** Every observation, every action, every result.

## 8.2 Finished Water Turbidity High

Turbidity is your primary real-time indicator of treatment performance. High turbidity at the tap means something upstream is failing.

### Rapid diagnosis flowchart



## 8.3 Filter Turbidity High (Filter Breakthrough)

| Possible Cause                                         | How to Confirm                                                                | Fix                                                                                            |
|--------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Filter is due for backwash                             | Check headloss and run time                                                   | Initiate backwash                                                                              |
| Poor coagulation upstream — floc passing through media | Check settled water turbidity (if low, coagulation is fine; if high, see 8.4) | Fix coagulation; consider backwash to reset filter                                             |
| Media compaction / "mudball" formation                 | Visual inspection of media surface during backwash                            | Increase backwash intensity; air scour if available; if persistent, media may need replacement |
| Filter ran too long and exhausted adsorption capacity  | Check run time vs. normal                                                     | Backwash immediately; re-evaluate max run time                                                 |
| Damage to media or underdrains                         | Inspect during backwash — turbidity spike that doesn't clear                  | Take filter offline; inspect; repair                                                           |

| Possible Cause                 | How to Confirm                                                    | Fix                                                  |
|--------------------------------|-------------------------------------------------------------------|------------------------------------------------------|
| Filter ripening after backwash | Turbidity elevated in first 15–30 min after backwash, then clears | Normal — route to waste until turbidity meets target |

## 8.4 Settled Water Turbidity High (Clarifier/Sedimentation Problem)

| Possible Cause                                                    | How to Confirm                                                                         | Fix                                                                                 |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Under-dosing of coagulant                                         | Settled turbidity rising; pinpoint or no floc visible                                  | Increase coagulant dose; run jar test                                               |
| Over-dosing of coagulant (charge reversal)                        | Also causes high turbidity but floc may be very small; check dose vs. jar test optimum | Decrease dose; re-run jar test                                                      |
| pH outside optimal range for coagulant                            | Check post-coagulation pH vs. optimal range for your chemical                          | Adjust pH first, then re-optimize dose                                              |
| Raw water alkalinity too low (coagulant consuming all alkalinity) | Test alkalinity after coagulant addition                                               | Add lime or sodium bicarbonate                                                      |
| Sludge blanket too deep — solids carrying over weir               | Measure sludge blanket depth                                                           | Increase sludge drain frequency                                                     |
| Sudden increase in raw water turbidity (storm event)              | Check raw turbidity vs. recent trend                                                   | Increase coagulant dose; run jar test immediately                                   |
| Mixing problem (short-circuiting in flocculation)                 | Observe flow patterns; dye test if needed                                              | Check weirs/baffles; ensure mixers are running                                      |
| Temperature change (cold water — poor coagulation)                | Check raw water temperature vs. recent weeks                                           | Consider switching from alum to PACl (better cold-water performance); increase dose |

## 8.5 Chlorine Residual Low in Finished Water

| Possible Cause                                                         | How to Confirm                                     | Fix                                                                                   |
|------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------|
| Chlorine dose too low                                                  | Check dosing pump output; check tank level         | Increase dose; calibrate pump                                                         |
| High chlorine demand in raw water (organics, iron, manganese, ammonia) | Test raw water; check NOM levels                   | Increase dose to meet demand; investigate source                                      |
| Chlorine dosing pump failed                                            | Visual inspection; check tank level hasn't changed | Switch to backup pump; repair primary                                                 |
| High turbidity protecting demand (chlorine "hidden" by particles)      | Check turbidity — if high, fix turbidity first     | Fix coagulation/filtration; once turbidity resolved, chlorine residual often recovers |
| Warm water increases decay rate                                        | Check temperature                                  | Increase dose; adjust for seasonal conditions                                         |
| Long hydraulic detention time (low flow through contact chamber)       | Check flow rate                                    | Ensure minimum flow maintained; do not reduce flow too drastically                    |
| Chlorine solution degraded (old/warm NaOCl)                            | Check tank: dark brown/yellow = degraded bleach    | Order fresh chemical; improve storage (cool, dark, sealed)                            |

## 8.6 pH Out of Range

| Symptom                              | Possible Cause                                                 | Fix                                                                      |
|--------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------|
| pH too low (acidic) — finished water | Coagulant consuming alkalinity; high CO <sub>2</sub> in source | Add lime or caustic; add alkalinity; reduce CO <sub>2</sub> via aeration |
| pH too high — finished water         | Over-addition of lime or caustic                               | Reduce dose; check pH probe calibration first                            |
| pH fluctuating erratically           | Dosing pump running inconsistently; instrument error           | Check pump for air in lines; calibrate probe                             |

| Symptom                              | Possible Cause                                             | Fix                                                                               |
|--------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------|
| pH drop at filter effluent vs. inlet | CO <sub>2</sub> release from biological activity in filter | Normal if small; large drops suggest biological fouling — review filter operation |

## 8.7 Pump Problems

### Pump Not Starting

| Possible Cause              | Check                                                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Tripped breaker or overload | Check MCC / panel; reset if tripped. If it trips again immediately, do NOT keep resetting — investigate the cause |
| E-stop engaged              | Check E-stop buttons — are any pressed in?                                                                        |
| Control signal missing      | Check the PLC/SCADA output; check local/remote switch position                                                    |
| Pump seized                 | Try to rotate shaft manually (only when confirmed de-energized and LOTO applied)                                  |

### Pump Running But No/Low Flow

| Possible Cause                                | Check                                                                                                               |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Discharge valve closed                        | Walk to the pump and check all valves in the discharge line                                                         |
| Air lock (lost prime)                         | Check suction line for air; open vent valve                                                                         |
| Impeller clogged                              | Take pump offline (LOTO); inspect and clear obstruction                                                             |
| Pump running backwards (wrong phase sequence) | Compare rotation direction to arrow on casing; if reversed, swap two electrical phases (qualified electrician only) |

### Pump Vibration / Noise

| Symptom               | Cause                       | Action                                                             |
|-----------------------|-----------------------------|--------------------------------------------------------------------|
| Grinding/gravel sound | Cavitation                  | Check suction pressure, reduce discharge restriction, reduce speed |
| Rhythmic thumping     | Misalignment or imbalance   | Check alignment; inspect impeller for damage                       |
| Bearing whine         | Bearing failure approaching | Schedule bearing replacement; increase monitoring frequency        |
| Loud knocking         | Water hammer                | Add slow-closing valve; check check valve                          |

## 8.8 Chemical Dosing Problems

| Problem                                       | Likely Cause                                                         | Fix                                                                 |
|-----------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|
| Dose dropping despite no pump changes         | Tank is getting low                                                  | Refill; switch to backup tank                                       |
| Dosing pump running but no chemical delivered | Air in tubing; cracked tubing; failed check valve on injection point | Prime pump; inspect and replace tubing; check injection check valve |
| Dose going up but no change in treatment      | Chemical has degraded (NaOCl especially)                             | Test chemical concentration; replace if degraded                    |
| Crystals forming around injection point       | Chemical concentration too high; slow flush after injection          | Flush injection line; dilute if needed; increase flush flow         |
| Peristaltic tubing failed (chemical spraying) | Tubing age / UV damage / incompatible material                       | Immediate LOTO; replace tubing; clean spill                         |

## 8.9 Alarm Response Quick Reference

| Alarm                 | Immediate Check                                | Escalate if                                                     |
|-----------------------|------------------------------------------------|-----------------------------------------------------------------|
| High filter turbidity | Manual grab sample; check upstream coagulation | Turbidity > permit limit; cannot reduce within 15 min           |
| Low chlorine residual | Manual test; check dosing pump                 | Residual below regulatory minimum; cannot restore within 15 min |

| Alarm                     | Immediate Check                               | Escalate if                                                   |
|---------------------------|-----------------------------------------------|---------------------------------------------------------------|
| High raw water turbidity  | Check source conditions; check coagulant dose | Turbidity > 10× normal or cannot achieve settled water target |
| Pump fail / trip          | Check for E-stop, tripped breaker, overload   | After 2 reset attempts; any sign of mechanical failure        |
| High clear well level     | Check all flows; check distribution pump      | Risk of overflow                                              |
| Low clear well level      | Check all flows; check raw water pump output  | < 20% of capacity (risk of running out of water)              |
| Power failure (UPS alarm) | Confirm generator started                     | Generator does not start automatically within 30 sec          |

*Last updated: 2026-06-04 | Source: Initial framework*

# Section 9 — Safety & PPE

Water treatment plants have real hazards. People are seriously injured and killed in this industry every year — almost always because a shortcut was taken or a procedure was skipped. Safety procedures are not bureaucratic obstacles. They are the distillation of hard lessons learned by people who got hurt.

## 9.1 The Non-Negotiable Safety Rules

- 1. **Never bypass LOTO (Lockout/Tagout).** Energized equipment kills.
- 2. **Never enter a confined space alone or without a permit.**
- 3. **Always wear the required PPE.** If you don't know what's required, don't start the task.
- 4. **If you're unsure, stop and ask.** The question you feel embarrassed to ask is usually the most important one.
- 5. **Report every near-miss.** Near-misses are accidents that didn't quite happen — they predict future accidents.

## 9.2 Site Hazard Overview

| Hazard Category         | Where You'll Find It                                                                     | Consequence                         |
|-------------------------|------------------------------------------------------------------------------------------|-------------------------------------|
| Chemical burns / splash | Chemical storage rooms, dosing areas, any injection point                                | Skin burns, eye damage, lung damage |
| Toxic gas               | Chlorine room, confined spaces (H <sub>2</sub> S, CO <sub>2</sub> , low O <sub>2</sub> ) | Unconsciousness, death              |
| Drowning                | Open basins, wet wells, clear well                                                       | Death                               |
| Electrocution           | MCC rooms, panels, pump controls, exposed wiring                                         | Death, severe burns                 |
| Crushing / entanglement | Pumps, mixers, scrapers, conveyors with moving parts                                     | Loss of limb, death                 |
| Slips, trips, falls     | Wet floors near pumps and basins, ladders, grates                                        | Fractures, concussions              |
| Manual handling         | Chemical drums, bags of lime, equipment                                                  | Back injuries, hernias              |

## 9.3 Personal Protective Equipment (PPE) Guide

### When to wear what

| Task                                                                              | Required PPE                                                                                                              |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| General plant rounds                                                              | Safety glasses, hard-toed boots, hard hat in construction zones                                                           |
| Working near open water (basins)                                                  | Life jacket if working within 1m of open water; buddy system                                                              |
| Handling liquid chemicals (alum, PACl, NaOCl, fluoride)                           | Chemical splash goggles (not just safety glasses), chemical-resistant gloves (nitrile minimum; neoprene for acids), apron |
| Handling concentrated acids (fluorosilicic acid, H <sub>2</sub> SO <sub>4</sub> ) | Face shield + chemical splash goggles, acid-resistant gloves (thick), acid apron, boots with no metal                     |
| Handling dry lime or powdered chemicals                                           | N95 dust mask, goggles, gloves                                                                                            |
| Electrical work                                                                   | Electrical insulating gloves (appropriate voltage rating), arc flash PPE per electrical risk assessment                   |
| Entering confined space                                                           | Follow full confined space entry procedure (Section 9.6)                                                                  |
| Chlorine room / gas chlorine systems                                              | SCBA (Self-Contained Breathing Apparatus), chemical suit — trained responders only                                        |
| Noise > 85 dB (large pump rooms)                                                  | Hearing protection (earplugs or earmuffs)                                                                                 |
| Working at height > 1.2 m (ladders, catwalks)                                     | Fall arrest harness attached to anchor point                                                                              |

### How to put on (don) and remove (doff) chemical PPE

**Donning (putting on — clean hands first):**

1. Apron
2. Gloves (check for holes by inflating)
3. Goggles / face shield

**Doffing (removing — assume contaminated):**

1. Gloves (peel off inside-out so contaminated surface faces in)
2. Goggles (handle by the strap)
3. Apron (roll away from body)
4. Wash hands immediately

---

## 9.4 WHMIS / GHS — Understanding Hazard Labels

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Every chemical container in a water plant must have a GHS-compliant label. Know these symbols:

| Symbol                                | Meaning                                       |
|---------------------------------------|-----------------------------------------------|
| Flame                                 | Flammable                                     |
| Skull and crossbones                  | Acutely toxic (can kill)                      |
| Exclamation mark                      | Irritant, less severe toxicity                |
| Corrosion                             | Destroys skin and metal                       |
| Gas cylinder                          | Compressed gas                                |
| Exploding bomb                        | Explosive                                     |
| Flame over circle                     | Oxidizer (increases fire risk)                |
| Health hazard (person with starburst) | Carcinogen, reproductive hazard, organ damage |
| Environment                           | Environmental hazard                          |

**Safety Data Sheets (SDS):**

Every chemical must have an SDS on file and accessible at all times. The SDS tells you:

- What the chemical is and its hazards
- First aid measures
- Firefighting measures
- Spill response procedures
- Handling and storage requirements
- Exposure limits

[ACTION] Locate the SDS binder or station for your site. Know exactly where it is — you may need it in an emergency.

---

## 9.5 Lockout / Tagout (LOTO) Procedure

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Also described in Section 7.3 (Maintenance). Repeated here because it's critical enough to deserve two mentions.

**The 6 steps:**

| Step            | What you do                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------|
| 1. Prepare      | Identify all energy sources (electrical, pneumatic, hydraulic, gravity, stored pressure, chemical) |
| 2. Notify       | Tell all affected workers that equipment is being locked out                                       |
| 3. Shut down    | Turn off the equipment through normal stopping procedure                                           |
| 4. Isolate      | Open all disconnects, close all isolation valves, bleed stored pressure                            |
| 5. Lock and tag | Place YOUR lock on each isolation point. Attach a tag with your name and the date                  |
| 6. Verify       | Try to start the equipment (it should not start). Check all pressure gauges read zero.             |



**When LOTO is removed:**

1. Confirm all workers are clear of the equipment.
2. Remove all tools, materials, and covers.
3. Each person removes ONLY their own lock.
4. Notify all affected workers.
5. Start up equipment and verify normal operation.

## 9.6 Confined Space Entry

Confined spaces at a water plant include: clear wells, sedimentation basins, filter underfloor access, wet wells, chemical storage vaults, and any other enclosed space with limited entry/exit not designed for continuous occupancy.

**Hazards specific to confined spaces:**

- Oxygen deficiency (O<sub>2</sub> displaced by CO<sub>2</sub>, N<sub>2</sub>, or methane)
- Toxic gases (H<sub>2</sub>S from biological decomposition, chlorine)
- Flooding
- Engulfment (sludge or water)

**Permit-required confined space entry procedure:**

 **WARNING:** Do NOT enter a confined space without a written entry permit and trained attendant. This is non-negotiable.

1. **Hazard assessment:** Identify all hazards before opening the space.
2. **Atmospheric testing:** Test for O<sub>2</sub> (must be 19.5–23.5%), flammable gases (< 10% LEL), and toxic gases (H<sub>2</sub>S < 1 ppm, CO < 25 ppm for entry without supplied air) using a calibrated gas monitor. Test before entry AND continuously during entry.
3. **Ventilation:** Force-ventilate the space for at least 5 minutes before entry; continue ventilating during entry.
4. **Entry permit:** Complete and sign the entry permit. It must list: hazards, control measures, rescue plan, PPE required, atmospheric test results.
5. **Attendant:** A trained, equipped attendant must remain at the entry point at all times. They do not enter the space. Their job is to monitor and summon rescue if needed.
6. **Rescue plan:** Know exactly how you will rescue an incapacitated entrant (tripod and lifeline, or other). Never do a non-entry rescue attempt (no one has ever "quickly jumped in to save someone" in a confined space and had it go well — you become the second victim).
7. **Communication:** Maintain continuous contact between entrant and attendant.

## 9.7 Electrical Safety

- Only qualified electricians work on live circuits.
- Operators should know the location of all main disconnects and emergency shutoffs.
- **Arc flash PPE:** High-voltage panels have arc flash hazard labels. Do not open these panels without the correct arc flash PPE (based on the incident energy level on the label).
- Never reset a tripped breaker more than twice without investigating the cause.
- Never use water near electrical equipment without de-energizing first.
- Report any unusual heat, smell, discolouration, or sparking from electrical equipment immediately.

## 9.8 Chemical Safety — First Aid Summary

Always consult the specific SDS for the chemical involved. These are general first aid guidelines.

| Exposure Type                        | Immediate Action                                                                                                                                               |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Skin contact with corrosive chemical | Immediately flush with large amounts of water for at least 20 minutes. Remove contaminated clothing while flushing. Seek medical attention.                    |
| Eye contact with any chemical        | Immediately flush at the eyewash station for at least 15 minutes. Hold eyelids open. Seek medical attention even if no pain.                                   |
| Inhalation of chlorine gas           | Move to fresh air immediately. If breathing is affected, call 911. Do NOT re-enter the contaminated area without SCBA.                                         |
| Ingestion of any chemical            | Do NOT induce vomiting unless SDS specifically directs it. Call Poison Control (1-800-222-1222 in the US; check your regional number). Seek medical attention. |

Eyewash station use:

1. Pull the activation handle (this starts the water flow).
2. Hold eyelids open with fingers (your instinct is to squeeze them shut — override it).
3. Roll eyes in all directions.
4. Flush for a full 15 minutes — set a timer; it feels much longer than it is.
5. Call for help — don't flush alone if you can avoid it.

9.9 Hydrogen Peroxide Spill — Specific Procedure

Hydrogen peroxide requires different spill treatment than most chemicals. **Do not just flush with water for a large spill.**

Per the Brenntag Canada MSDS (Date of Revision: 2014 March 31):

"Apply cautiously a dilute solution of a reducing agent such as sodium sulphite or sodium bisulphite to the contained spill. Neutralization is expected to be exothermic. Vigorous effervescence results. Flush spill area with water."

- [ACTION] For a H<sub>2</sub>O<sub>2</sub> spill:
1. Put on full PPE: full face shield, chemical splash goggles, neoprene gloves, apron.
  2. Contain the spill — stop it spreading to floor drains or toward combustible material (wood, rags, organic material).
  3. Apply dilute sodium sulphite or sodium bisulphite solution cautiously. (Confirm neutralizer is stocked on site — check with supervisor.)
  4. The reaction will bubble/effervesce — this is expected. Stand back.
  5. Flush the neutralized area thoroughly with water.
  6. Dispose of waste material as hazardous waste.
  7. **Never use sawdust, rags, or organic absorbents on concentrated H<sub>2</sub>O<sub>2</sub> — fire risk.**

**i** WHY: H<sub>2</sub>O<sub>2</sub> is an oxidizer. Strong enough concentrations will ignite organic materials on contact. Flushing alone just spreads it. The reducing agent (sodium sulphite/bisulphite) converts H<sub>2</sub>O<sub>2</sub> into water and oxygen — safe end products.

CO<sub>2</sub> inhalation (confirmed from Air Liquide MSDS):

- BC TWA (8-hour): 5,000 ppm / 9,000 mg/m<sup>3</sup>
- STEL (15-min): 30,000 ppm / 54,000 mg/m<sup>3</sup>
- IDLH (immediately dangerous): 40,000 ppm
- CO<sub>2</sub> has no smell — you will feel dizziness, irregular heartbeat, and shortness of breath before losing consciousness
- If CO<sub>2</sub> leak suspected in an enclosed area: evacuate immediately, ventilate, do not re-enter without SCBA

9.10 Emergency Phone Numbers

| Contact                                                   | Number                       |
|-----------------------------------------------------------|------------------------------|
| Emergency services (fire/ambulance)                       | 911                          |
| Poison control (Canada)                                   | 1-800-268-9017               |
| SUMAS Environmental Services                              | 1-250-374-4151               |
| Air Liquide (CO <sub>2</sub> cylinder emergency)          | 1-800-817-7697               |
| Brenntag Canada (H <sub>2</sub> O <sub>2</sub> emergency) | (confirm from SDS Section 1) |
| Site supervisor — Brennan                                 | (confirm number)             |
| On-call operators — Rob, Glen                             | (confirm numbers)            |
| Regulatory authority (BC MoE 24-hr)                       | (confirm with supervisor)    |

## Section 10 — Emergency Response

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Emergencies are rare but their consequences can be catastrophic. Your job during an emergency is simple: protect public health, protect your people, and communicate. Read this section before an emergency happens — not during one.

### 10.1 Universal Emergency Protocol

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Regardless of the type of emergency, your first four actions are always the same:

**STEP 1: ASSESS**

Is anyone in immediate danger?  
Are you in immediate danger?

**STEP 2: PROTECT**

Evacuate or isolate people from the hazard.  
If finished water quality is in question – STOP distribution.

**STEP 3: COMMUNICATE**

Call 911 if life safety is at risk.  
Call your supervisor immediately.  
Notify the regulatory authority as required.

**STEP 4: DOCUMENT**

Start recording from the moment you notice the emergency.  
Time of first observation, actions taken, readings, who was called and when.

### 10.2 Chemical Spill

---

#### Small spill (within your training and PPE capabilities)

A small spill is one you can safely control with your spill kit — typically < 20 L of a diluted chemical.

**[ACTION]**

1. Put on full PPE (face shield, splash goggles, chemical gloves, apron) before approaching.
2. Eliminate ignition sources if the chemical is flammable (most water treatment chemicals are not, but check SDS).
3. Stop the source of the spill if possible (close valve, upright container).
4. Contain the spread: use absorbent berms or pigs from your spill kit to stop the spill from reaching floor drains.
5. Absorb the spill using appropriate spill absorbent (check SDS — some chemicals require neutralization before disposal).
6. Collect all contaminated material in a labelled, sealed waste container.
7. Decontaminate the area with clean water.
8. Dispose of waste per your facility's chemical waste procedures.
9. Report the spill in the logbook and to your supervisor, even if it was minor.

#### Large spill or any chlorine gas release

**[ACTION]**

1. Do NOT attempt to control a large spill yourself or any chlorine gas release.
2. **Evacuate** the area immediately — upwind and uphill.
3. **Alert** all personnel in the building.
4. **Call 911** — emergency responders (HAZMAT team) will manage the scene.
5. Call your supervisor.
6. Meet emergency responders at the entry point and provide them with the SDS and a description of what happened.
7. Do NOT re-enter until the all-clear is given by emergency responders.

**i WHY:** Chlorine gas at even low concentrations (3 ppm) causes severe respiratory distress. At 30 ppm, it is life-threatening within 30 minutes. You cannot fight it without SCBA and training. Call the professionals.

---

## 10.3 Power Failure

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### In the first 60 seconds:

#### [CHECK]

- Did the backup generator start automatically? (You should hear it within 15–30 seconds.)
- Are critical systems (pumps, dosing, instrumentation) on the backed-up circuit?
- Is the SCADA/HMI screen live or black?

### If generator auto-starts successfully:

#### [ACTION]

1. Confirm all critical processes are running on generator power.
2. Check all chemical dosing pumps — some may need to be manually restarted after a power blip.
3. Note the time of the outage in the logbook.
4. Contact the utility company to report the outage and get an estimated restoration time.
5. Monitor generator fuel level — know how long your fuel supply lasts at current load.

### If generator does NOT start automatically:

#### [ACTION]

1. Attempt manual start at the generator (per your generator's SOP).
2. If unsuccessful, call maintenance immediately.
3. Assess how long the plant can coast on the clear well storage before water supply is affected.
4. Notify the regulatory authority and your supervisor.
5. Begin load-shedding: prioritize disinfection and distribution pumps above everything else.

### Returning to grid power:

When utility power is restored:

- Do not switch back to grid power during a power surge — wait for stable voltage.
- Transfer back to grid in coordination with the generator's transfer switch procedure.
- After transfer, inspect all equipment for restart — some pumps and instruments may need manual reset.

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## 10.4 Equipment Failure — Critical Equipment

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### Raw water pump failure

#### [ACTION]

1. Confirm the failure (trip alarm, no flow on meter, breaker tripped).
2. Start the standby raw water pump (you do have a standby, right? — confirm during your orientation).
3. Investigate the failed pump (see Section 8 — Troubleshooting).
4. If both pumps fail: notify supervisor; manage on clear well storage; initiate emergency repair.

### Filter failure (turbidity spike)

#### [ACTION]

1. Take the affected filter offline (close inlet valve).
2. Divert flow to other filters (increase their load, staying within design limits).
3. Do NOT send high-turbidity filter water to the clear well.
4. Backwash the failed filter; if turbidity doesn't recover, investigate (see Section 8.3).

### Chlorine dosing failure (no chlorine residual)

#### [ACTION]

1. **Immediately** close the distribution system entry valve. Under-disinfected water must not leave the plant.
  2. Diagnose and fix the dosing failure (see Section 8.5).
  3. Once repaired, confirm residual is restored.
  4. DO NOT resume distribution until the finished water free chlorine meets the minimum residual requirement.
  5. Notify the regulatory authority per your permit requirements (this is often a mandatory reporting event).
-

## 10.5 Contamination Event / Adverse Water Quality

This covers: detection of an unusual taste or odour in finished water, an unexpected turbidity spike, a microbiological positive, or a chemical detection event.

### Immediate response:

#### [ACTION]

1. **Stop distribution** if the contaminant poses a health risk or if you are uncertain. Do not wait for confirmation to stop.
2. **Collect and preserve samples** from all points in the treatment train immediately — these are your evidence.
3. **Notify your supervisor** and the regulatory authority immediately. Most jurisdictions require notification within a few hours of a contamination event — check your permit.
4. **Do not flush the system yet** — preserve the evidence first.
5. Document everything: first sign of the problem, all test results, all actions taken, all people contacted.

### Investigation:

Work backward through the treatment train:

- Where in the process did the contamination enter or originate?
- Was there an upstream event? (algal bloom, agricultural runoff, spill in the watershed, infrastructure failure?)
- Was there a treatment failure? (missed dose, equipment failure, bypassed process?)
- Was there contamination in the distribution system? (backflow event, main break, unauthorized connection?)

### Communication during a Boil Water Advisory:

If a Boil Water Advisory (BWA) is issued by the regulatory authority:

- You are a technical resource; the regulatory authority issues the advisory and manages public communication.
- Cooperate fully with the authority's investigation.
- Document every step of the corrective action.
- BWA lifting requires: corrective actions complete, adequate disinfection restored, bacteriological samples confirming safety (typically two consecutive clean samples, 24 hours apart).

## 10.6 Flooding or Extreme Weather Events

#### [ACTION]

1. Monitor Environment Canada / local weather forecasts during severe weather seasons.
2. If flooding is possible:
  - Raise or sandbag critical electrical equipment
  - Ensure backup generator is fuelled and tested
  - Pre-position equipment and chemical supplies
  - Ensure you have contact information for all emergency services
3. If intake turbidity increases rapidly during a storm event: immediately run a jar test, increase coagulant dose, and increase monitoring frequency.
4. Extreme cold: ensure all chemical lines (especially sodium hypochlorite, which freezes near 0°C), instrument sample lines, and water lines are protected from freezing.

## 10.7 Emergency Contacts and Reporting Requirements

(Fill in with your site and jurisdiction-specific information.)

| Event                                         | Who to Notify                         | When                                    |
|-----------------------------------------------|---------------------------------------|-----------------------------------------|
| Finished water turbidity exceeds permit limit | Supervisor + Regulatory authority     | Per your permit — often within 24 hours |
| Chlorine residual below minimum               | Supervisor + Regulatory authority     | Per permit — often within hours         |
| Confirmed positive E. coli in distribution    | Regulatory authority + Public Health  | Immediately (same business day)         |
| Boil Water Advisory issued                    | Follow regulatory authority direction | As directed                             |
| Chemical spill > reportable quantity          | Regulatory authority + SPILLS line    | Immediately                             |
| Power failure > X hours                       | Supervisor                            | Per your protocol                       |

| Event                                       | Who to Notify | When        |
|---------------------------------------------|---------------|-------------|
| Major equipment failure affecting treatment | Supervisor    | Immediately |

**Regulatory reporting hotline:** *(TBD — check your operating permit)*

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*Last updated: 2026-06-04 | Source: Initial framework*

# Section 11 — Regulations & Reporting

You are operating a public utility that directly affects human health. Regulations exist because people got sick when plants were operated carelessly. Understanding the regulatory framework tells you what the minimum standard is — your goal is always to exceed it.

## 11.1 The Regulatory Framework (Canadian Context)

This section uses Canadian references as the default. Update with your specific jurisdiction and permit details.

### Federal Level

**Health Canada** sets the *Guidelines for Canadian Drinking Water Quality* — the national science-based benchmarks. These are not legally enforceable by Health Canada but are adopted by provinces as the basis for their regulations.

### Provincial/Territorial Level

Each province has its own drinking water act and regulations (e.g., Ontario's *Safe Drinking Water Act, 2002*; Alberta's *Water Act*). These set:

- Legal maximum contaminant levels (MCLs)
- Monitoring and reporting requirements
- Operator certification requirements
- Operating permit requirements

### Your Operating Permit

The single most important document for your plant. It specifies:

- Your plant's specific treatment requirements
- Sampling and monitoring frequencies
- Reporting requirements and deadlines
- Permit limits (your specific MCLs — may be stricter than provincial guidelines)
- Any site-specific conditions

**[ACTION]** Get a copy of your operating permit. Keep it in the control room. Read it cover to cover.

## 11.2 Key Parameters and Limits

The table below shows Health Canada guideline values. Your permit limits may be stricter. Fill in your site-specific limits once you have your permit.

| Parameter                             | Health Canada Guideline      | Your Permit Limit | Operational Target |
|---------------------------------------|------------------------------|-------------------|--------------------|
| Turbidity (treated)                   | ≤ 1 NTU                      | (TBD)             | (TBD)              |
| E. coli                               | 0 CFU/100 mL                 | 0 CFU/100 mL      | 0 CFU/100 mL       |
| Total coliforms                       | 0 CFU/100 mL (treated)       | (TBD)             | 0 CFU/100 mL       |
| Free chlorine residual (distribution) | Maintain measurable residual | (TBD)             | (TBD)              |
| pH                                    | 6.5–8.5                      | (TBD)             | (TBD)              |
| Fluoride                              | ≤ 1.5 mg/L                   | (TBD)             | (TBD)              |
| Trihalomethanes (THMs)                | ≤ 0.1 mg/L (100 µg/L)        | (TBD)             | Minimize           |
| Haloacetic acids (HAAs)               | ≤ 0.08 mg/L (80 µg/L)        | (TBD)             | Minimize           |
| Lead                                  | ≤ 0.005 mg/L (5 µg/L)        | (TBD)             | Minimize           |
| Nitrate (as N)                        | ≤ 10 mg/L                    | (TBD)             | —                  |

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## 11.3 Monitoring Requirements

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Your permit will specify:

- **What** to test (parameters)
- **Where** to sample (sample points)
- **How often** to sample (continuous, daily, weekly, monthly, annually)
- **Who must analyze** the sample (operator, accredited lab)
- **Holding times** (how fast samples must reach the lab)

### Microbiological monitoring

Bacteriological samples (E. coli, total coliform) must typically be analyzed by a certified/accredited laboratory. These cannot be done in-house without accreditation. Know your lab's name, pickup schedule, and whether they have emergency testing capabilities.

### THM/HAA monitoring

Usually quarterly or annually. Must be sent to an accredited lab. Sample collection and preservation must follow strict protocols (check with your lab — typically requires a special pre-dosed container and 4°C preservation).

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## 11.4 Reporting Requirements

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### Routine reporting

Most provinces require:

- Monthly operating reports submitted to the regulatory authority
- Annual operating reports
- Records kept for a minimum of 5–10 years (check your permit)

Your monthly report typically includes:

- Flow totals
- Treatment chemical usage
- All water quality test results
- Any permit exceedances with explanation of corrective action
- Maintenance activities

### Adverse event reporting

Certain events trigger mandatory reporting to the regulatory authority:

- Turbidity exceedance above permit limit
- Chlorine residual below minimum
- Any positive E. coli or total coliform result
- Treatment system failure affecting water quality
- Chemical spills
- Equipment failures with water quality implications

**These are typically required within hours of the event — check your permit for timelines.** Late reporting is itself a compliance violation. When in doubt, report immediately.

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## 11.5 Operator Certification

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### Why certification matters

Water treatment and distribution operators in Canada must hold a valid licence issued by the provincial authority. Operating without a licence (or letting an unlicensed person operate independently) is illegal and puts the utility at risk.

### Certification levels



Most provinces have 4–5 levels based on plant size and complexity:

- **Class 1 / Level 1:** Small, simple plants. Entry point for new operators.
- **Class 2 / Level 2:** Medium-sized plants.
- **Class 3 / Level 3:** Large, complex plants.
- **Class 4 / Level 4 (some provinces):** Very large or complex systems.

Your plant is classified at a specific level — the operator-in-charge must hold a licence at or above that level.

## Getting certified

Requirements vary by province but typically include:

1. Minimum age (usually 18)
2. Application and fee
3. Written examination
4. Work experience (accumulated as a trainee/apprentice under a licensed operator)
5. Continuing education (to maintain certification — e.g., points every 3 years)

**Talk to your supervisor about the certification pathway at your plant.**

## Keeping your certification current

- Track your continuing education hours.
- Renew your licence before it expires — lapsed licences require re-examination in many jurisdictions.
- Any criminal charges or convictions may affect your licence.

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## 11.6 Record Keeping Best Practices

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### Why records matter:

- Prove to regulators that you were compliant
- Protect YOU personally if there is ever a contamination event or legal action
- Help diagnose recurring problems over time
- Required for accreditation audits

### What to keep:

- All operator log sheets (all shifts, all days)
- All laboratory test results (original, not copies)
- All calibration records for instruments
- All maintenance records
- All chemical delivery and inventory records
- All incident reports and corrective action records
- All regulatory correspondence

**For how long:** Check your permit, but 5–10 years is standard. Some parameters (lead/copper, certain microbiological results) may require longer retention.

**Where to keep them:** Organized, labelled, accessible. If an inspector shows up unannounced, you should be able to produce any record from the last 5 years within 30 minutes.

## Section 12 — Glossary

Every term you need to know, in plain English. When you see a word you don't recognize in this manual or from your supervisor, look here first.

| Term                                   | Plain-English Explanation                                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adsorption</b>                      | A process where dissolved molecules stick to the surface of a solid material (like activated carbon). Different from absorption — adsorption is surface sticking, not soaking in.                        |
| <b>Alkalinity</b>                      | A measure of water's ability to resist changes in pH (its "buffering capacity"). Expressed as mg/L as CaCO <sub>3</sub> . Low alkalinity means pH can swing dramatically when you add a coagulant.       |
| <b>Alum</b>                            | Common name for aluminum sulphate — the most widely used coagulant in water treatment. A white crystalline solid or clear liquid solution.                                                               |
| <b>Backwash</b>                        | Reversing the flow through a filter (or sending water upward through the bed) to clean out trapped particles and restore filter capacity.                                                                |
| <b>Backwash expansion</b>              | How much the filter bed "fluffs up" during backwashing. Typically targeted at 20–30% above the settled bed depth. Too little and the bed doesn't clean; too much and media washes out.                   |
| <b>Biofilm</b>                         | A community of bacteria that attaches to pipe walls or equipment surfaces and forms a slimy layer. Biofilm in distribution pipes can shelter pathogens from disinfectants and consume chlorine residual. |
| <b>BOD (Biochemical Oxygen Demand)</b> | A measure of how much oxygen microorganisms will consume to break down organic matter in a water sample. High BOD = high organic content.                                                                |
| <b>Boil Water Advisory (BWA)</b>       | An official public notice issued when drinking water may be microbiologically unsafe. Issued by the regulatory authority, not the operator.                                                              |
| <b>Buffer / Buffering capacity</b>     | A solution's ability to resist pH changes. Water with high alkalinity is well-buffered — adding acid or base doesn't change pH much. Low alkalinity = poorly buffered = pH swings easily.                |
| <b>Cavitation</b>                      | A damaging phenomenon in pumps: low-pressure zones cause water to vaporize into tiny bubbles, which then collapse violently. Sounds like gravel in the pump. Erodes impeller metal over time.            |
| <b>Chloramine</b>                      | A compound formed when chlorine reacts with ammonia. A weaker disinfectant than free chlorine but longer-lasting residual in distribution. Some utilities add ammonia intentionally ("chloramination").  |
| <b>Chlorine demand</b>                 | The amount of chlorine consumed by organic matter, iron, manganese, and other substances in water before any residual is established. Demand must be satisfied before a residual can exist.              |
| <b>Chlorine residual</b>               | The amount of chlorine remaining in treated water after the demand has been met. "Free chlorine residual" = hypochlorous acid + hypochlorite ion.                                                        |
| <b>Clarifier</b>                       | A tank or basin that uses settling (and sometimes chemical treatment) to remove suspended solids from water. Same basic idea as a sedimentation basin.                                                   |
| <b>Coagulant</b>                       | A chemical (typically alum or PACl) that neutralizes the electrical charge on suspended particles, allowing them to clump together.                                                                      |
| <b>Coagulation</b>                     | The process of neutralizing particle charges so they can stick together. The first step before flocculation.                                                                                             |
| <b>Coliform bacteria</b>               | A group of bacteria used as indicators of faecal contamination. Total coliform = broad group (some are non-faecal). E. coli = specifically faecal — a positive E. coli result is a serious concern.      |
| <b>Combined chlorine</b>               | Chlorine that has reacted with ammonia to form chloramines. Total chlorine – free chlorine = combined chlorine.                                                                                          |
| <b>Confined space</b>                  | An enclosed or partially enclosed space not designed for continuous occupancy, with limited entry/exit, that may have an atmospheric hazard. Examples: clear well, wet well, buried valve vault.         |
| <b>Contact time (CT)</b>               | Concentration of disinfectant (mg/L) × time (minutes) = CT value. Regulators specify minimum CT to achieve pathogen inactivation. Higher CT = better disinfection.                                       |
| <b>Cryptosporidium</b>                 | A protozoan parasite resistant to chlorine at practical doses. Causes cryptosporidiosis (severe diarrhea). Removed by filtration and inactivated by UV.                                                  |
| <b>DBPs (Disinfection Byproducts)</b>  | Compounds formed when chlorine reacts with natural organic matter (NOM). Primary DBPs of concern: trihalomethanes (THMs) and haloacetic acids (HAAs). Some are probable carcinogens.                     |
| <b>Differential pressure (DP)</b>      | The difference in pressure between two points — across a filter, valve, or meter. As a filter clogs, its DP increases. Used as the primary indicator of when to backwash.                                |

| Term                                                  | Plain-English Explanation                                                                                                                                                                       |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DPD</b>                                            | N,N-Diethyl-p-phenylenediamine — the reagent used in the colorimetric test for free and total chlorine. DPD-1 = free chlorine; DPD-3 = total chlorine.                                          |
| <b>E. coli (Escherichia coli)</b>                     | A faecal coliform bacterium used as the primary indicator of sewage or faecal contamination. Any detectable E. coli in finished water is unacceptable and triggers mandatory reporting.         |
| <b>Effluent</b>                                       | Water leaving a treatment process. "Filter effluent" = water that has passed through the filter.                                                                                                |
| <b>Floc</b>                                           | The clumps of particles formed during flocculation. Healthy floc looks like white or grey snowflakes. Good floc settles quickly and is easy to filter.                                          |
| <b>Flocculation</b>                                   | The process of gently mixing coagulated water to cause particles to collide and stick together into larger, heavier floc. Follows coagulation.                                                  |
| <b>Flow meter</b>                                     | An instrument that measures the volume of water passing a point per unit time (L/min, m³/hour, MGD). Essential for controlling chemical doses.                                                  |
| <b>Free chlorine</b>                                  | Chlorine in its active disinfecting forms: hypochlorous acid (HOCl) and hypochlorite ion (OCl⁻). HOCl is 80× more effective as a disinfectant than OCl⁻.                                        |
| <b>G-value</b>                                        | A measure of mixing intensity (turbulence) in units of s⁻¹. Rapid mix uses high G (300–1000 s⁻¹); flocculation uses low G (10–75 s⁻¹).                                                          |
| <b>GHS (Globally Harmonized System)</b>               | The international standard for chemical hazard communication — standardizes hazard labels and SDS formats worldwide.                                                                            |
| <b>Giardia</b>                                        | A protozoan parasite. More resistant to chlorine than bacteria but less so than Cryptosporidium. Removed by filtration; inactivated by UV or sufficient CT. Causes giardiasis ("beaver fever"). |
| <b>HAA (Haloacetic Acids)</b>                         | A class of chlorination disinfection byproducts (DBPs). Five HAAs (HAA5) are regulated. Formed when chlorine reacts with NOM.                                                                   |
| <b>Head / Headloss</b>                                | "Head" is a way of expressing pressure using the height of a water column. "Headloss" is the pressure drop across a filter or pipe — caused by resistance. Measured in metres or feet.          |
| <b>HOCl (Hypochlorous acid)</b>                       | The most effective form of free chlorine. Forms at lower pH. At pH 7.5, roughly 50% of free chlorine is HOCl; at pH 6.5, ~90% is HOCl. This is why pH control matters for disinfection.         |
| <b>Hydraulic detention time (HDT)</b>                 | How long water spends in a basin or chamber. HDT = Volume ÷ Flow rate. Longer HDT = more settling or more disinfection contact time.                                                            |
| <b>Hypochlorite (NaOCl)</b>                           | The weaker ionic form of free chlorine (OCl⁻). Less effective as a disinfectant than HOCl but more prevalent at higher pH. Also the name for liquid bleach.                                     |
| <b>Influent</b>                                       | Water entering a treatment process. "Filter influent" = water about to enter the filter. Opposite of effluent.                                                                                  |
| <b>Jar test</b>                                       | A small-scale laboratory simulation of coagulation/flocculation to determine the optimal chemical dose before making changes to the full-scale process.                                         |
| <b>kW / kWh</b>                                       | Kilowatt (power) and kilowatt-hour (energy). Track these for operational efficiency and cost management. Large pumps are the biggest energy consumers.                                          |
| <b>Lamella settler / tube settler / plate settler</b> | Inclined plates or tubes inside a clarifier that dramatically increase the effective settling area without increasing the basin footprint.                                                      |
| <b>LOTO (Lockout/Tagout)</b>                          | The procedure for isolating equipment from all energy sources before maintenance. Prevents workers from being injured by unexpected startup.                                                    |
| <b>MCL (Maximum Contaminant Level)</b>                | The legal maximum concentration of a contaminant in drinking water. Set by provincial regulations.                                                                                              |
| <b>Membrane filtration</b>                            | Filtration through semi-permeable polymer membranes with very small pores. Types: microfiltration (MF), ultrafiltration (UF), nanofiltration (NF), reverse osmosis (RO).                        |
| <b>mg/L</b>                                           | Milligrams per litre. The standard unit for expressing chemical concentrations in water. 1 mg/L ≈ 1 ppm (part per million).                                                                     |
| <b>µg/L</b>                                           | Micrograms per litre. Used for trace contaminants. 1 µg/L = 0.001 mg/L = 1 ppb (part per billion).                                                                                              |
| <b>NOM (Natural Organic Matter)</b>                   | Dissolved organic material from decomposing vegetation, algae, and soil. Causes colour and taste in water. Reacts with chlorine to form DBPs.                                                   |
| <b>NTU (Nephelometric Turbidity Unit)</b>             | The unit of turbidity measurement. Named after the nephelometric measurement method (measuring scattered light).                                                                                |
| <b>Operating permit</b>                               | The legally binding document from your regulatory authority that specifies exactly how your plant must operate.                                                                                 |
| <b>PACI (Polyaluminium Chloride)</b>                  | A pre-polymerized aluminum coagulant. Works better than alum at cold temperatures and over a wider pH range.                                                                                    |
| <b>PAC (Powdered Activated Carbon)</b>                | A fine carbon powder used to adsorb taste and odour compounds (geosmin, MIB) during algal events. Added upstream of coagulation.                                                                |
| <b>Pathogen</b>                                       | A disease-causing microorganism. In water treatment, primary concerns are bacteria (E. coli, Salmonella), viruses (norovirus, hepatitis A), and protozoa (Giardia, Cryptosporidium).            |

| Term                                                            | Plain-English Explanation                                                                                                                                                                                                  |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>pH</b>                                                       | A scale (0–14) measuring how acidic or basic a solution is. pH 7 = neutral; < 7 = acidic; > 7 = basic. Critical for coagulation effectiveness and chlorine disinfection.                                                   |
| <b>P&amp;ID (Piping and Instrumentation Diagram)</b>            | A technical drawing showing all pipes, equipment, valves, and instruments in a process and how they interconnect. Your roadmap to the plant.                                                                               |
| <b>PPE (Personal Protective Equipment)</b>                      | Equipment worn to protect against workplace hazards. Includes gloves, goggles, face shields, respirators, hard hats, etc.                                                                                                  |
| <b>Rapid mix</b>                                                | The high-speed, short-duration mixing step immediately after coagulant addition. Disperses the coagulant uniformly before it reacts unevenly.                                                                              |
| <b>Residual</b>                                                 | What's left after demand has been satisfied. "Chlorine residual" = chlorine remaining after demand is met. "Residual pressure" = pressure remaining at the end of a distribution pipe.                                     |
| <b>SCADA (Supervisory Control and Data Acquisition)</b>         | A computer-based control and monitoring system. Shows real-time readings from instruments, logs data, and allows remote control of equipment.                                                                              |
| <b>SDS (Safety Data Sheet)</b>                                  | A document provided by chemical manufacturers that contains hazard information, handling instructions, first aid, spill response, and disposal guidance. Required for every chemical on-site.                              |
| <b>Sedimentation</b>                                            | The removal of suspended particles from water by gravity settling. Particles denser than water sink to the bottom.                                                                                                         |
| <b>Short-circuiting</b>                                         | When water flows through a basin or contact chamber faster than the theoretical HDT — bypassing the full treatment time. Caused by uneven flow, wind, temperature stratification, or poor baffling.                        |
| <b>Sludge</b>                                                   | Settled particles removed from the bottom of sedimentation basins and clarifiers. Must be removed periodically to prevent it from re-suspending and carrying over to the filters.                                          |
| <b>Softening</b>                                                | Removal of hardness (calcium and magnesium) from water. Not always required; depends on source water and customer needs.                                                                                                   |
| <b>SOR (Surface Overflow Rate)</b>                              | Flow rate ÷ basin surface area. A key design parameter for sedimentation. If SOR is too high, particles don't have time to settle before they overflow.                                                                    |
| <b>SOP (Standard Operating Procedure)</b>                       | A written, step-by-step procedure for performing a task. This manual is a collection of SOPs.                                                                                                                              |
| <b>Surface water</b>                                            | Water from a lake, river, stream, or reservoir. Subject to runoff and seasonal variation. Requires more treatment than groundwater in most cases.                                                                          |
| <b>THM (Trihalomethane)</b>                                     | A class of chlorination disinfection byproducts. The four most common (TTHM) are chloroform, bromodichloromethane, dibromochloromethane, and bromoform. Probable human carcinogens at high exposure.                       |
| <b>Turbidity</b>                                                | A measure of water cloudiness, caused by suspended particles scattering light. Measured in NTU. High turbidity in filtered water = inadequate treatment. Turbidity can "shield" pathogens from disinfectants.              |
| <b>UV (Ultraviolet)</b>                                         | Electromagnetic radiation used to disinfect water by damaging pathogen DNA. Effective against Cryptosporidium (which chlorine cannot adequately inactivate at practical doses).                                            |
| <b>UVT</b>                                                      | See NOM.                                                                                                                                                                                                                   |
| <b>UVT (UV Transmittance)</b>                                   | The percentage of UV light that passes through a water sample at 254 nm wavelength. Higher UVT = more UV gets to pathogens = more effective disinfection. Turbidity and NOM reduce UVT.                                    |
| <b>Water hammer</b>                                             | A pressure surge in a pipe caused by a sudden change in flow velocity — e.g., rapidly closing a valve or starting/stopping a pump abruptly. Can crack pipes and damage valves.                                             |
| <b>WHMIS (Workplace Hazardous Materials Information System)</b> | Canada's hazard communication standard, now aligned with GHS. Covers labels and SDS requirements.                                                                                                                          |
| <b>Zeta potential</b>                                           | The electrical charge on suspended particles in water. Most particles are negatively charged, which keeps them apart (they repel each other). Coagulants neutralize this charge, allowing particles to approach and stick. |

## Site-Specific Terms (Added from Photos)

| Term                 | Plain-English Explanation                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AODD pump</b>     | Air-Operated Double-Diaphragm pump. Runs on compressed air, no electricity. Ideal for sludge and thick slurries. Makes a characteristic rhythmic "chunk-chunk" sound. |
| <b>Accu-Tab Blue</b> | Brand name for calcium hypochlorite tablet chlorinator system. Class 5.1 oxidizer. Used on this site for final disinfection.                                          |
| <b>Cal-Hypo</b>      | Short for calcium hypochlorite — a dry or tablet form of chlorine disinfectant. Stronger and more stable than liquid bleach (sodium hypochlorite).                    |

| Term                                     | Plain-English Explanation                                                                                                                                                                                                  |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Carbon vessel / GAC vessel</b>        | A pressure vessel filled with granular activated carbon (GAC). Removes dissolved organics, petroleum hydrocarbons, colour, and odour by adsorption. Cannot be regenerated on-site — media must be replaced when exhausted. |
| <b>CD30TIX</b>                           | Model name for the ozone generation system on this site. Generates ozone (O <sub>3</sub> ) from air for iron oxidation and disinfection.                                                                                   |
| <b>AERQUS</b>                            | Brand name for the ozone diffuser used to introduce ozone into the water stream.                                                                                                                                           |
| <b>CK-311</b>                            | Product name for the PACl coagulant on this site (Aluminum Hydroxide Chloride Solution), supplied by CARBONeT Holdings through SUMAS. pH 2.2–2.8.                                                                          |
| <b>CL-200</b>                            | Product name for the liquid polymer flocculant on this site (aqueous solution of water-soluble polymer), supplied by SUMAS. pH 6.5–7.5.                                                                                    |
| <b>Flopham</b>                           | Brand name for the dry polyacrylamide-based polymer on this site. Must be pre-dissolved in water before use.                                                                                                               |
| <b>HaloKlear LiquiFloc</b>               | Brand name for a chitosan acetate flocculant from HaloSource Inc. A natural biopolymer flocculant.                                                                                                                         |
| <b>Fe<sup>2+</sup> (ferrous iron)</b>    | Dissolved, invisible iron. Cannot be filtered. Must be oxidized (by H <sub>2</sub> O <sub>2</sub> or ozone) to Fe <sup>3+</sup> before it can be removed.                                                                  |
| <b>Fe<sup>3+</sup> (ferric iron)</b>     | Oxidized iron that forms reddish-brown iron hydroxide precipitate [Fe(OH) <sub>3</sub> ]. This is the orange staining you see everywhere. Can be filtered.                                                                 |
| <b>Filter cake</b>                       | The semi-dry slab of compressed sludge solids that comes out of the filter press after a press cycle. Regulated waste — must be removed by licensed contractor.                                                            |
| <b>Filter press</b>                      | Machine that squeezes water out of sludge by pressing it between a stack of filter plates lined with filter cloth. Produces dry filter cakes and clean filtrate.                                                           |
| <b>Flocculant</b>                        | A chemical that bridges between coagulated particles to make them grow into larger, heavier floc. CL-200 and Flopham are the flocculants on this site.                                                                     |
| <b>Frac tank</b>                         | Large portable above-ground steel tank (typically 100,000 L) used for water storage. Originally developed for the oil/gas industry. Used here to store dirty, in-treatment, and treated water.                             |
| <b>Filtrate</b>                          | The water that comes out of the filter press after pressing. Usually returned to the head of the treatment system.                                                                                                         |
| <b>Iron hydroxide</b>                    | Fe(OH) <sub>3</sub> — the reddish-brown solid that forms when dissolved ferric iron (Fe <sup>3+</sup> ) reacts with water. This is what causes all the orange staining at this facility.                                   |
| <b>Kontek APC5</b>                       | Model of the filter press on this site. Manufacturer: Kontek Ecology Systems Inc., Burlington, Ontario. 120V, 1-phase.                                                                                                     |
| <b>LES HALL filter</b>                   | Brand of filter bags used in the bag filter housing on this site. Available in 10-micron (fine) and 100-micron (coarse) ratings.                                                                                           |
| <b>OWS</b>                               | Oil/Water Separator. A chamber that allows petroleum products to float and be skimmed, while heavier water sinks. The Shell Canada system at this site is an OWS.                                                          |
| <b>PV SERIES</b>                         | Brand designation on the SRS-CV-15 carbon vessel. "PV" = pressure vessel.                                                                                                                                                  |
| <b>Sand filter / SRS-SF-11</b>           | The green pressure vessel sand filter on this site. Removes iron floc and suspended solids. Tagged SRS-SF-11.                                                                                                              |
| <b>Carbon vessel / SRS-CV-15</b>         | The blue PV Series pressure vessel carbon filter on this site. Tagged SRS-CV-15.                                                                                                                                           |
| <b>SUMAS Environmental Services Inc.</b> | The water treatment contractor managing this system. Contact: 1-250-374-4151. Located in BC.                                                                                                                               |
| <b>Separation cell</b>                   | Another name for the wet cell — the indoor concrete settling tanks.                                                                                                                                                        |
| <b>Stabilization pit / pond</b>          | The outdoor pond that receives oil-contaminated or heavily iron-laden water for primary settling and separation before further treatment.                                                                                  |
| <b>Wet cell</b>                          | The indoor multi-compartment concrete settling tanks where chemical flocculation occurs and sludge accumulates. A confined space.                                                                                          |

Last updated: 2026-06-05 | Source: Site photos (IMG\_2574–2631), equipment labels, MSDS documents, operational notes